

Computer Vision 1

1-2: Image Formation and Color

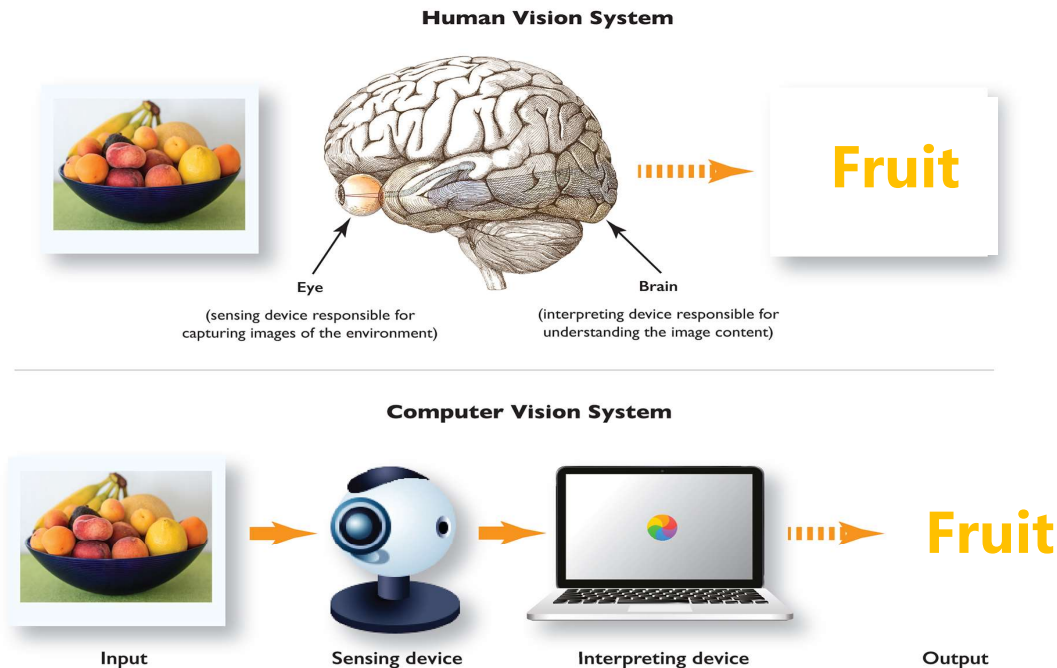
Master of AI – 2024/2025
University of Amsterdam

Arnoud Visser & Martin Oswald

Based on a presentation of Shaodi You,
augmented with material of Joe Redmons & many others

The computer vision problem

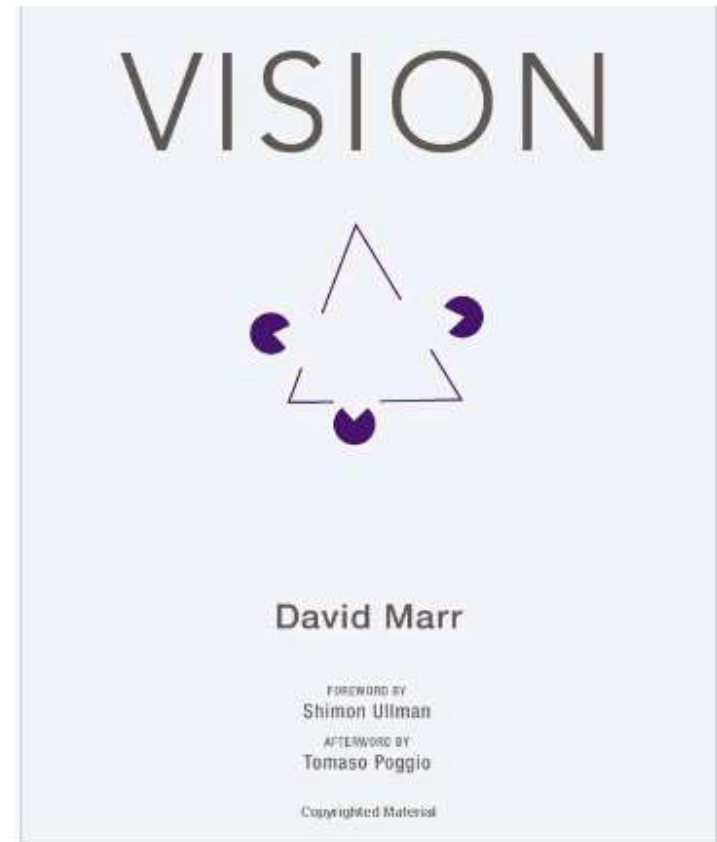
- Make a computer see and understand images



- We know it is physically possible –
we do it every day and effortlessly!

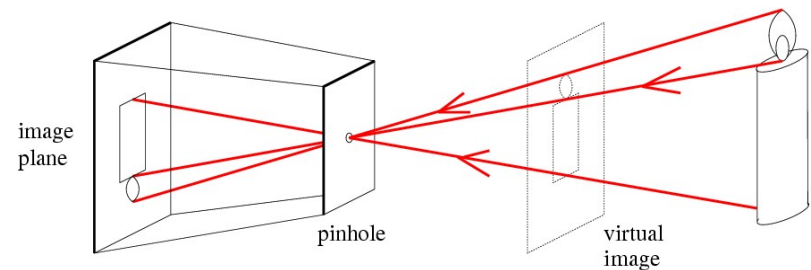
Three levels of vision processing

- Low-level vision:
 - image processing, denoising, filtering, image restoration, low-level feature extraction.
- Mid-level vision:
 - image analysis, image segmentation, contour extraction, perceptual organization, 2-1/2D representation, 3D information recovery.
- High-level vision:
 - visual detection, recognition, understanding, semantic labelling, activity, event detection.



A quick summary

- Pinhole camera model and camera projection matrix
- Homogeneous coordinates



$$\mathbf{x} = \mathbf{K} \begin{bmatrix} \mathbf{R} & \mathbf{t} \end{bmatrix} \mathbf{X}$$

$$(x, y) \Rightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Textbook

- 2.2,
- 2.3.1,
- 2.3.2

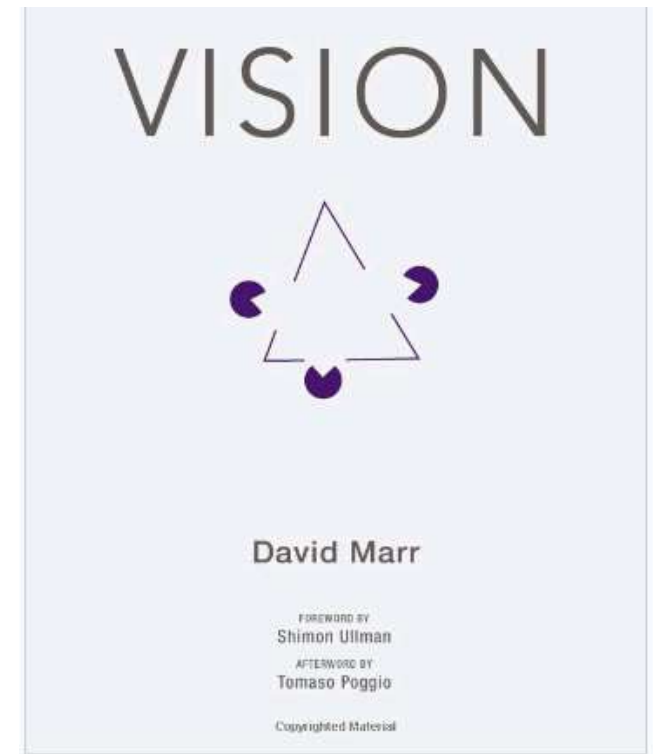
Outline

- Camera Model
 - A pinhole camera
 - Geometric image formation
 - Photometric image formation
 - Image representation
- Colour
 - Physical and Biological Model
 - Light source
 - Object
 - Observer
 - Tristimulus Theory
 - Colour Systems

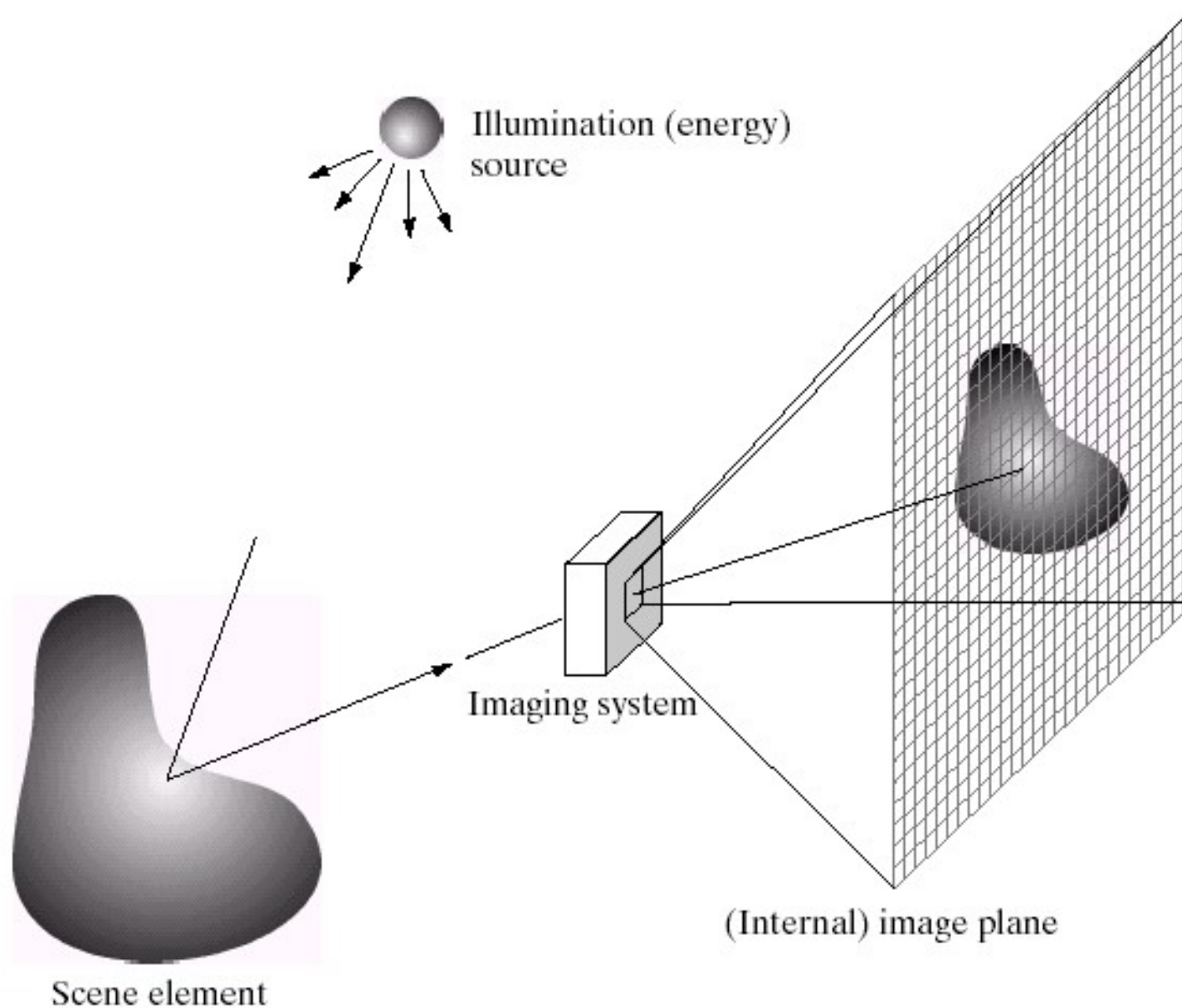


4 main factors which define intensity values

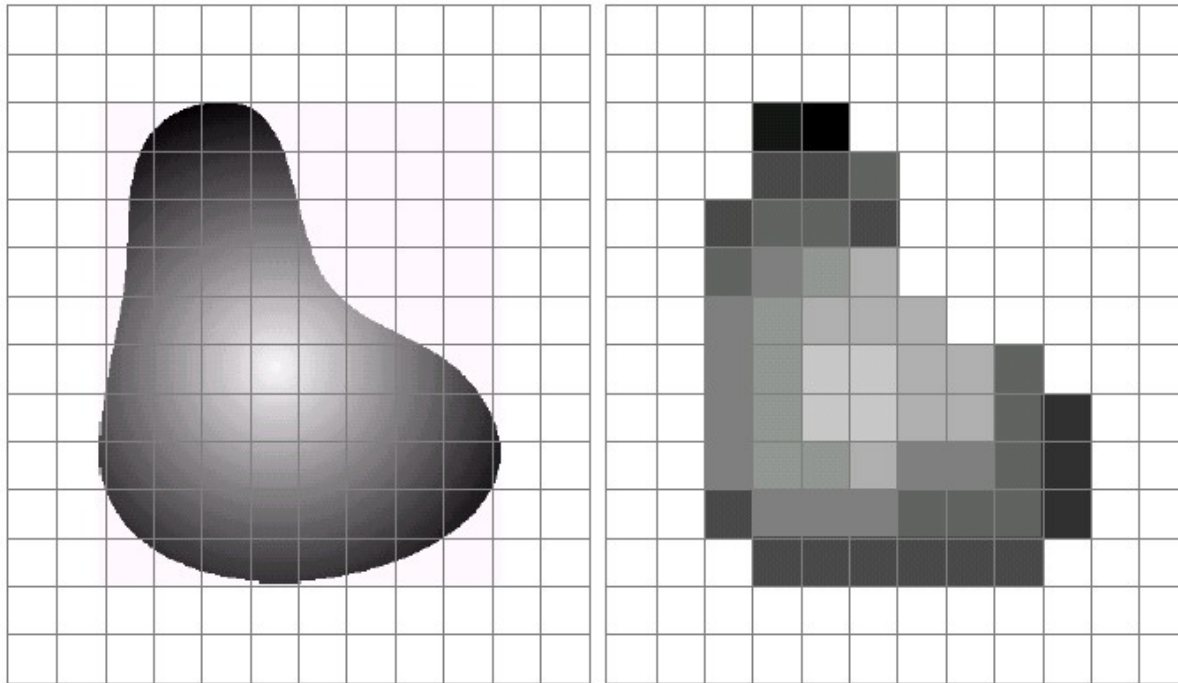
- the geometry of the scene
- the reflectances of the visible surface
- the illumination of the scene
- the viewpoint



Photometric Image Formation



Sensor Array



a b

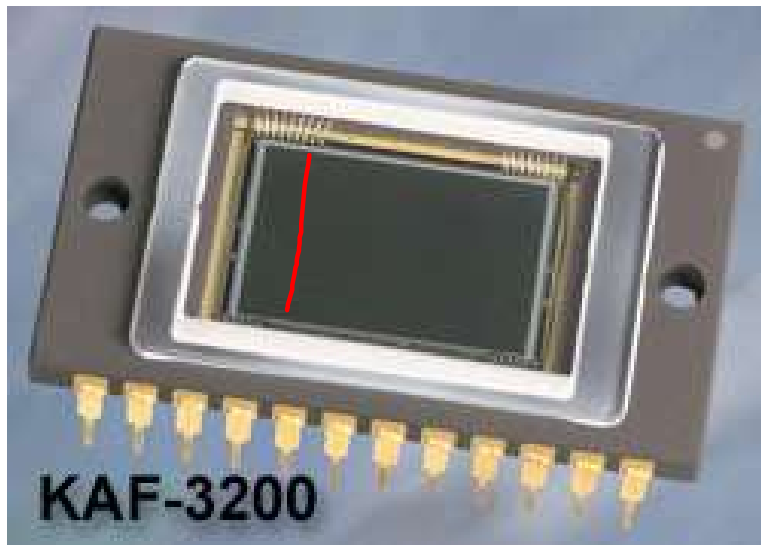
FIGURE 2.17 (a) Continuous image projected onto a sensor array. (b) Result of image sampling and quantization.



CCD/CMOS sensor

Image Sensors : Array Sensor

Charge-Coupled Device (CCD)



CCD KAF-3200E from Kodak.
(2184 x 1472 pixels,
Pixel size 6.8 microns²)

- Used for convert a continuous image into a digital image
- Contains an array of light sensors
- Converts photon into electric charges accumulated in each sensor unit

Not a perfect device



Low light



Dynamic Range



CCD overflow



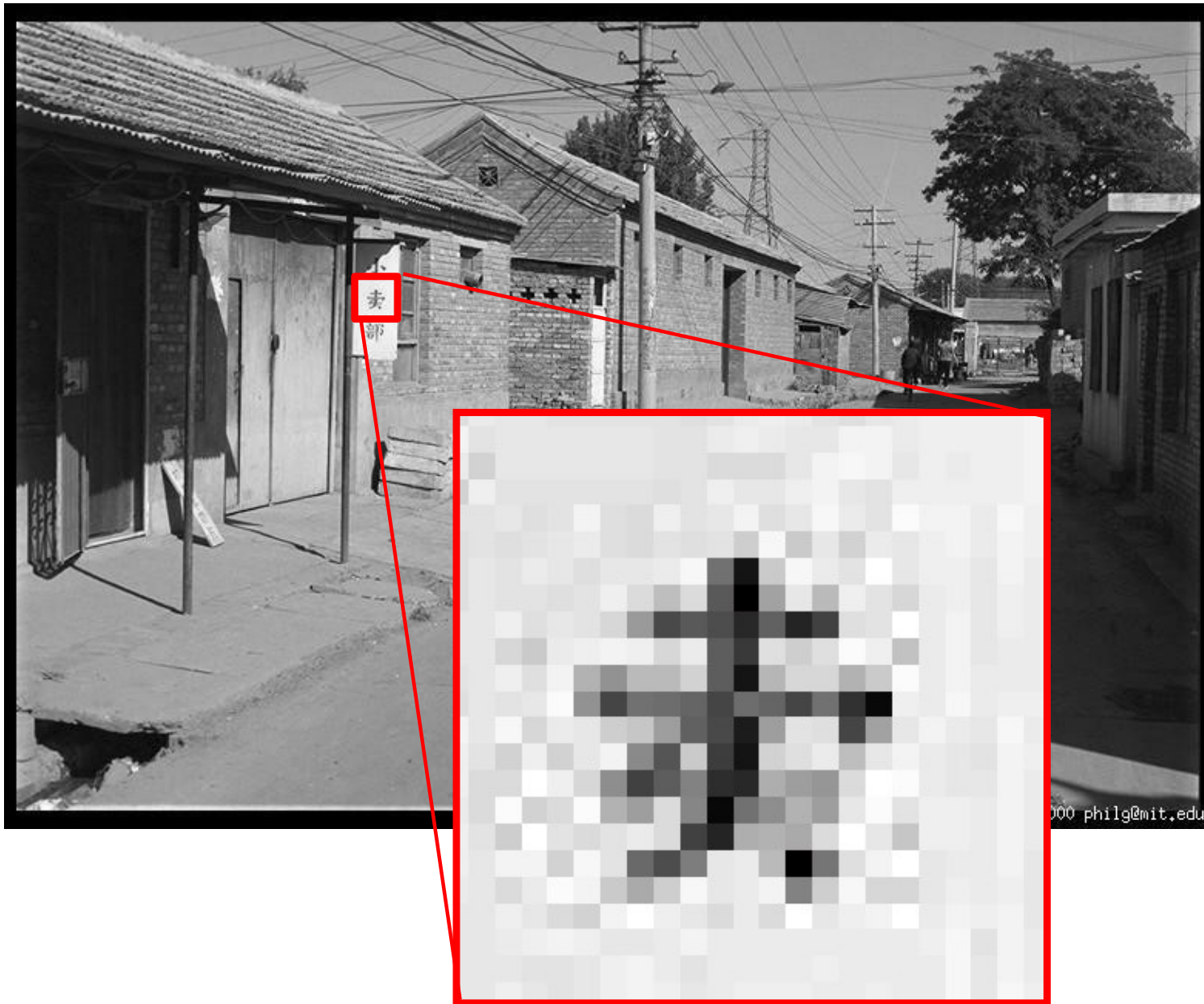
Rolling Shutter

Add more!

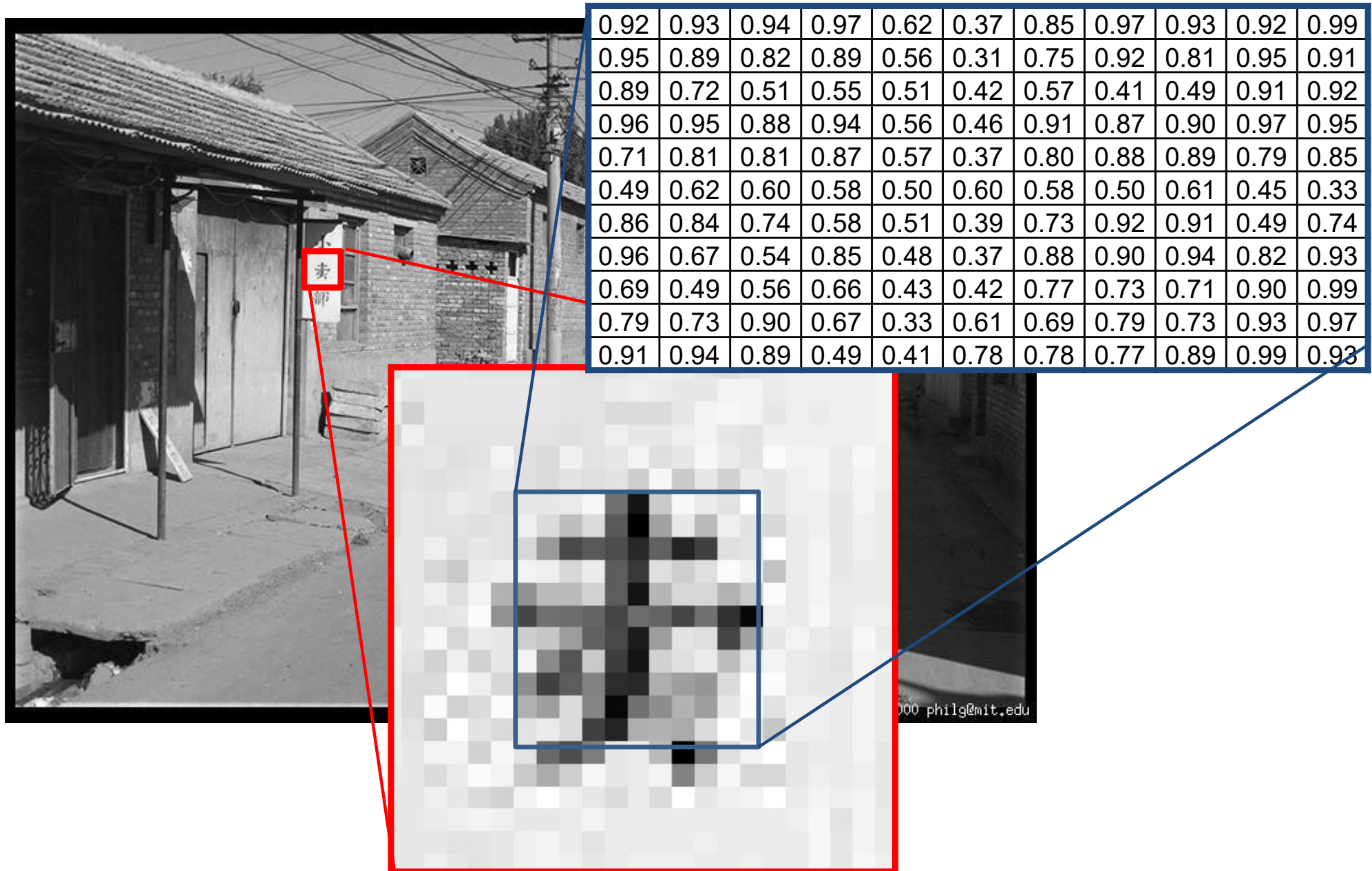
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The digital image (pixel matrix)



The digital image (pixel matrix)

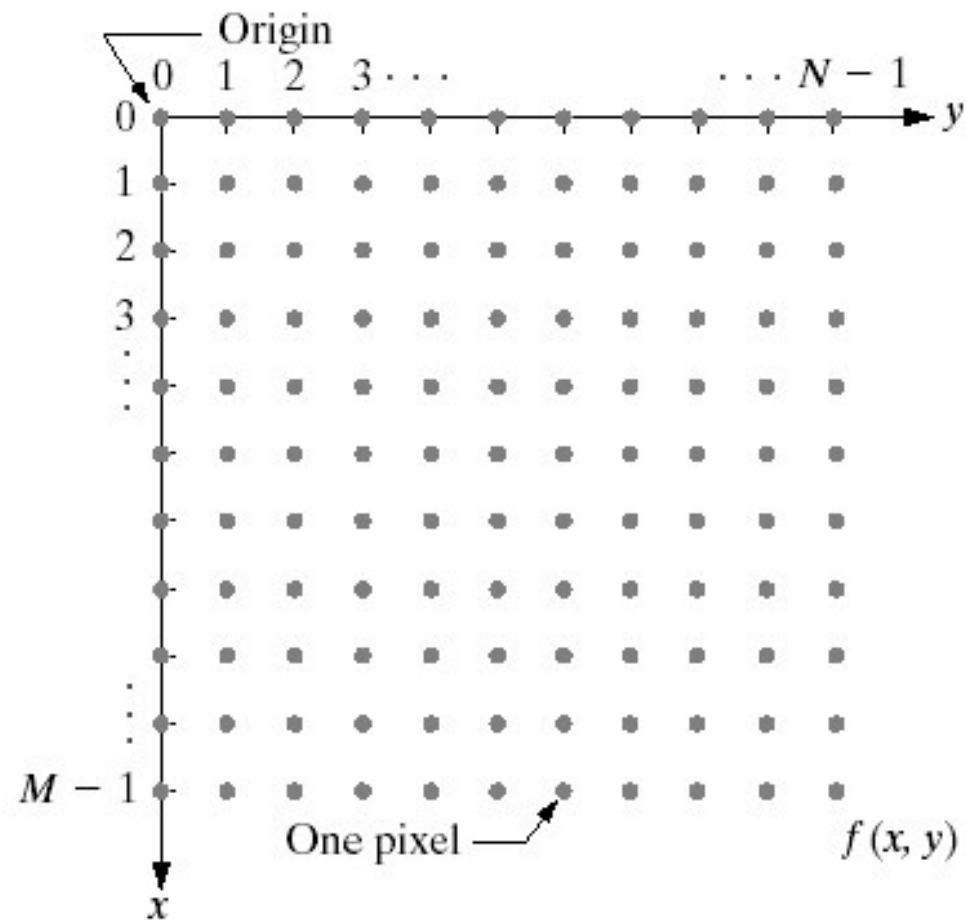


Fundamentals of Digital Images

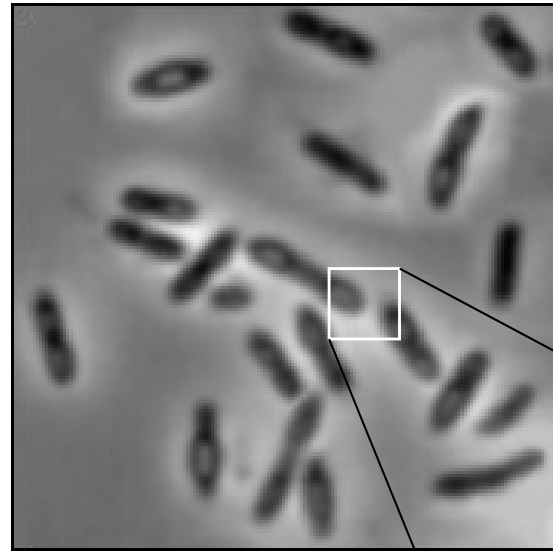


- ♦ **An image=a function of spatial coordinates.**
- ♦ **Spatial coordinate:** (x,y) for 2D case such as photograph,
 (x,y,z) for 3D case such as CT scan images
 (x,y,t) for video.
- ♦ The function f may represent intensity (for greyscale images) or color (for color images) or other associated values.

Conventional Coordinate for Image Representation

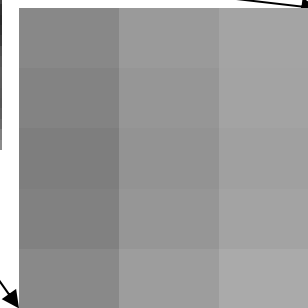
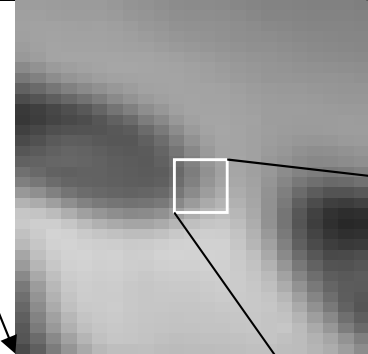


Gray-scale Image



Intensity image or grey-scale image

Each pixel corresponds to light intensity normally represented in gray scale, e.g. 0-black, 1-white (or 256 in 8-bit scale)



Gray scale values



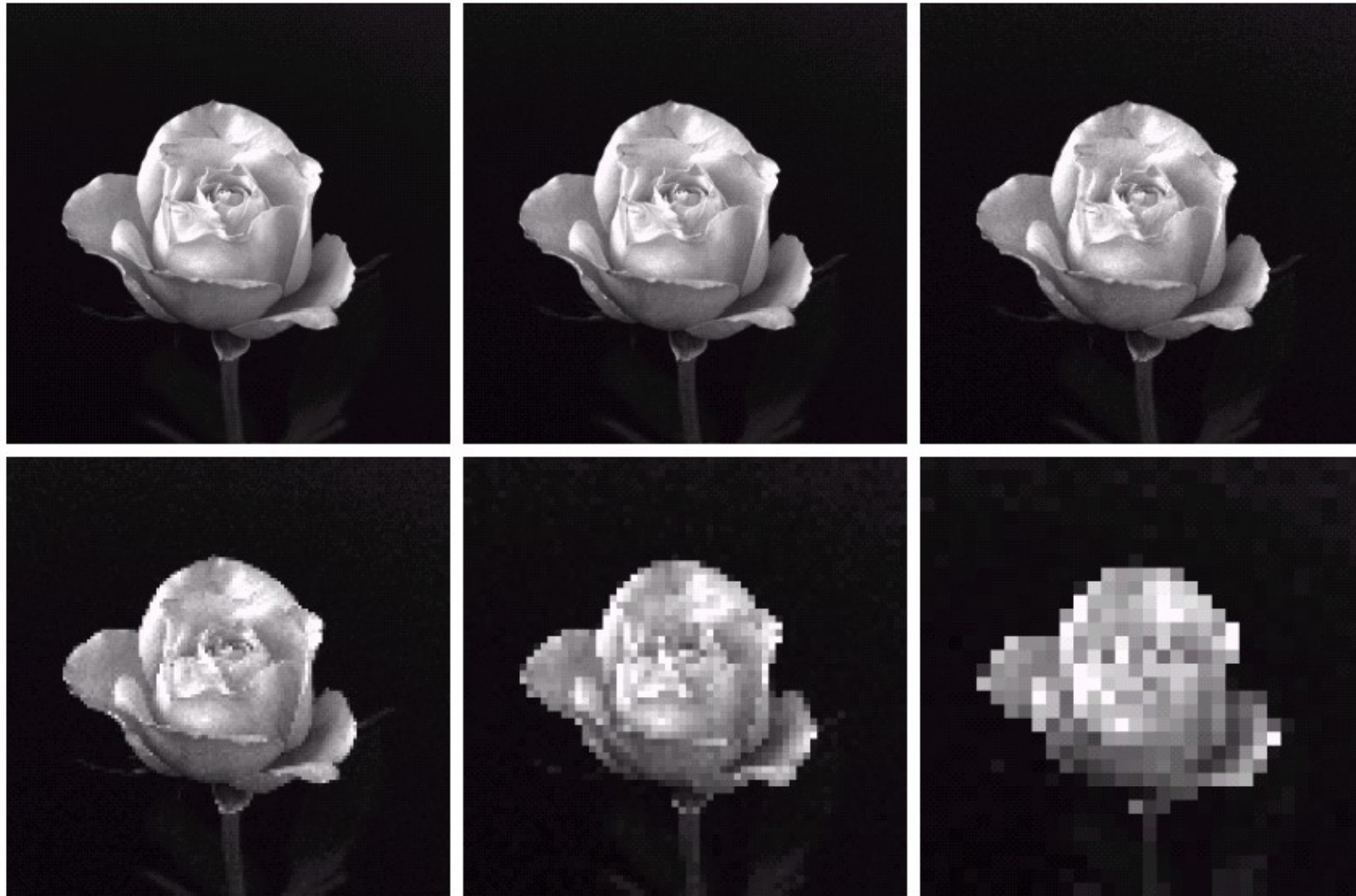
10	10	16	28
9	6	26	37
15	25	13	22
32	15	87	39

Effect of Spatial Resolution



FIGURE 2.19 A 1024×1024 , 8-bit image subsampled down to size 32×32 pixels. The number of allowable gray levels was kept at 256.

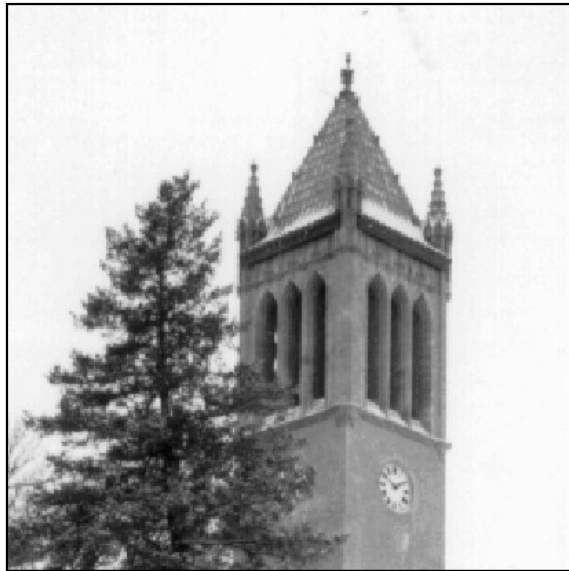
Effect of Spatial Resolution



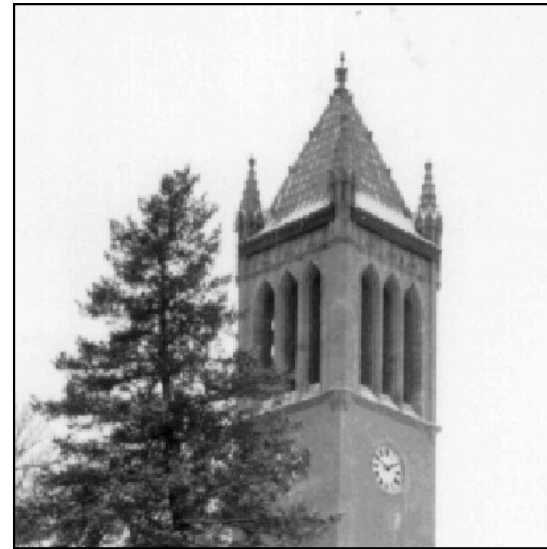
a	b	c
d	e	f

FIGURE 2.20 (a) 1024×1024 , 8-bit image. (b) 512×512 image resampled into 1024×1024 pixels by row and column duplication. (c) through (f) 256×256 , 128×128 , 64×64 , and 32×32 images resampled into 1024×1024 pixels.

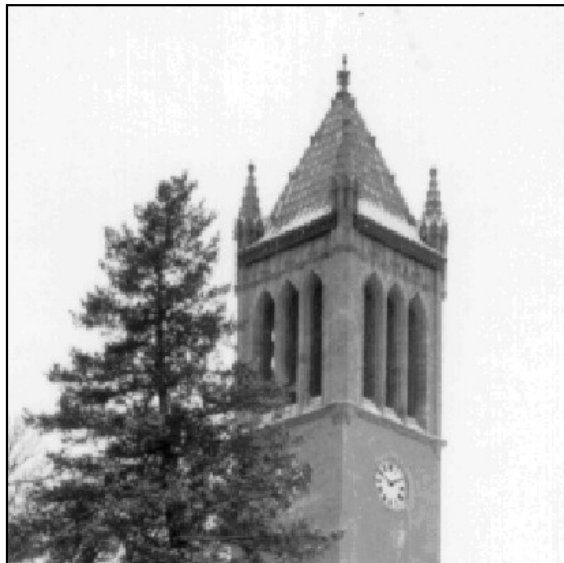
Effect of Quantization Levels



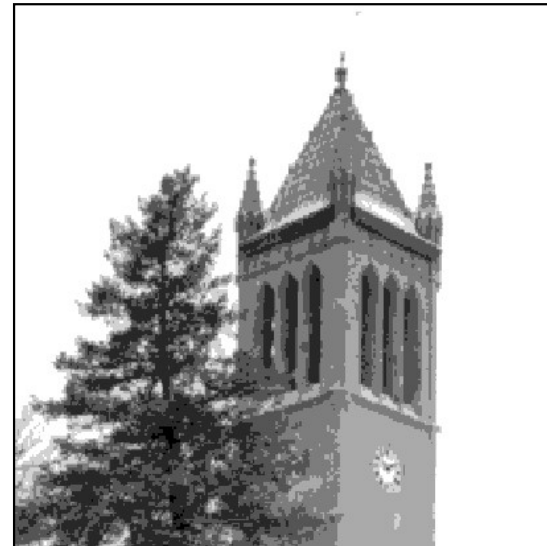
256 levels (8-bit)



128 levels (7-bit)

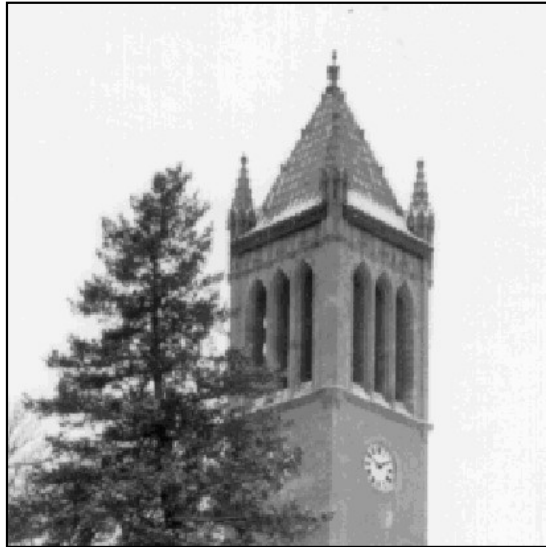


64 levels (6-bit)

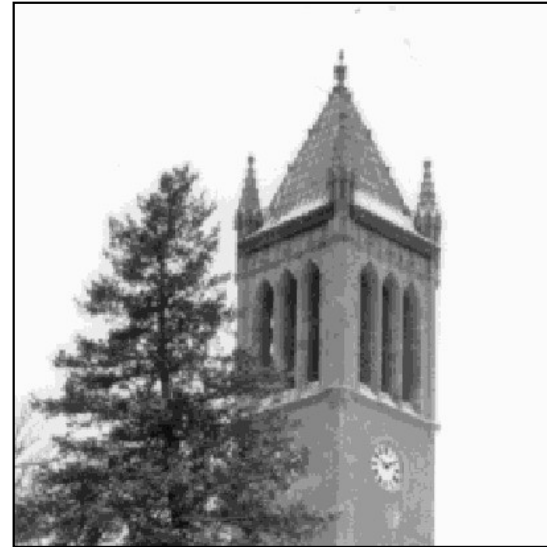


32 levels (5-bit)

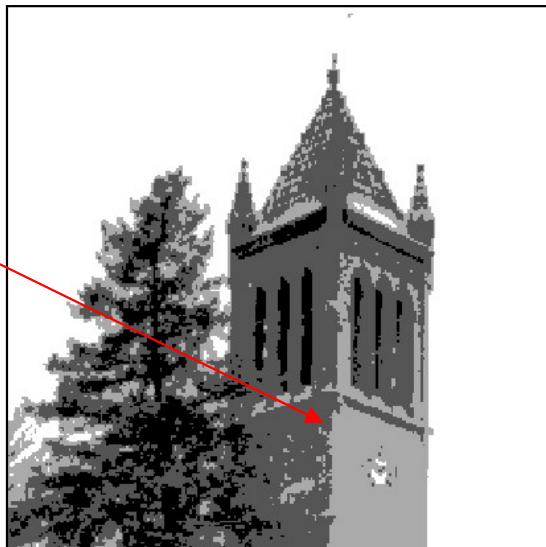
Effect of Quantization Levels (cont.)



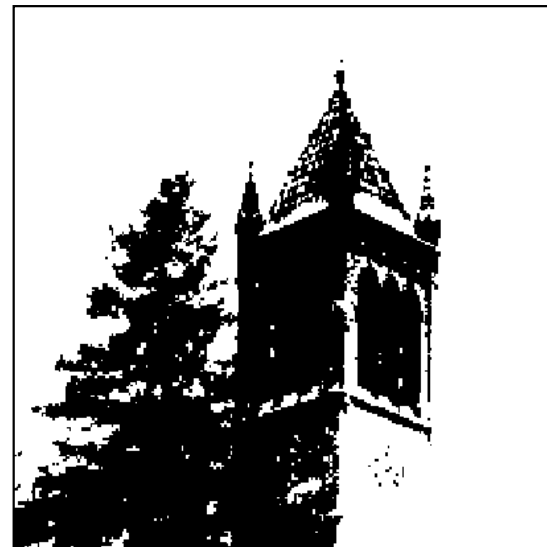
16 levels (4-bit)



8 levels (3-bit)



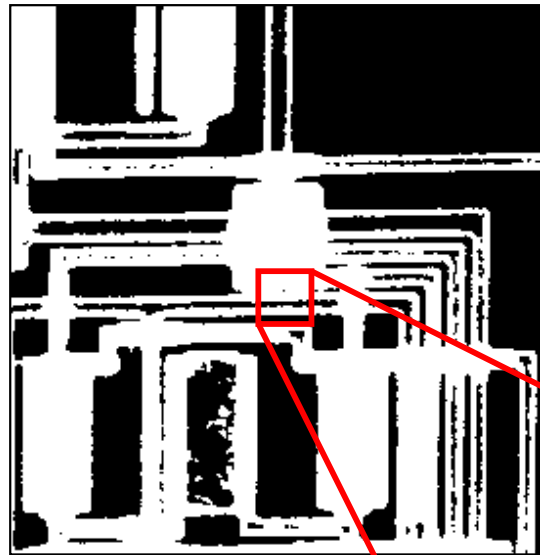
4 levels (2-bit)



2 levels (1-bit)

In this image,
it is easy to see
false contour.

Two-level image: binary image

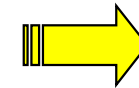
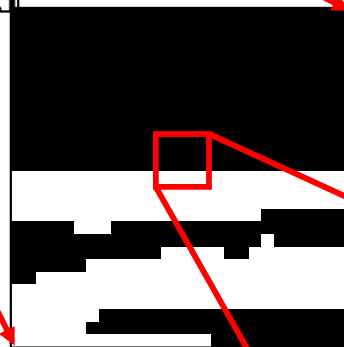


Binary image or black and white image

Each pixel contains one bit :

1 represent white

0 represents black

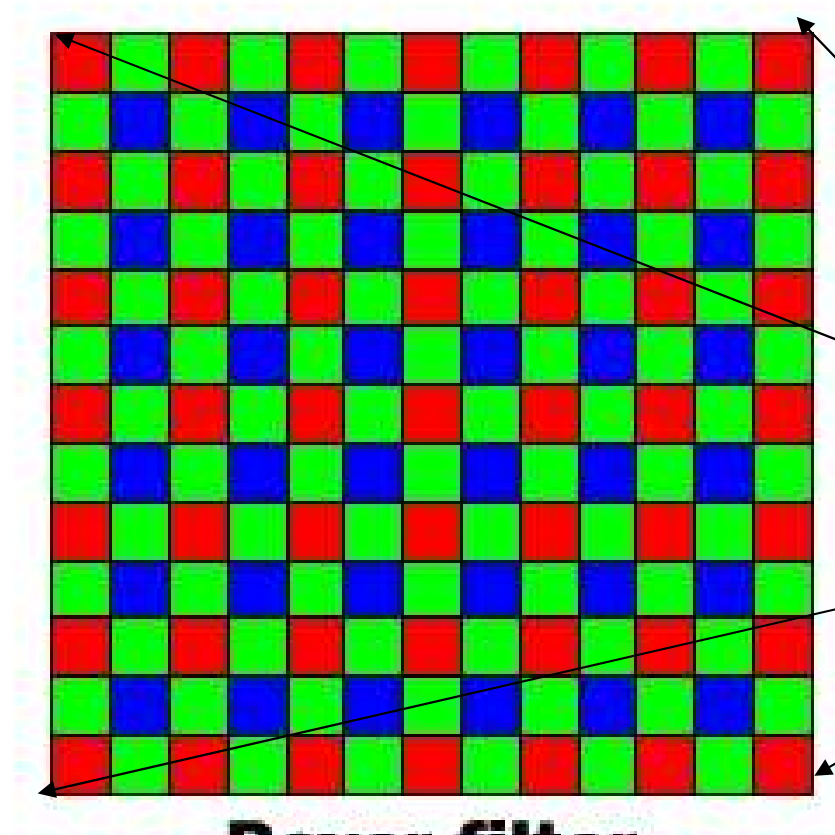


Binary data

0	0	0	0
0	0	0	0
1	1	1	1
1	1	1	1

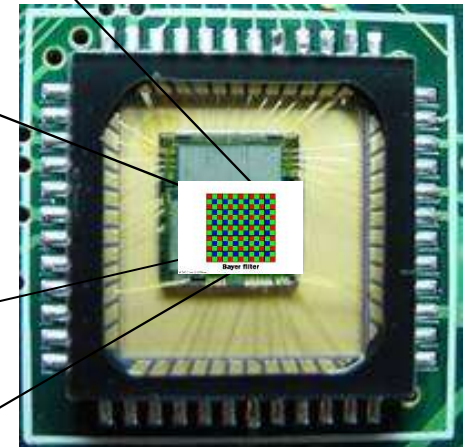
How to form colour images?

Bayer filter:
Green fills in half of
the checker board
and Red and Blue
fill the rest.

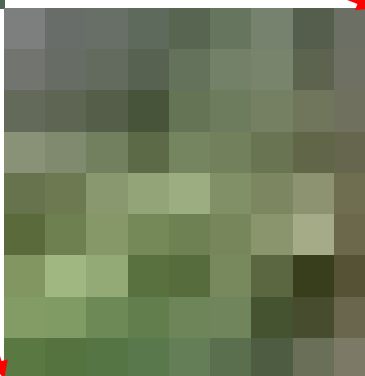


Bayer filter

© 2000 How Stuff Works



Colour Image



Color (RGB) image:

each pixel contains a vector representing red, green and blue components.

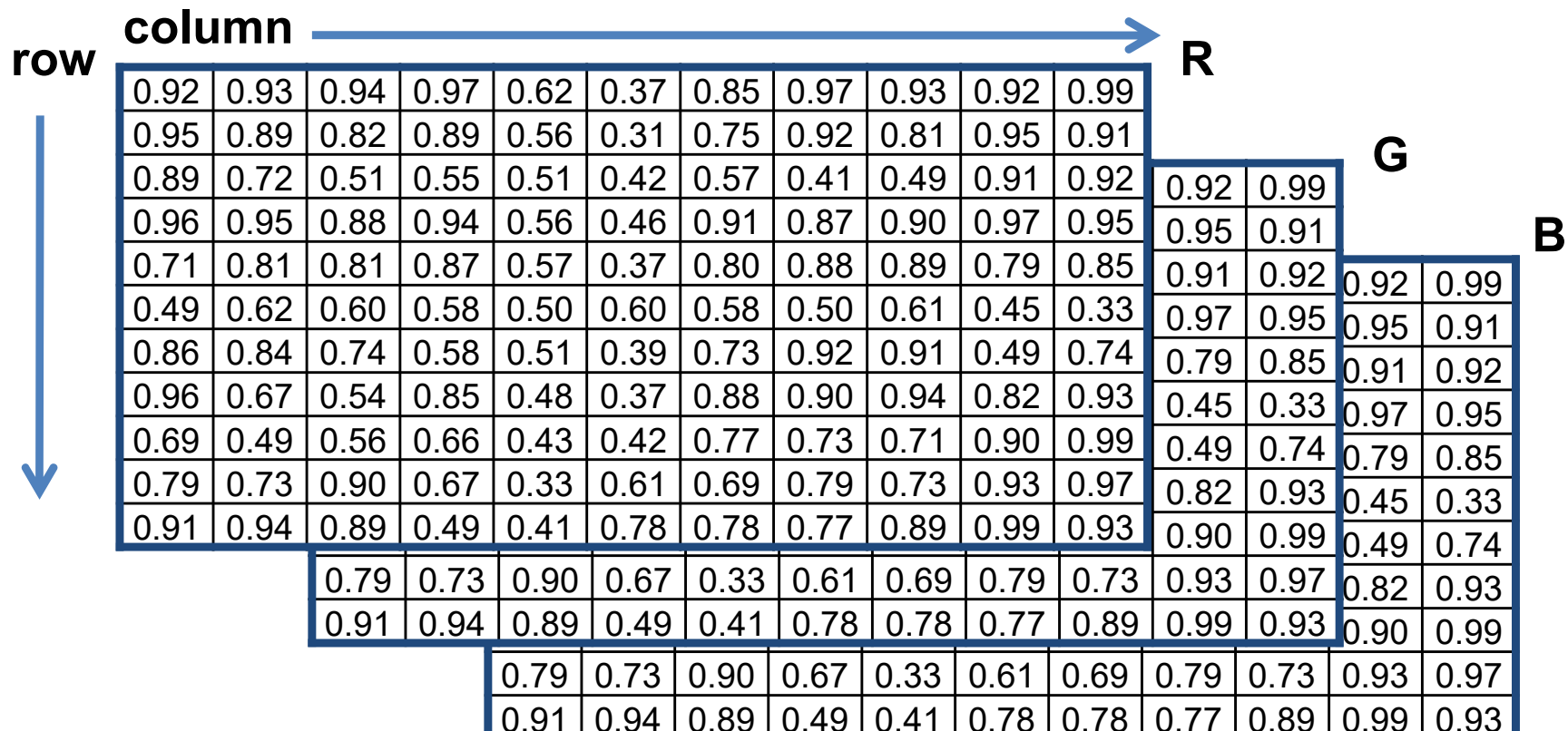
a multidimensional array of numbers (such as intensity image) or vectors (such as color image)

RGB components

10	10	16	28		
9	65	70	56	43	
15	32	99	70	56	78
32	21	60	90	96	67
	54	85	85	43	92
		32	65	87	99

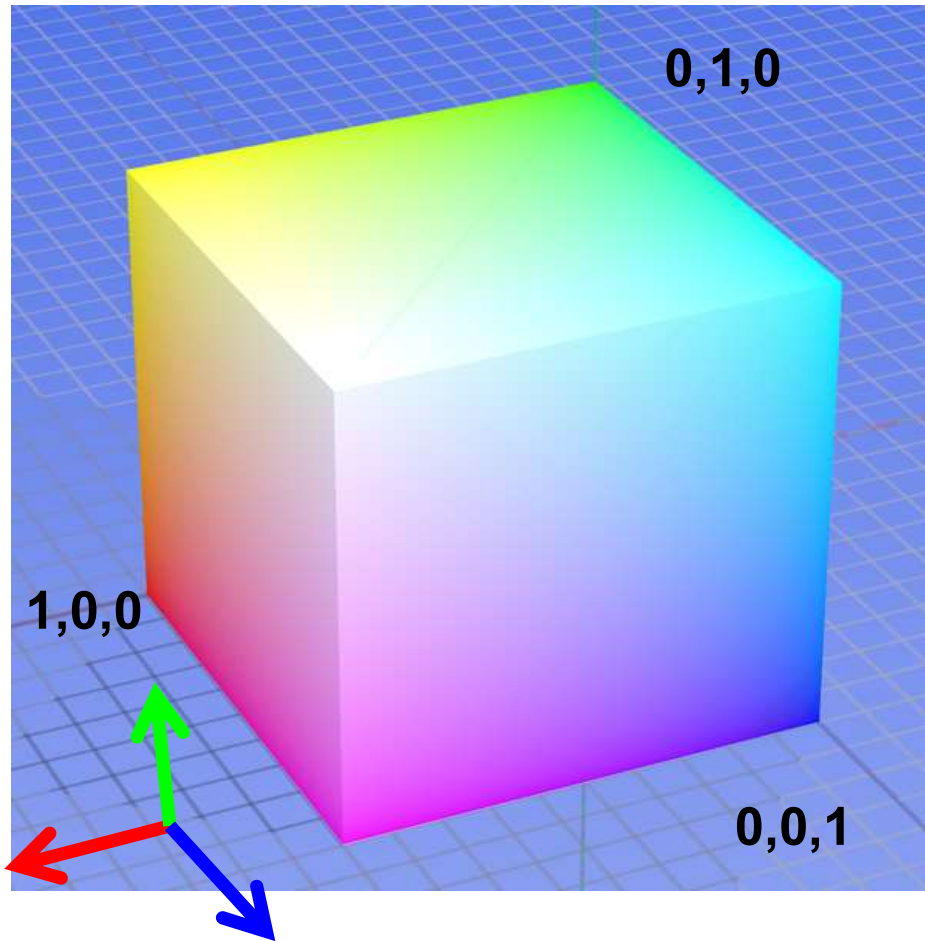
Colour Images in Matlab

- Images represented as a matrix
- Suppose we have a NxM RGB image called "im"
 - `im(1,1,1)` = top-left pixel value in R-channel
 - `im(y, x, b)` = y pixels down, x pixels to right in the bth channel
- `imread(filename)` returns a uint8 image (values 0 to 255)
 - **Convert to double format (values 0 to 1) with `im2double()`. (important !)**



RGB space

Default color space



Some drawbacks

- Strongly correlated channels
- Not perceptually meaningful



R
(G=0,B=0)

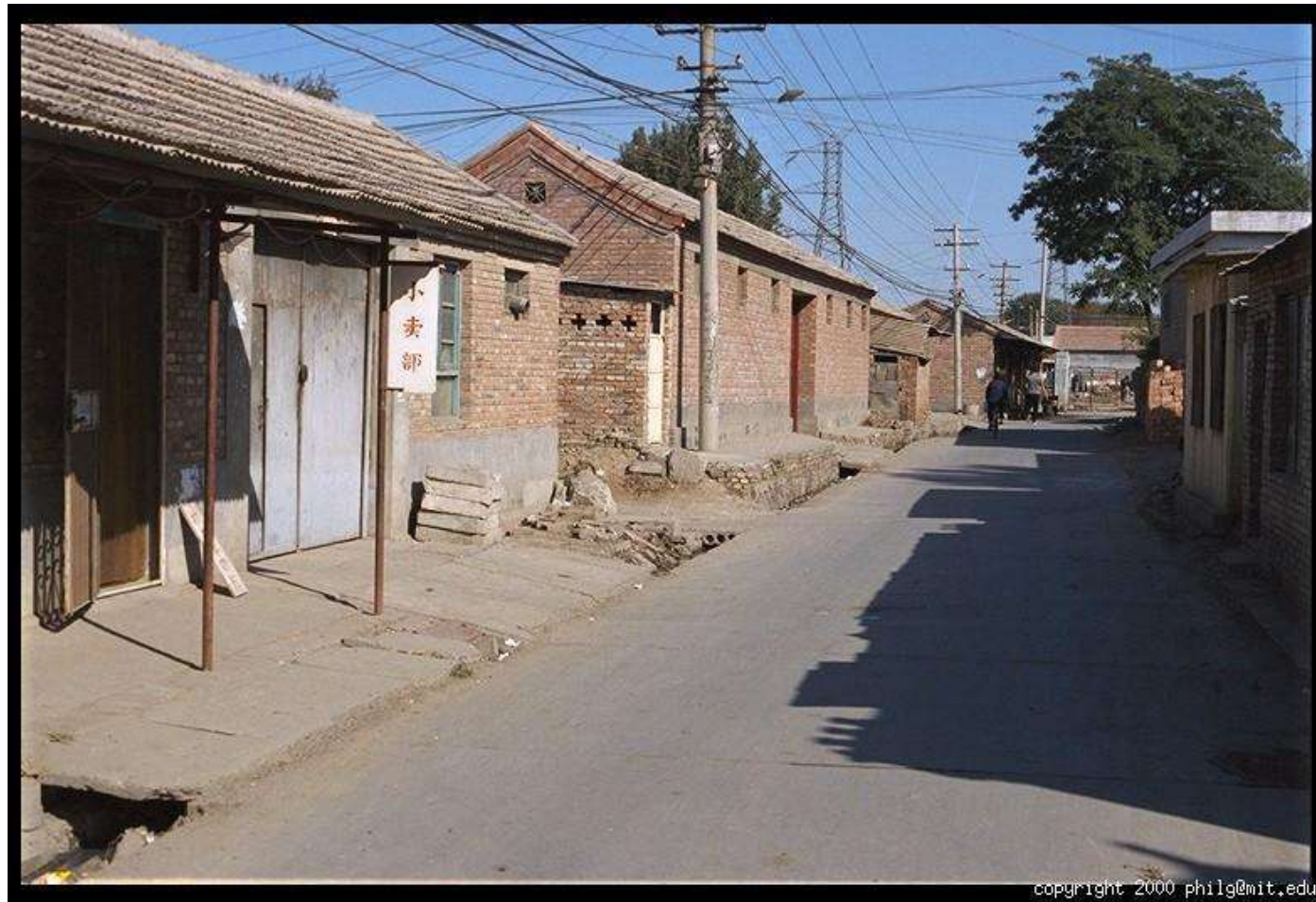


G
(R=0,B=0)



B
(R=0,G=0)

Most semantic information is however contained in the intensity band



Original image

Most information is contained in intensity channel



Greyscale image

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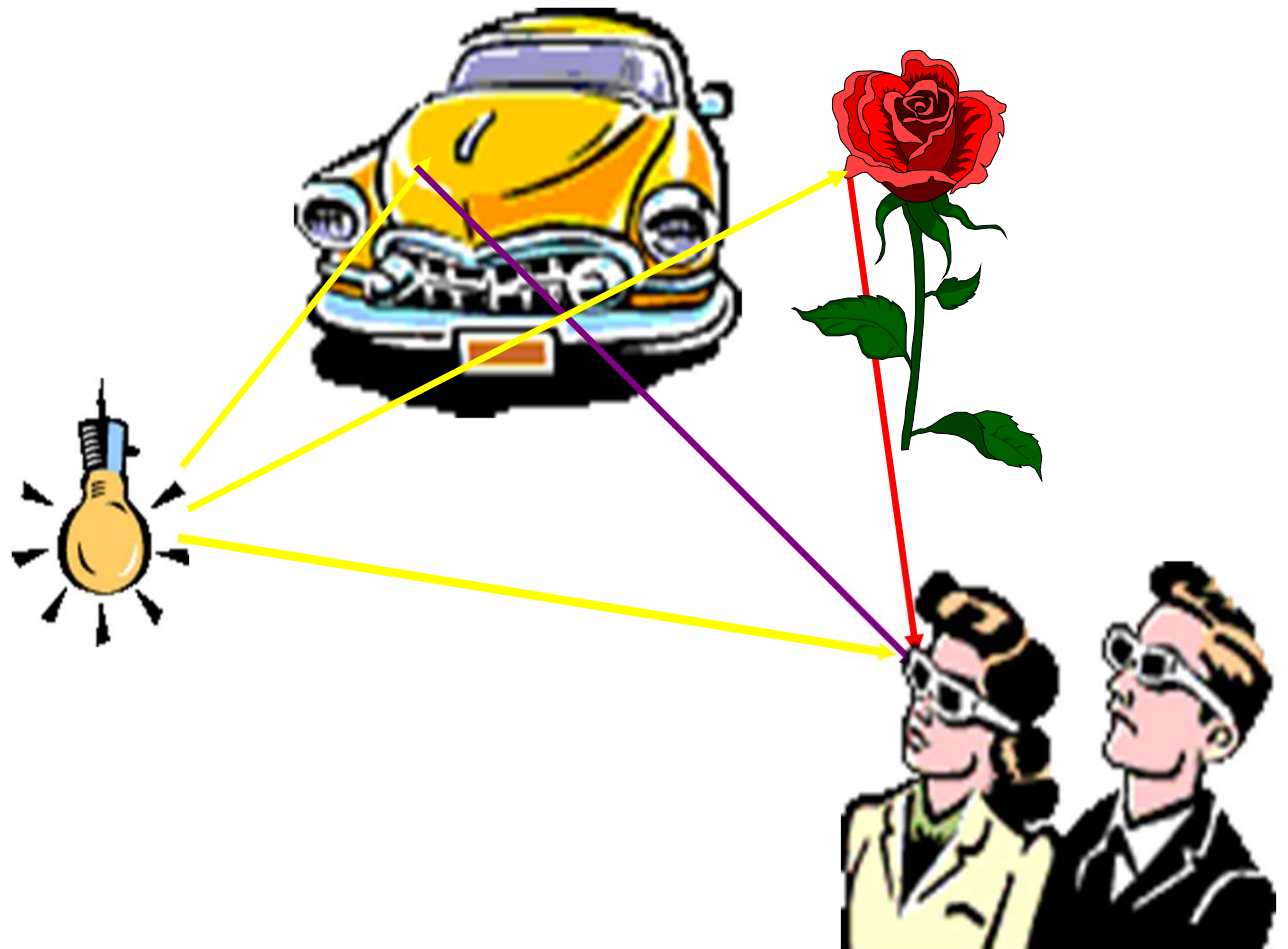
What makes an image?

the triplet light-objects-observer

Light source

Object(s)

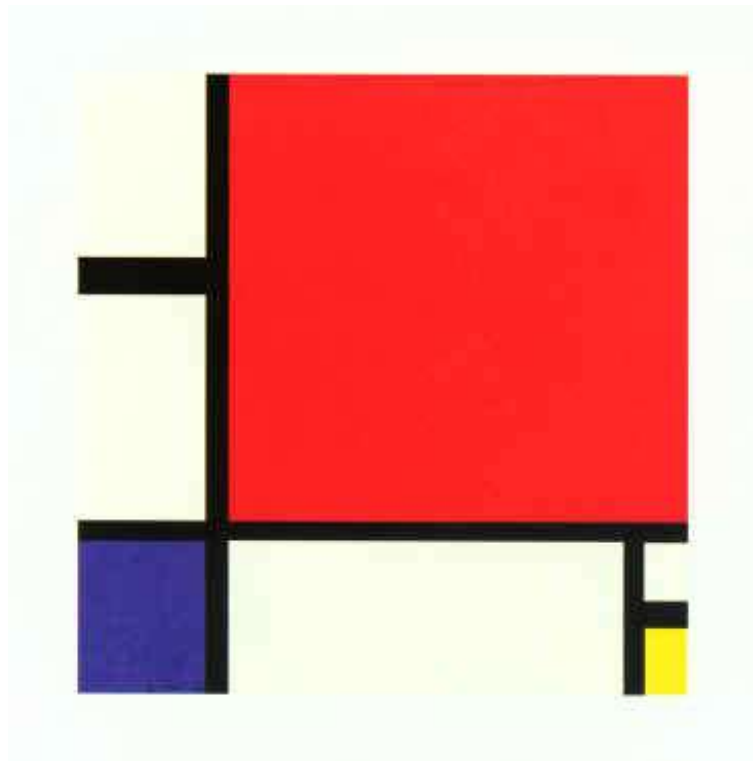
Sensor



The science of Light and Colour

Fundamentals of colour science

Hall of fame



Mondrian

Pythagoras: undulation theory

Aristoteles: curpus theory

Newton 1665 "Opticks"

Planck, Einstein and Bohr "Quantum mechanics"

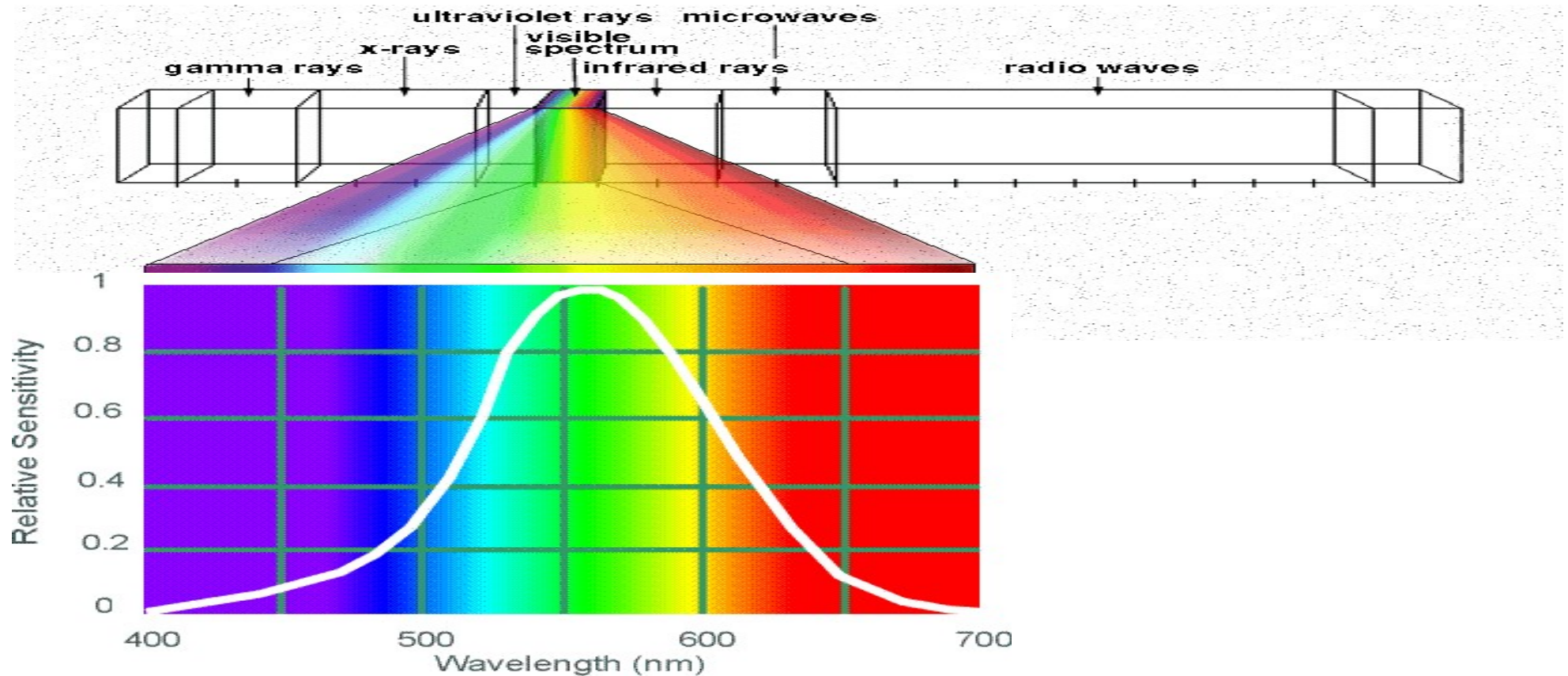
Goethe 1840 "Farbenlehre"

Munsell 1905 "A Colour Notation"

Descartes, Schopenhauer,

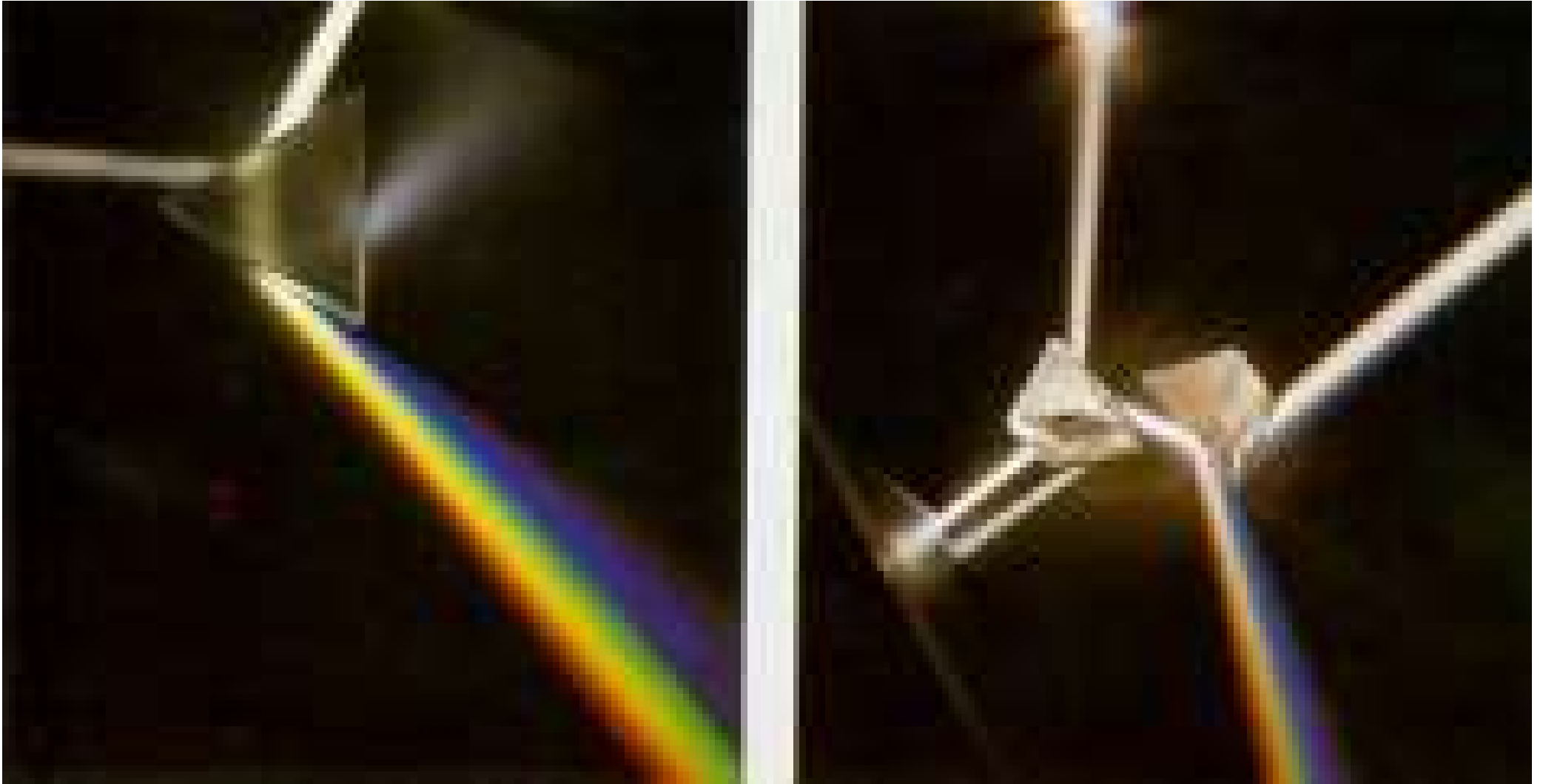
Hegel, Wittgenstein...and many others

Electromagnetic Spectrum

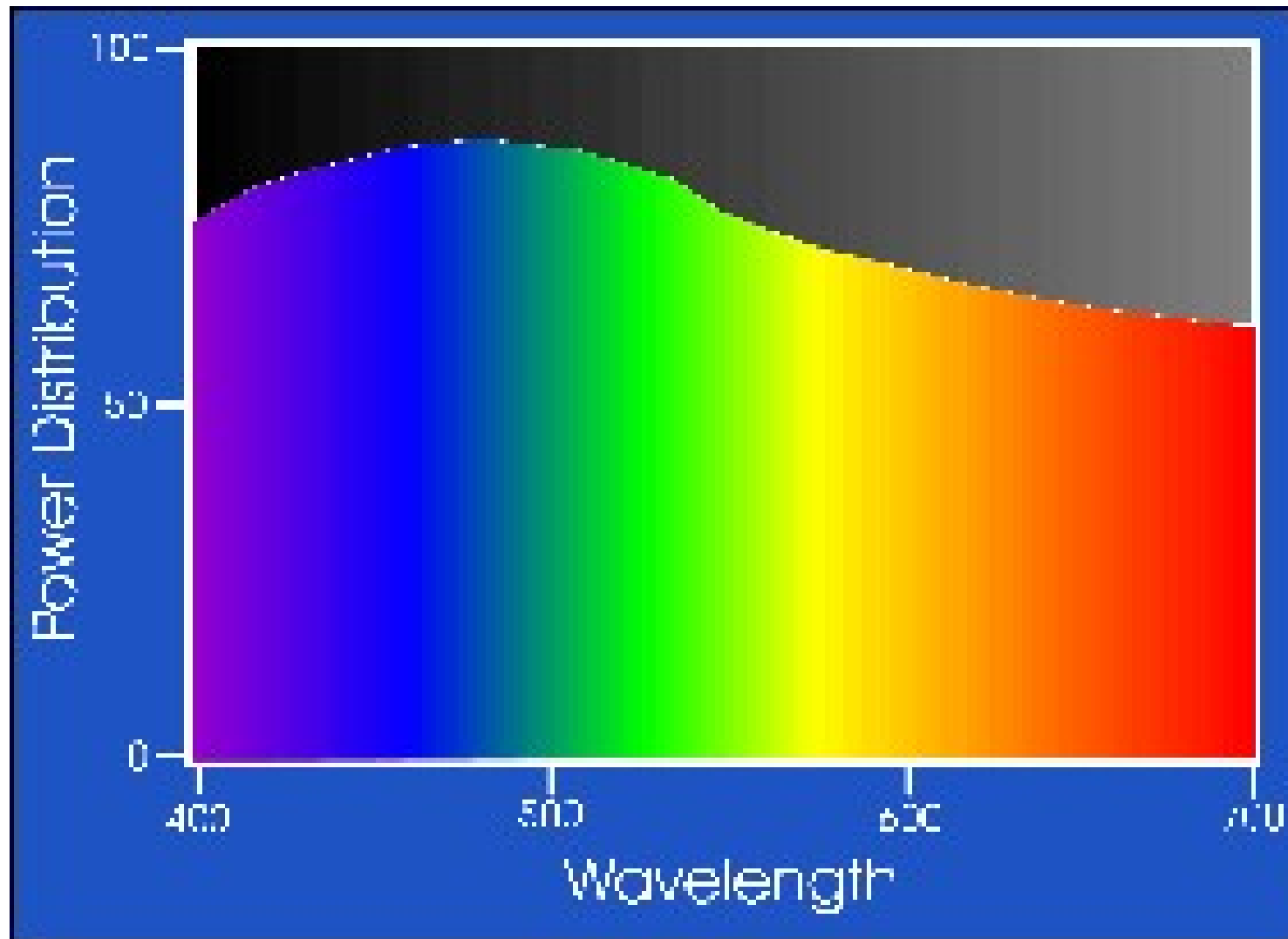


Human Luminance Sensitivity Function

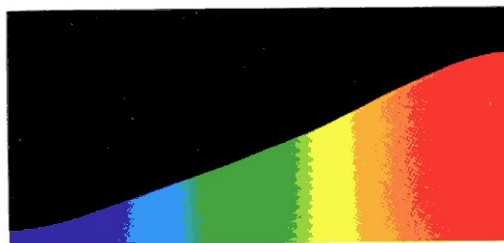
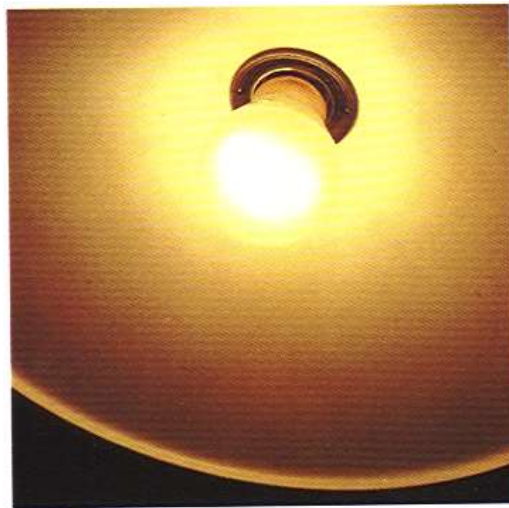
Electromagnetic Spectrum



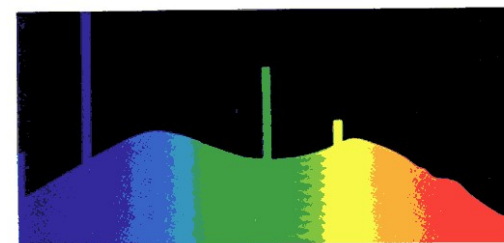
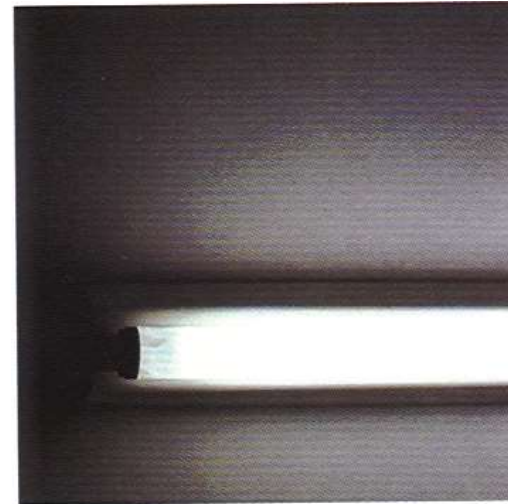
Spectral Power Distribution



Light sources and illuminants



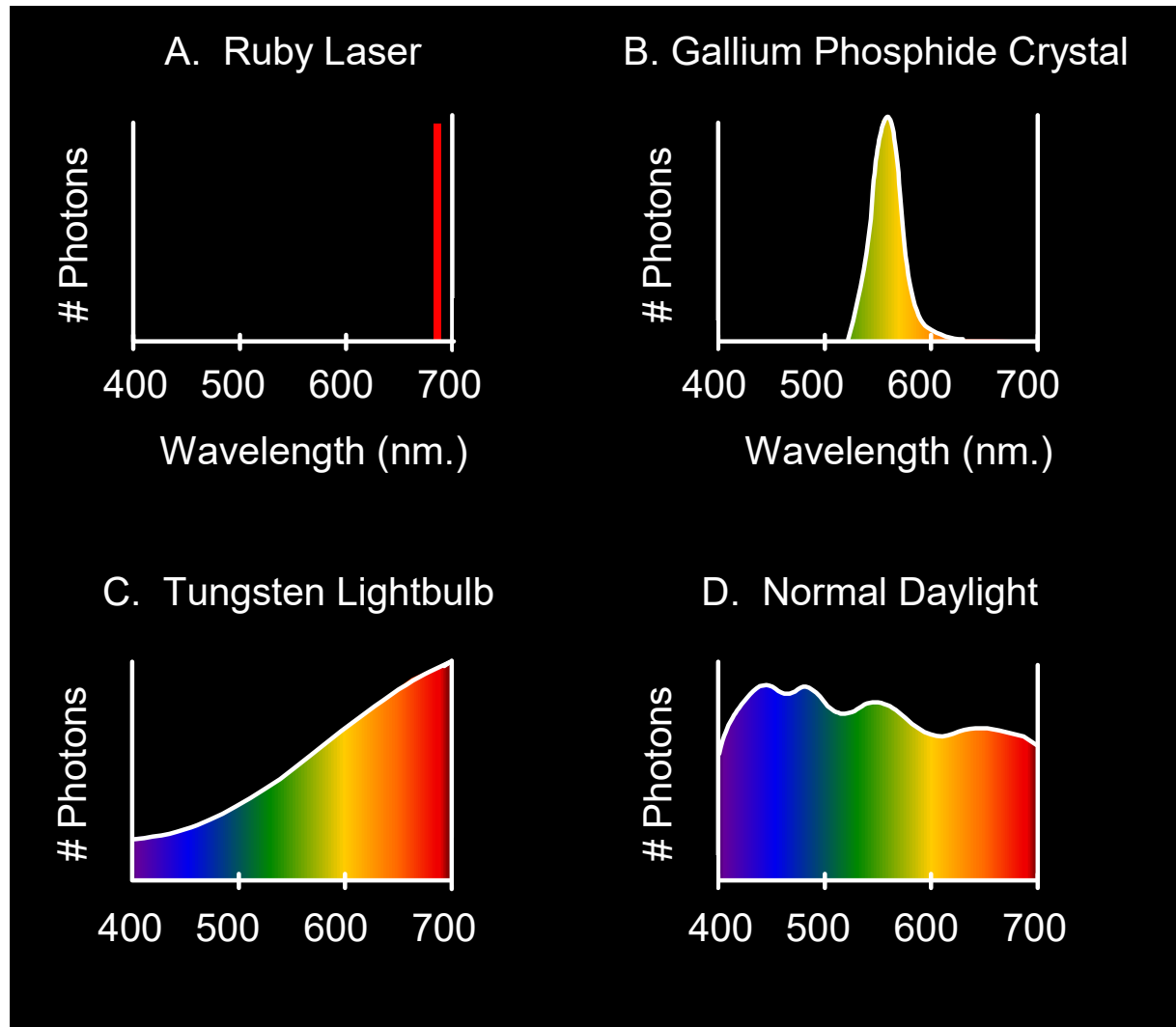
Incandescent lamp



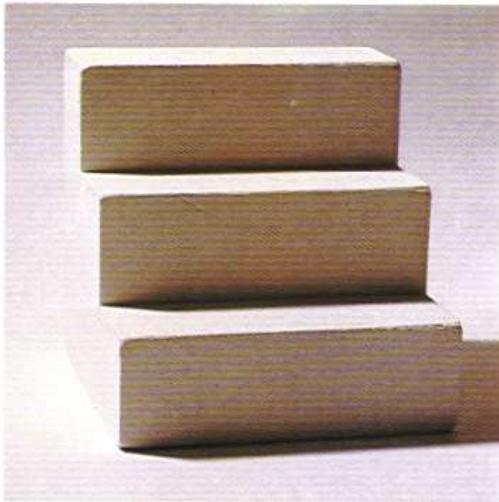
Fluorescent lamp

The Physics of Light

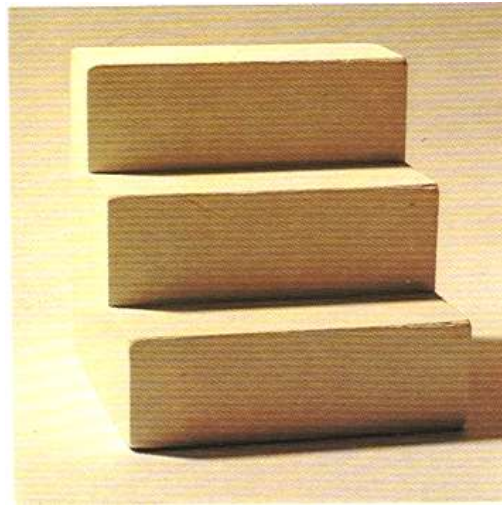
Some examples of the spectra of light sources



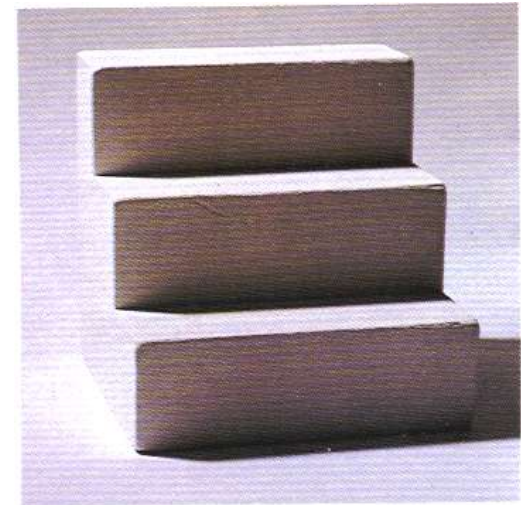
Influence of Light Sources



Average daylight

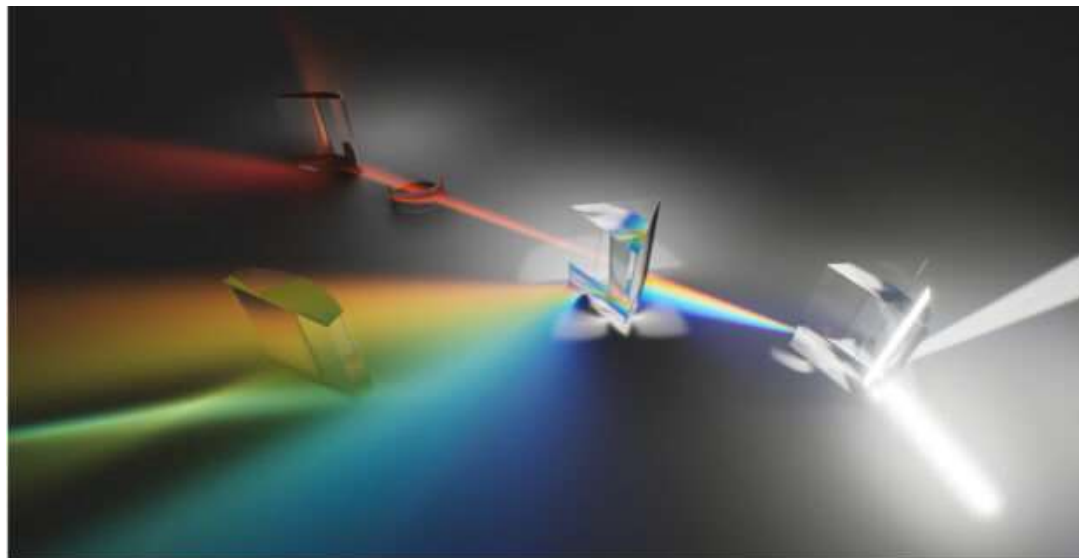


Incandescent lamp



Fluorescent lamp

Absorption and reflection



Nvidia, [Generating Ray-Traced Caustic Effects in Unreal Engine 4](#), Dec. 2020

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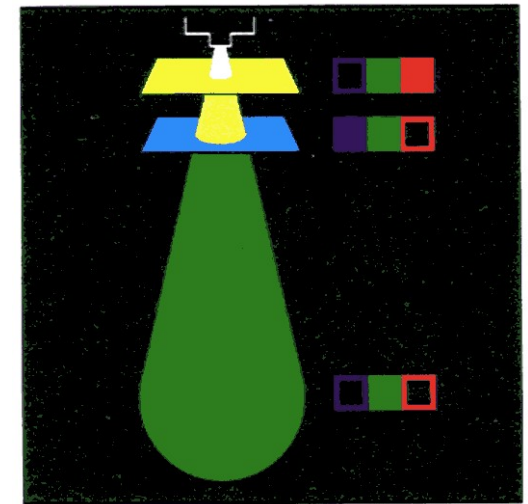
Object Colours

Materials:

Transparent

Opaque

Spectral Reflectance $\rho(\lambda)$

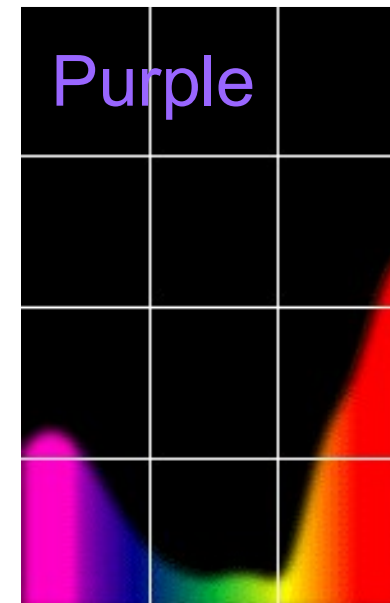
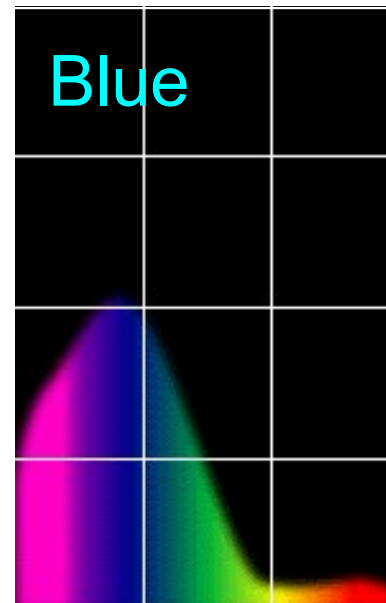
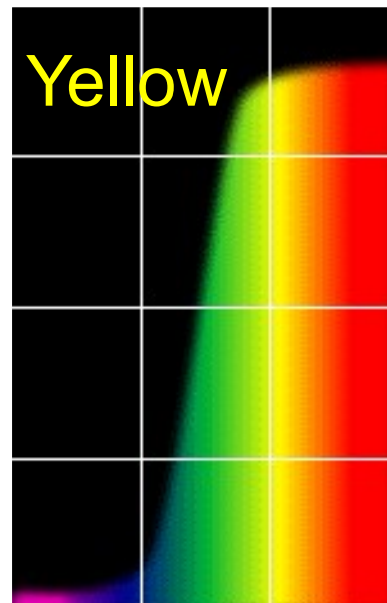
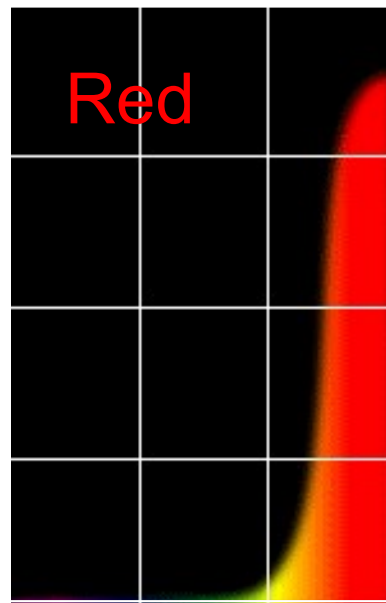


Object Colours

Some examples of the reflectance spectra of surfaces

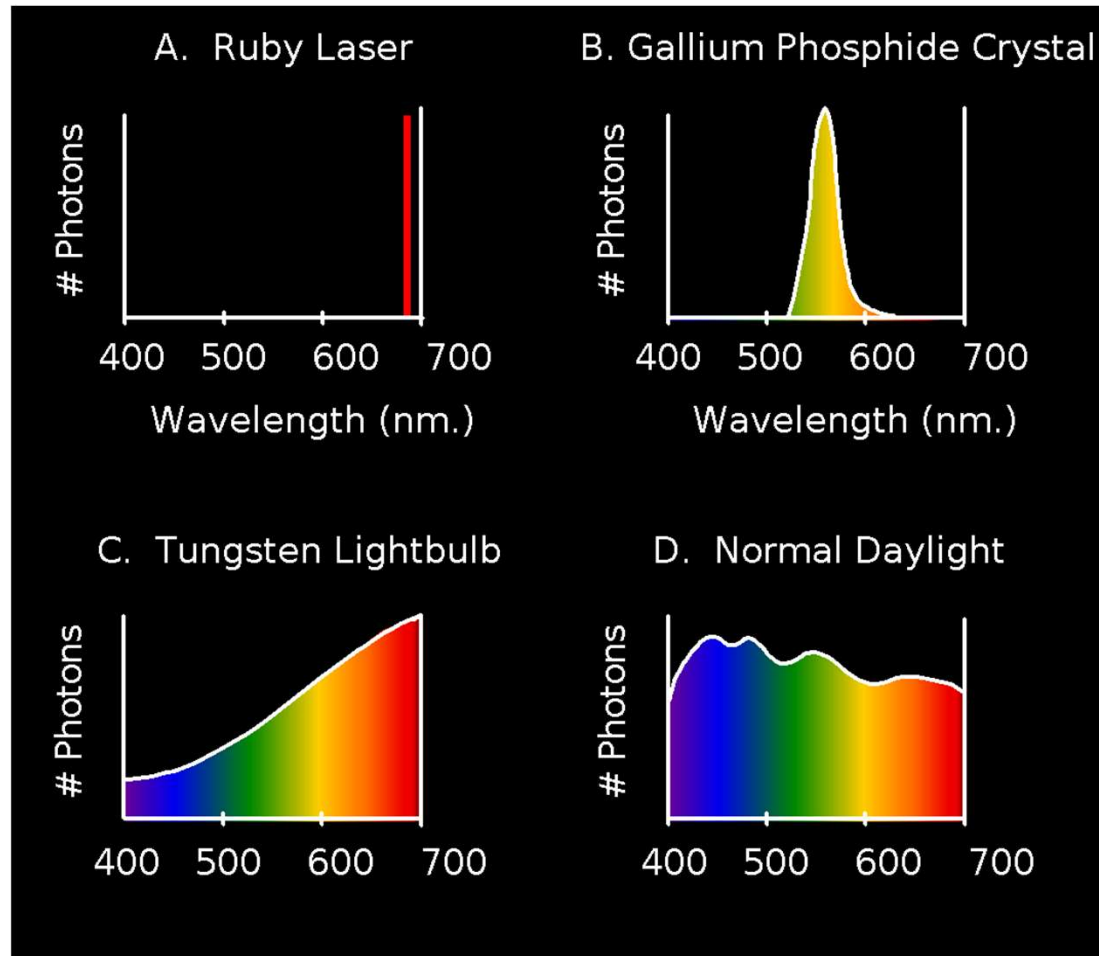


% Photons Reflected



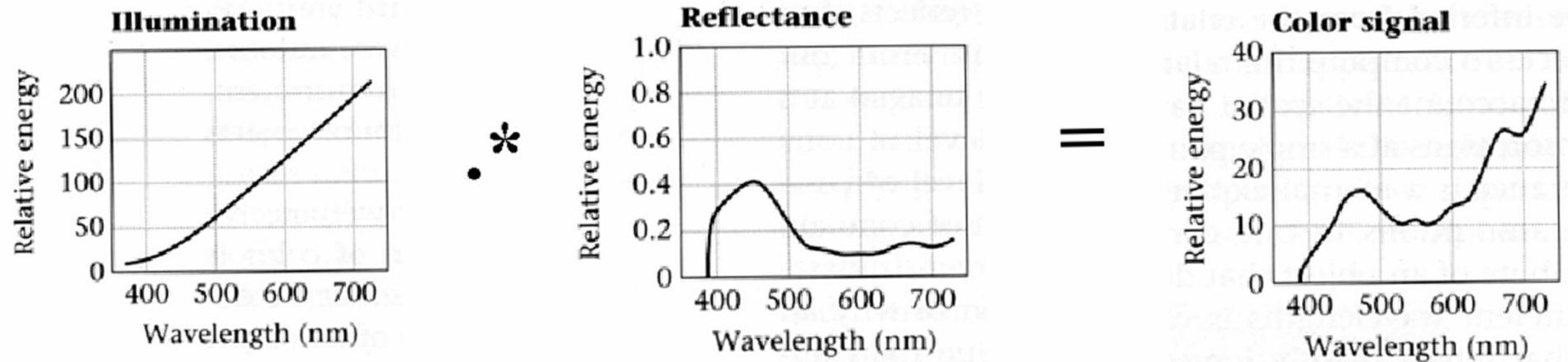
Wavelength (nm)

Sources of light are diverse!



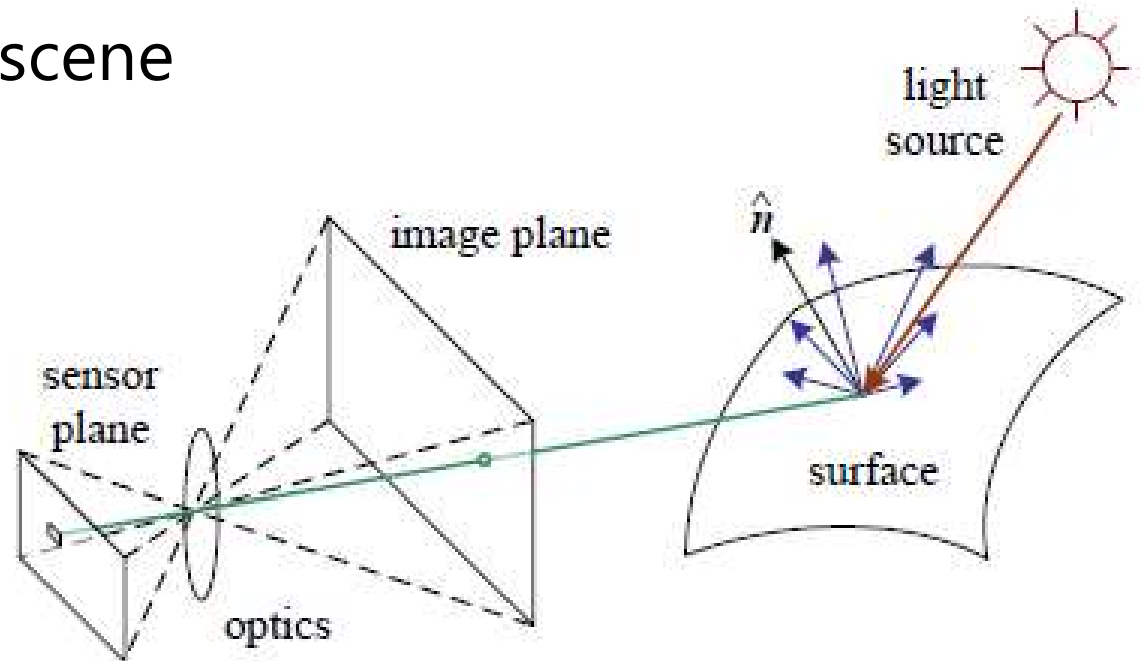
What color is the object?

- The “color” of an object depends on both the incident light and the objects reflectance:



4 main factors which define intensity values

- the geometry of the scene
- the reflectances of the visible surface
- the illumination of the scene
- the viewpoint

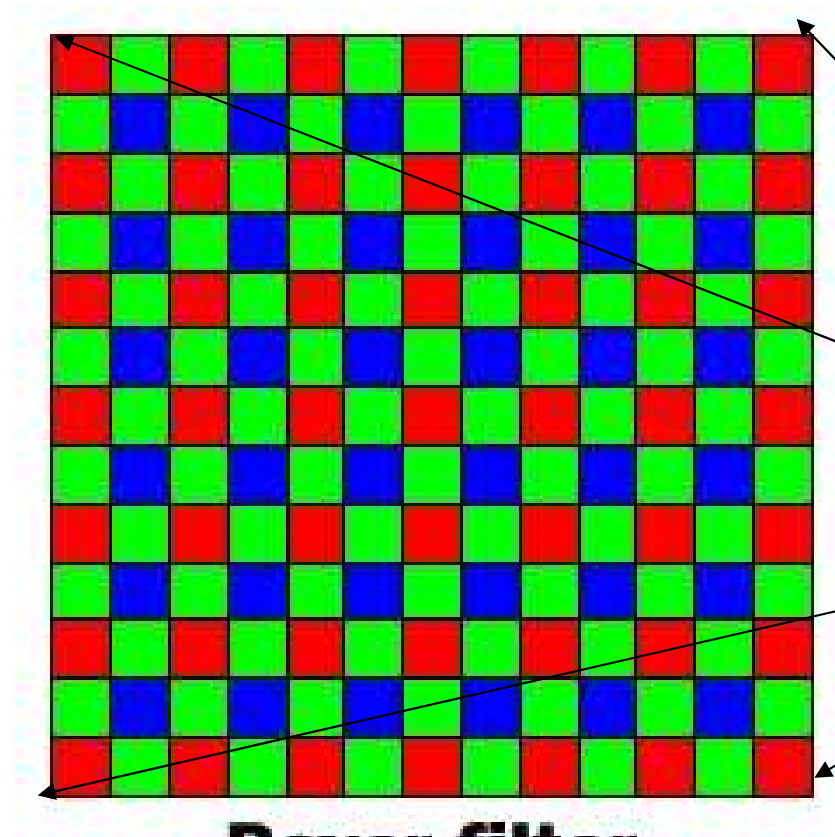


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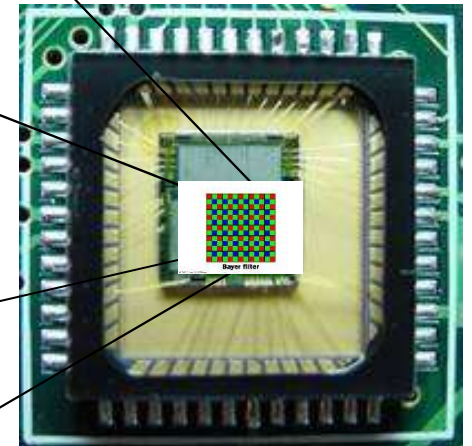
How to form colour images?

Bayer filter:
Green fills in half of
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and Red and Blue
fill the rest.



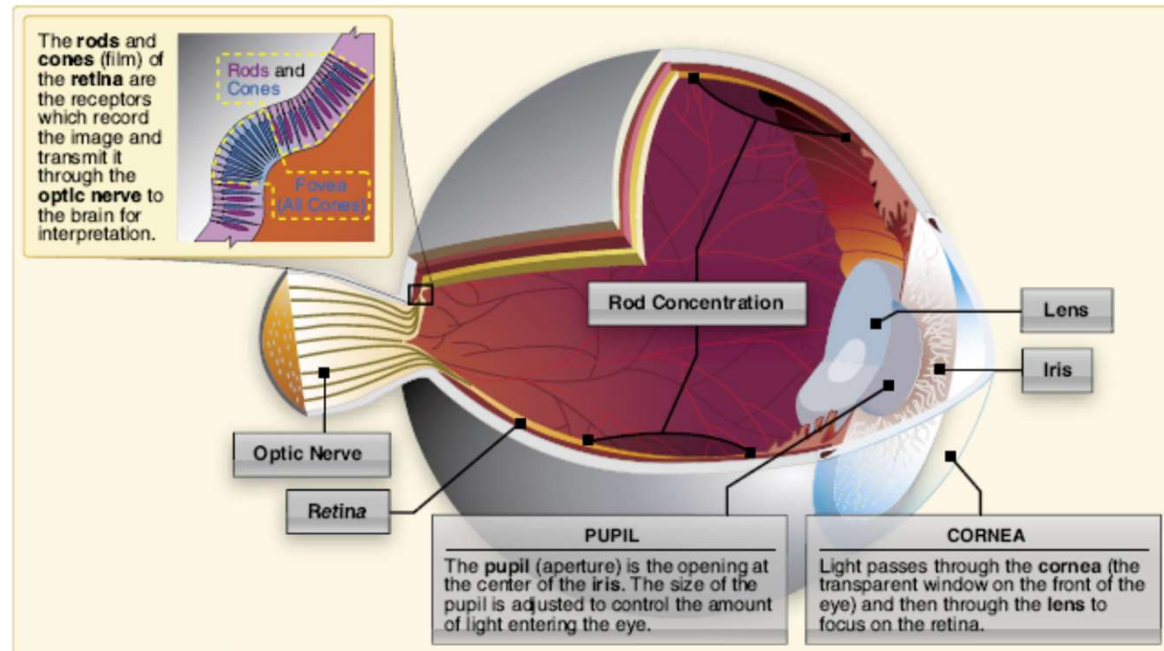
Bayer filter

© 2000 How Stuff Works



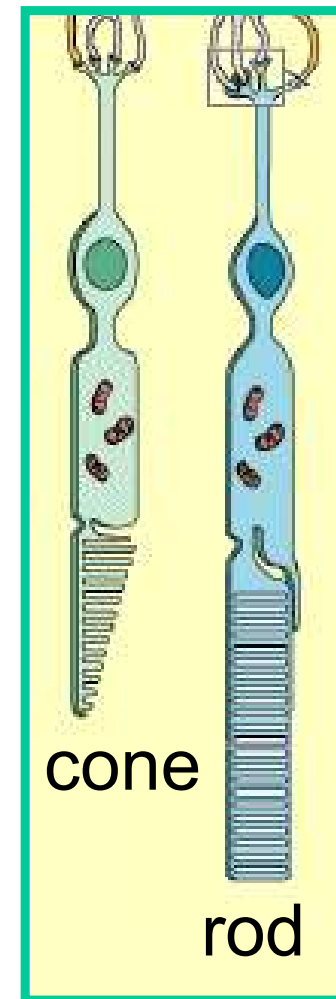
Observer: Human Eye

- Retina contains light sensitive cells that convert light energy into electrical impulses that travel through nerves to the brain.
- Brain interprets the electrical signals to form images.



Two Types of Light-Sensitive Receptors

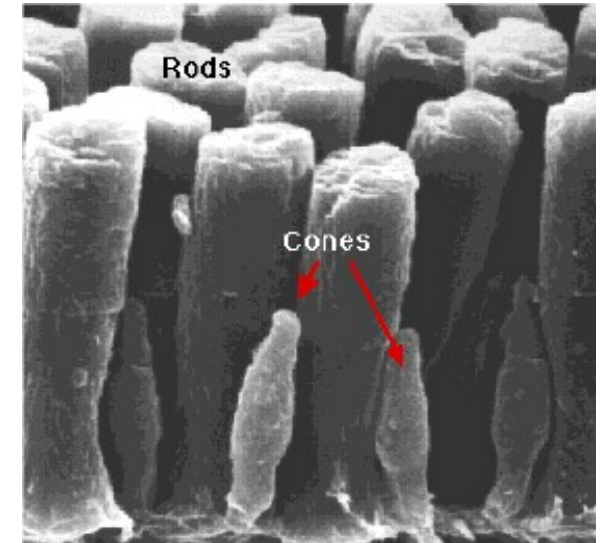
- Cones
 - cone-shaped
 - less sensitive
 - operate in high light
 - color vision
- Rods
 - rod-shaped
 - highly sensitive
 - operate at night
 - gray-scale vision



Light Detection: Rods and Cones

Rods:

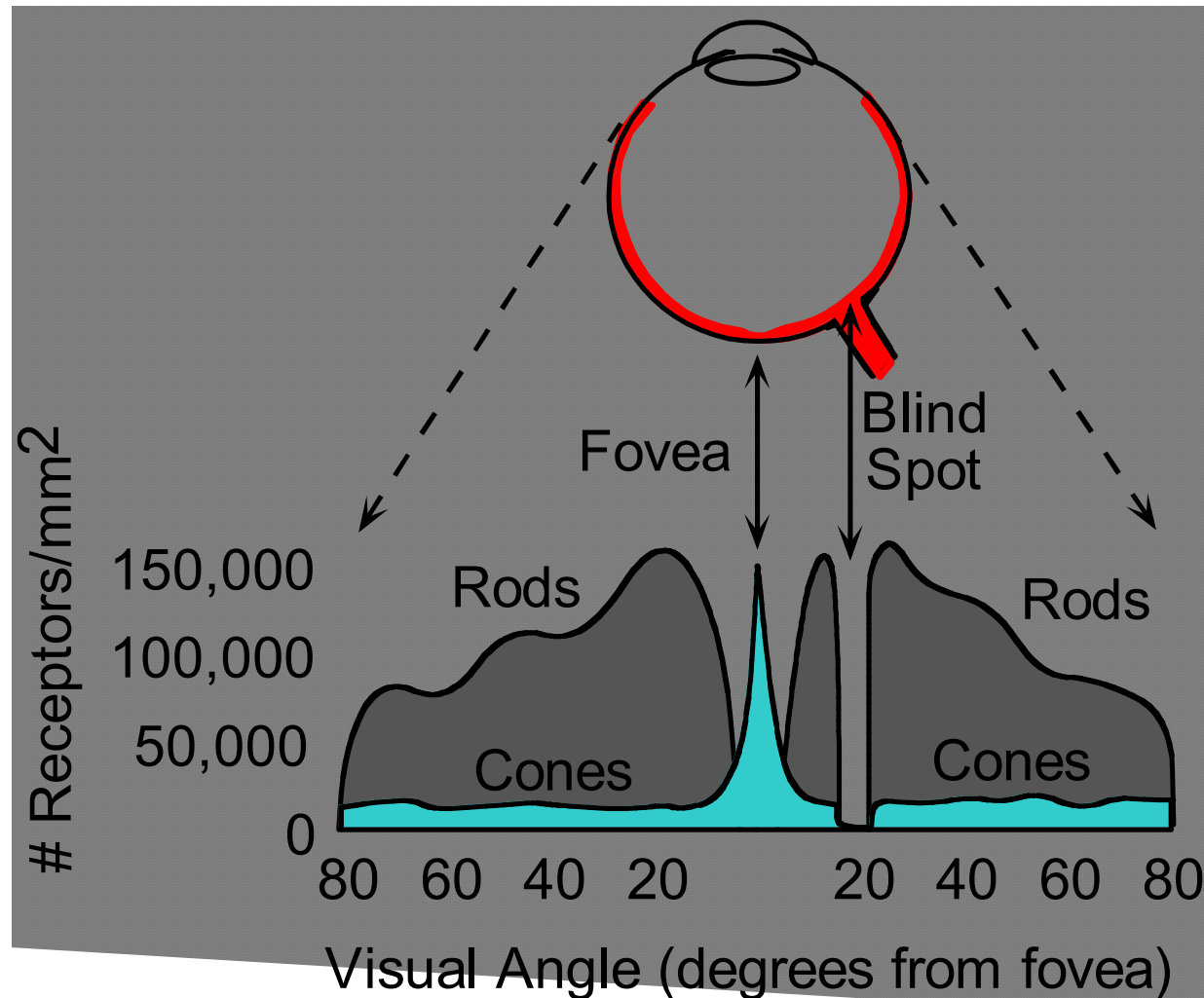
- 120 million rods in retina
- 1000X more light sensitive than Cones
- Discriminate B/W brightness in low illumination
- Short wave-length sensitive



Cons:

- 6-7 million cones in the retina
- Responsible for high-resolution vision
- Discriminate Colors
- Three types of color sensors: 64% red, 32% green, 2% blue)
- Sensitive to any combination of three colors

Distribution of Rods and Cones

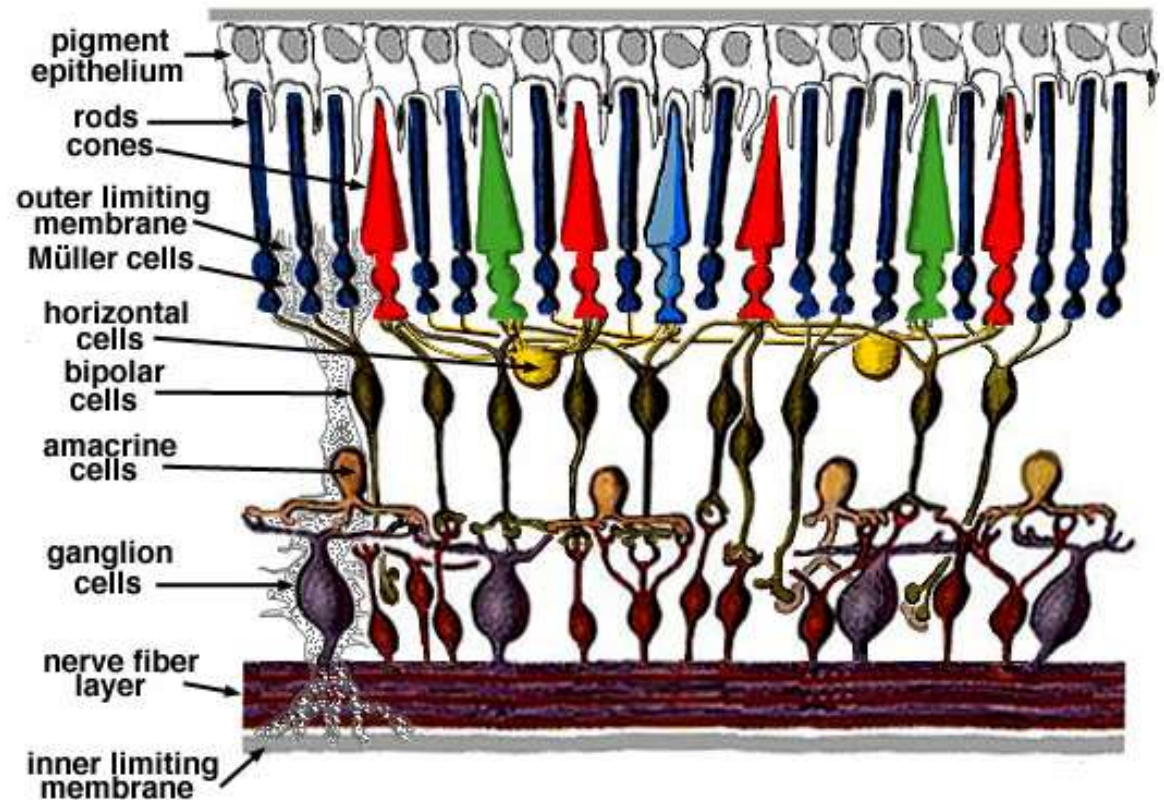


Night Sky: why are there more stars off-center?

Averted vision: http://en.wikipedia.org/wiki/Averted_vision

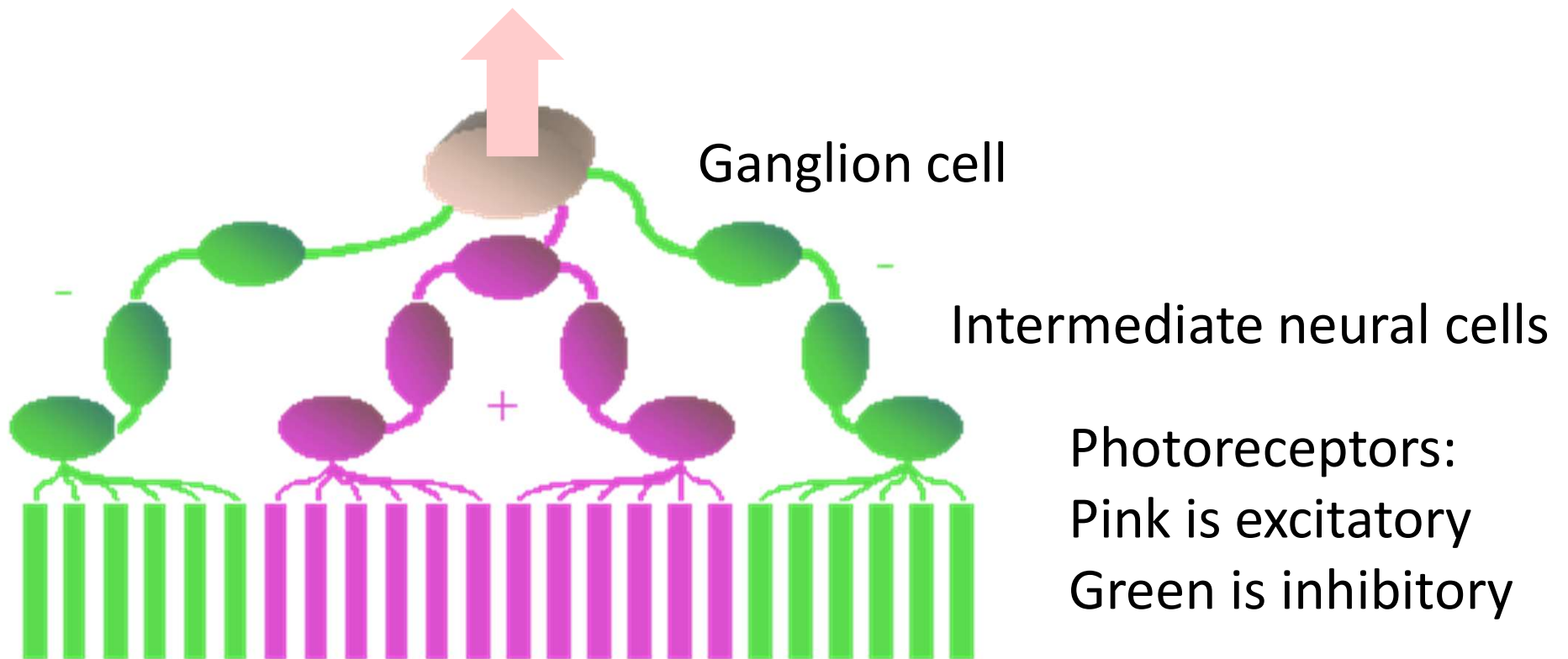
The Retina

- 0.5 mm thick
- The photosensors (the rods and cones) lie outermost in the retina.
- Interneurons
- Ganglion cells (the output neurons of the retina) lie innermost in the retina closest to the lens and front of the eye.

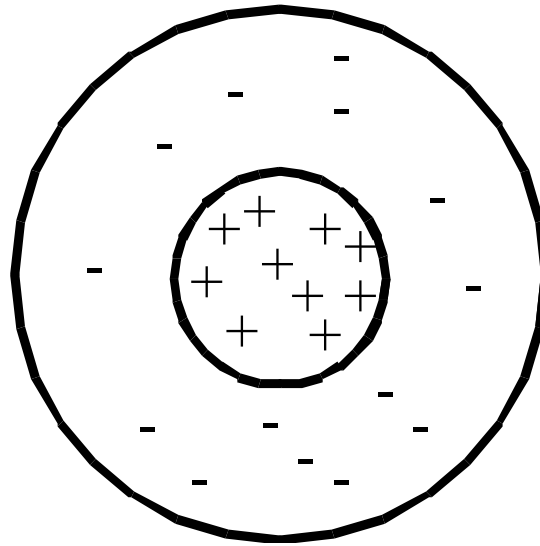


Receptive Fields

The ganglion cell produces some background response even when there is no light on its receptive field

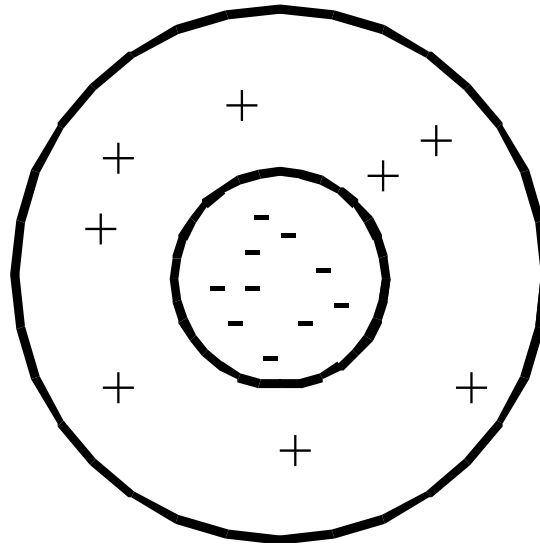


“On-Center” Ganglion Cell



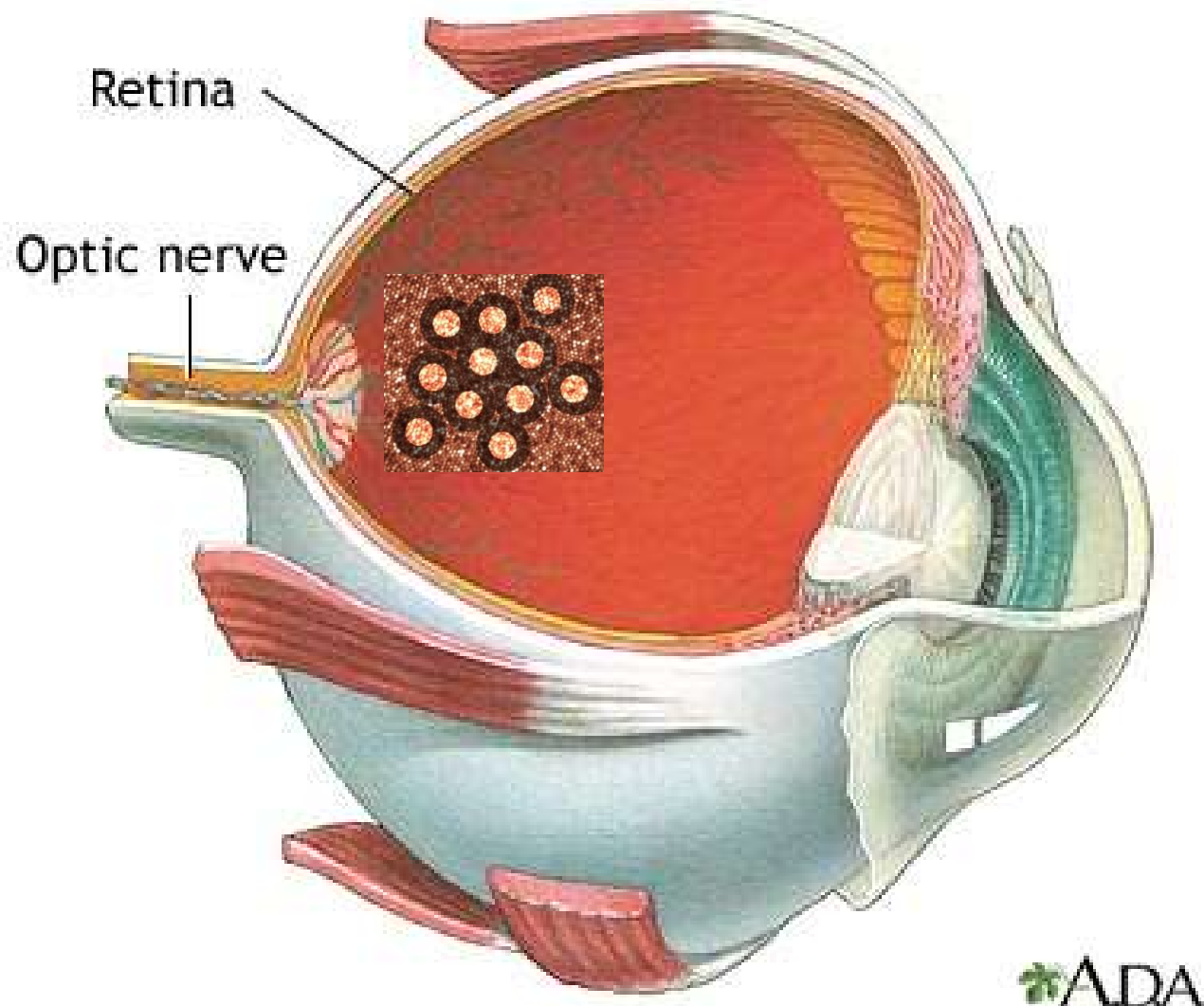
Responds maximally to light increments in the center,
and light decrements in the surround.

“Off-Center” Ganglion Cell



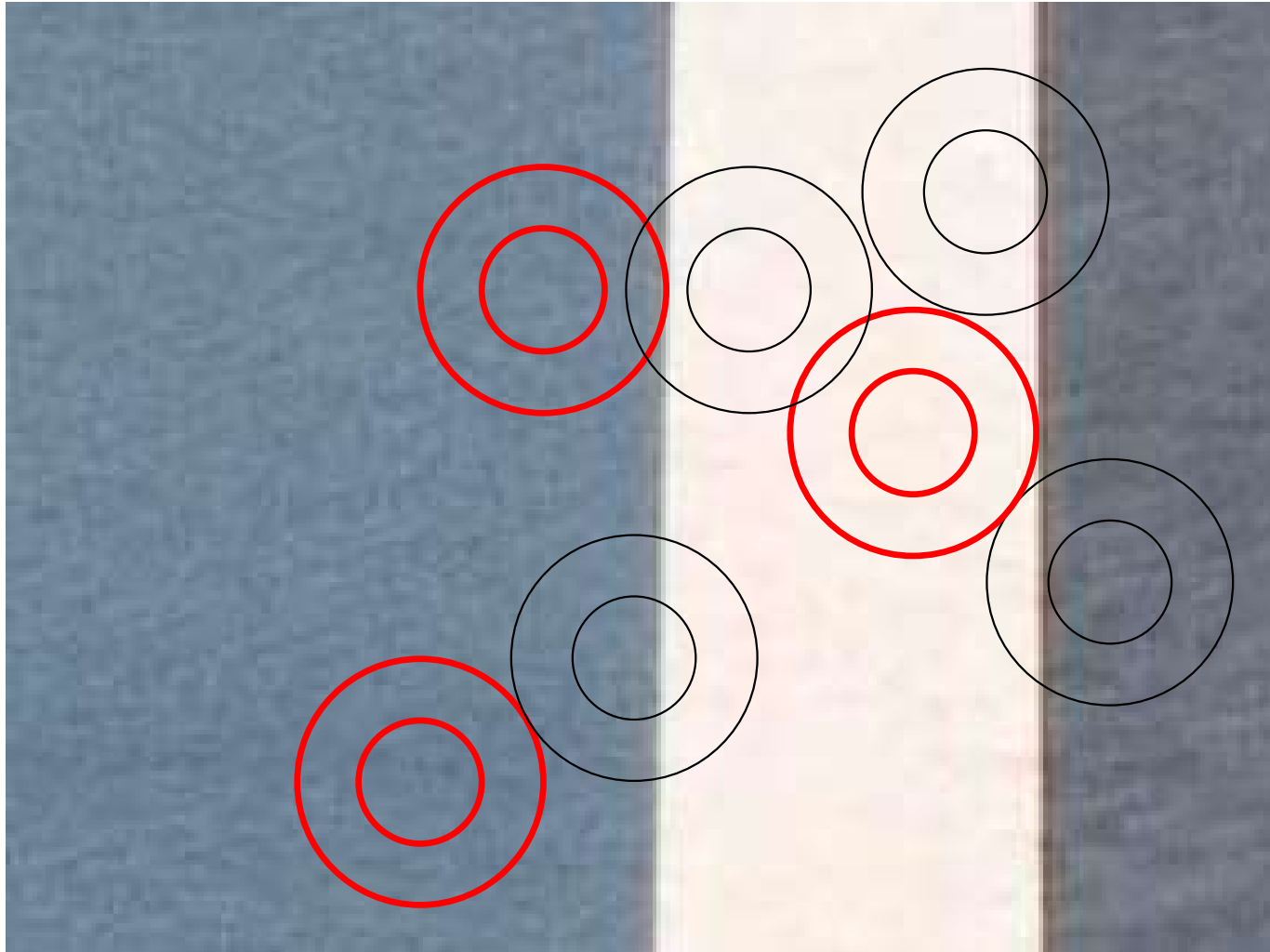
Responds maximally to light decrements in the center,
and light increments in the surround.

Receptive Fields on the Retina

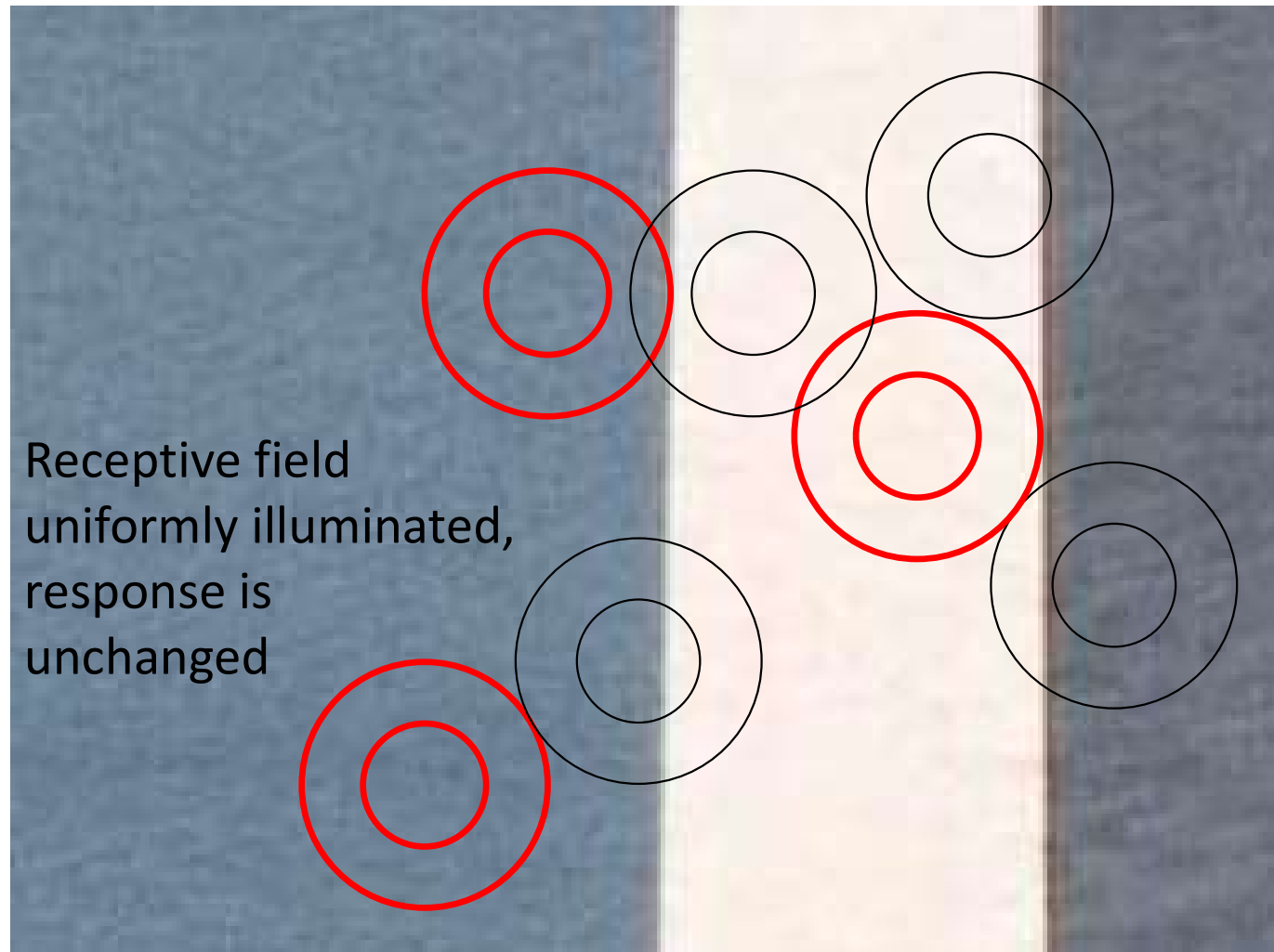


The size of the receptors and the receptive field are shown here much larger than actual size!

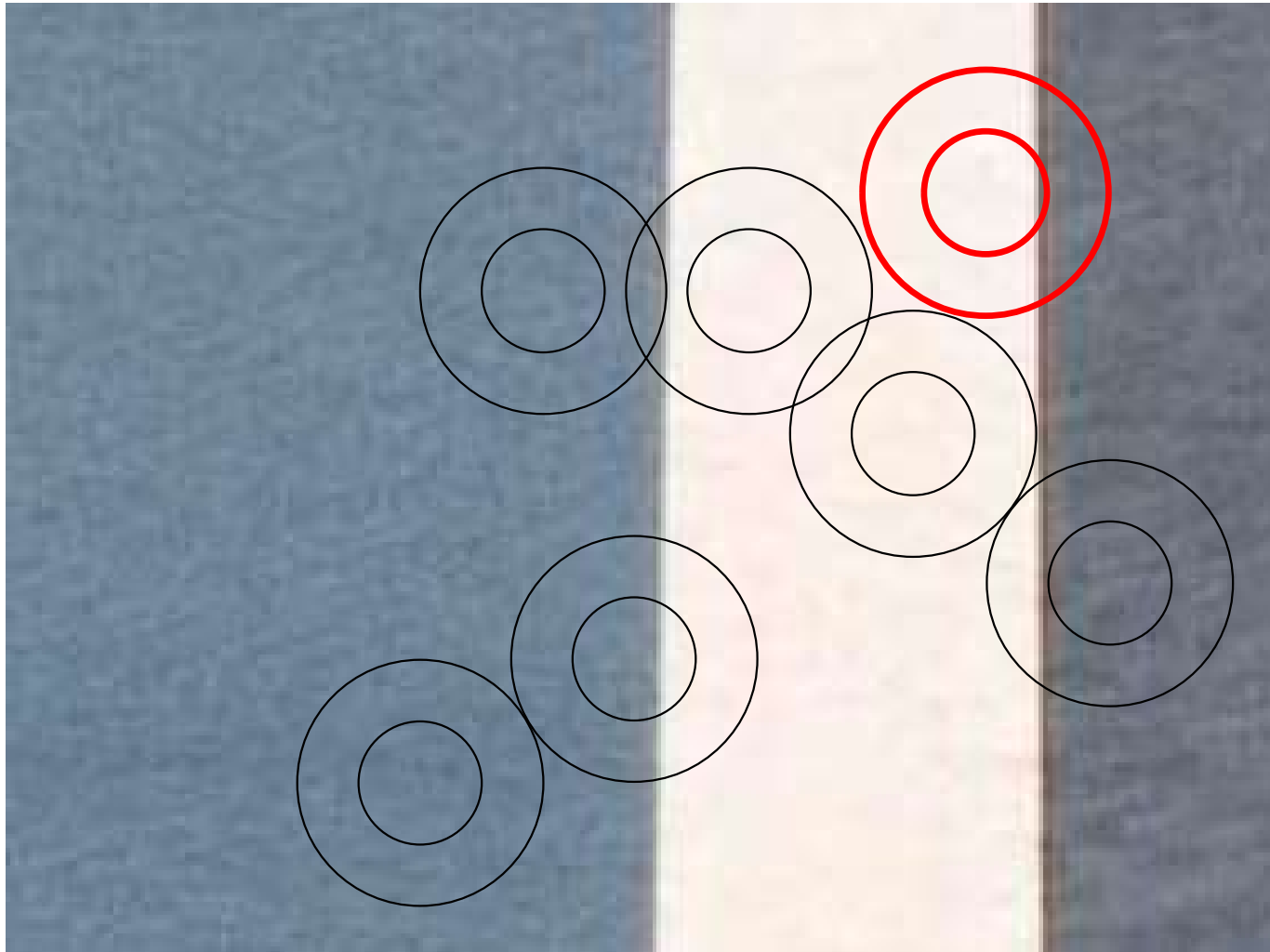
Ganglion Cells: Receptive Fields



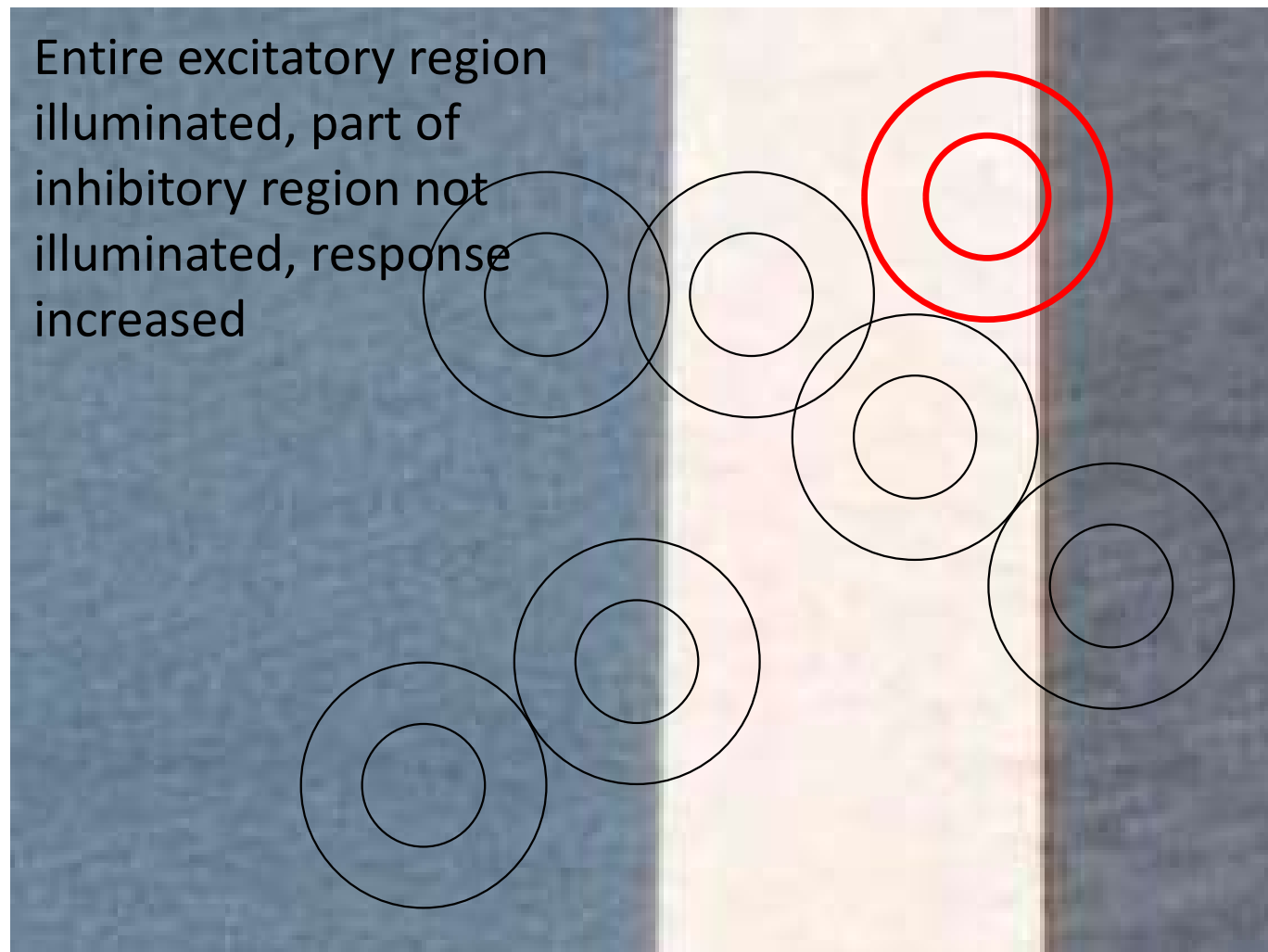
Ganglion Cells: Receptive Fields



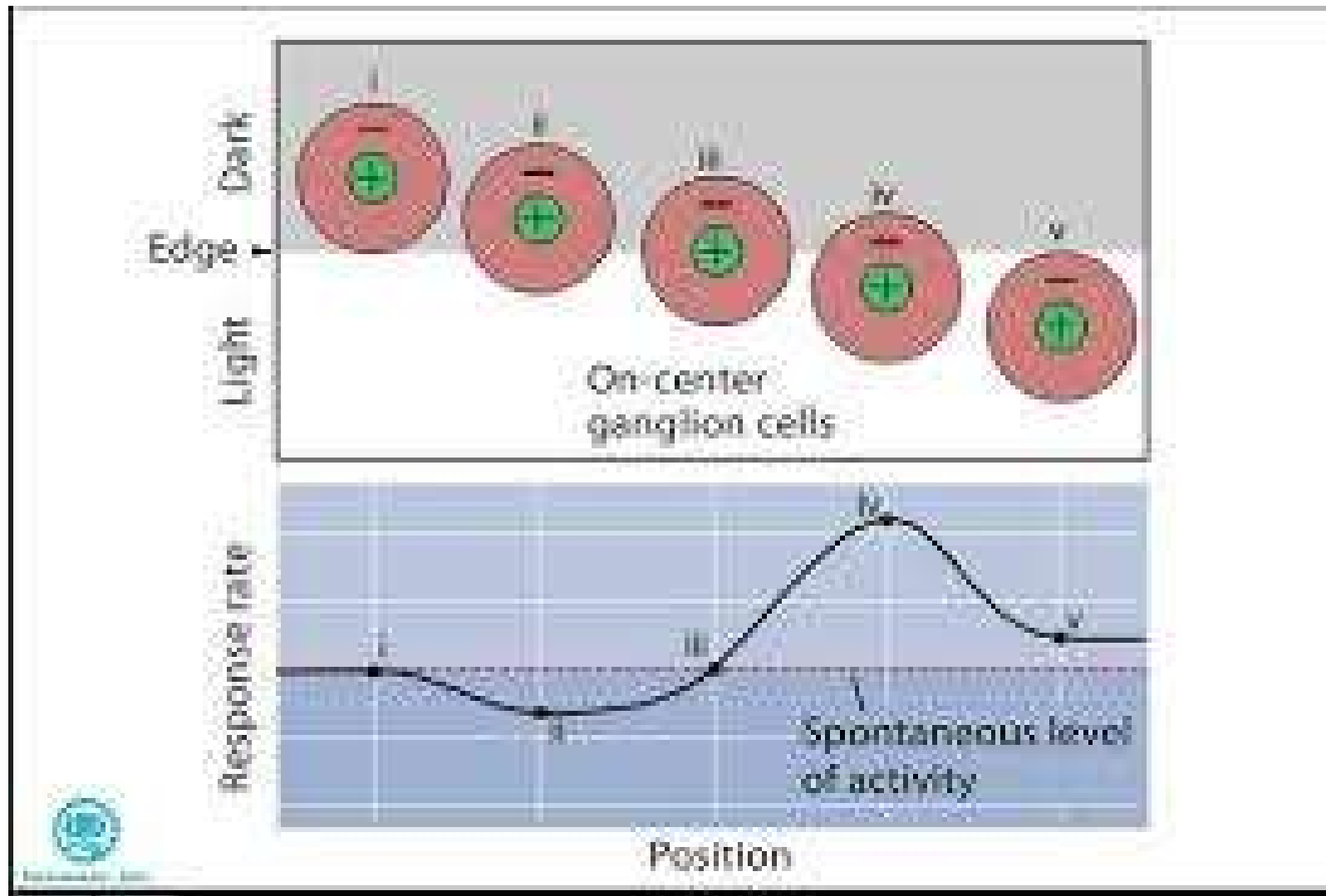
Ganglion Cells: Receptive Fields



Ganglion Cells: Receptive Fields

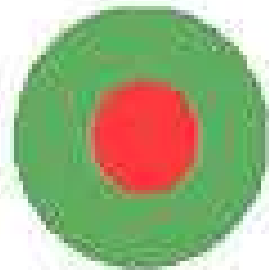


Responses: ON-Centre Ganglion Cells

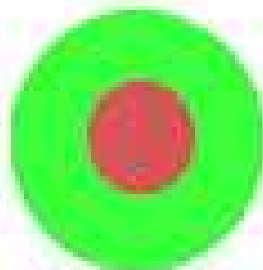


Ganglion Cells: Receptive Fields / Opponent Colors

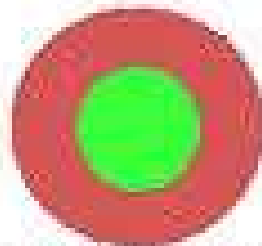
Color opponent ganglion cells



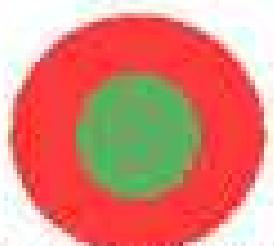
red ON/green OFF



red OFF/green ON

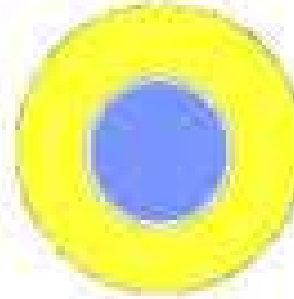


green ON/red OFF

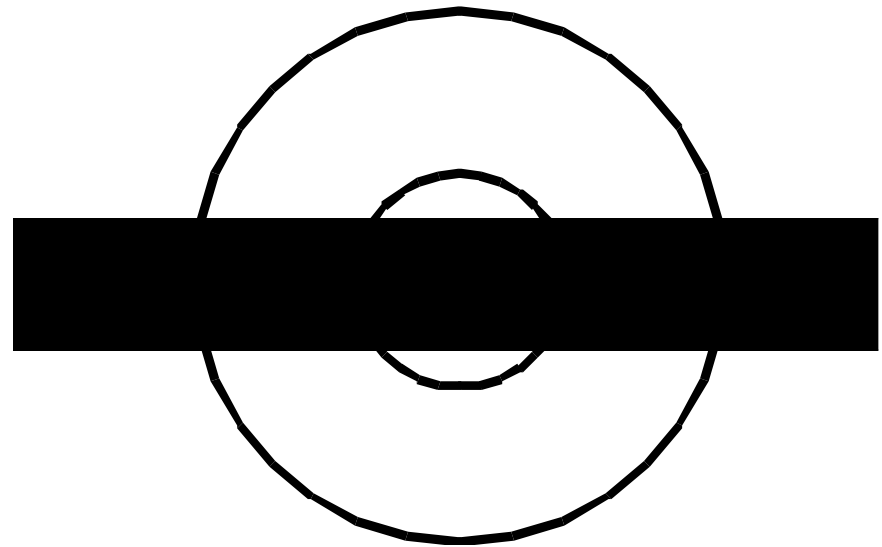
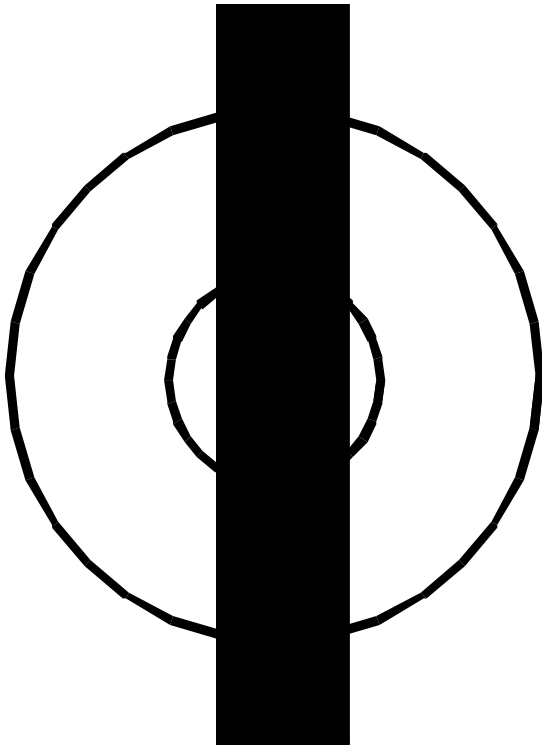


green OFF/red ON

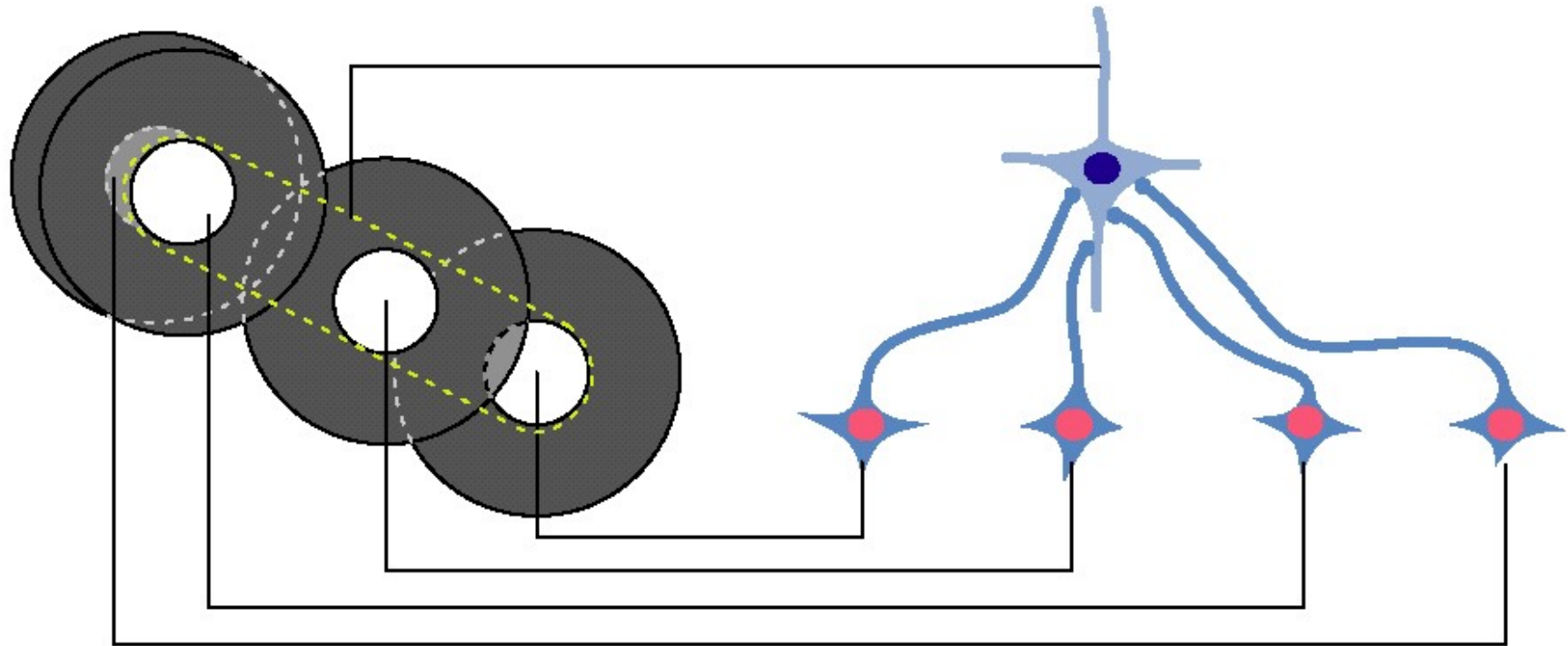
blue ON/yellow OFF



Ganglion cells have no orientation preference.



Model of how center-surround cells can be building blocks for simple cells.



Retinal ganglion cells respond to edges

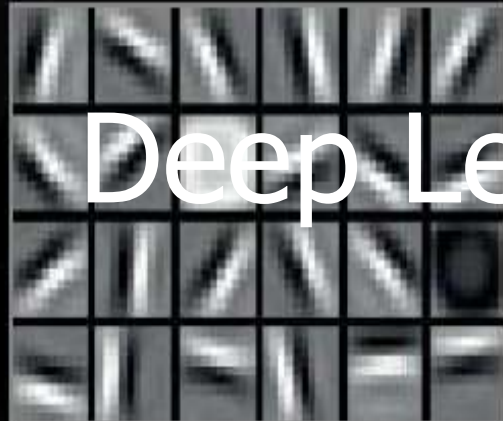
Input image
(cornea)



"Neural image"
(retinal ganglion cells)

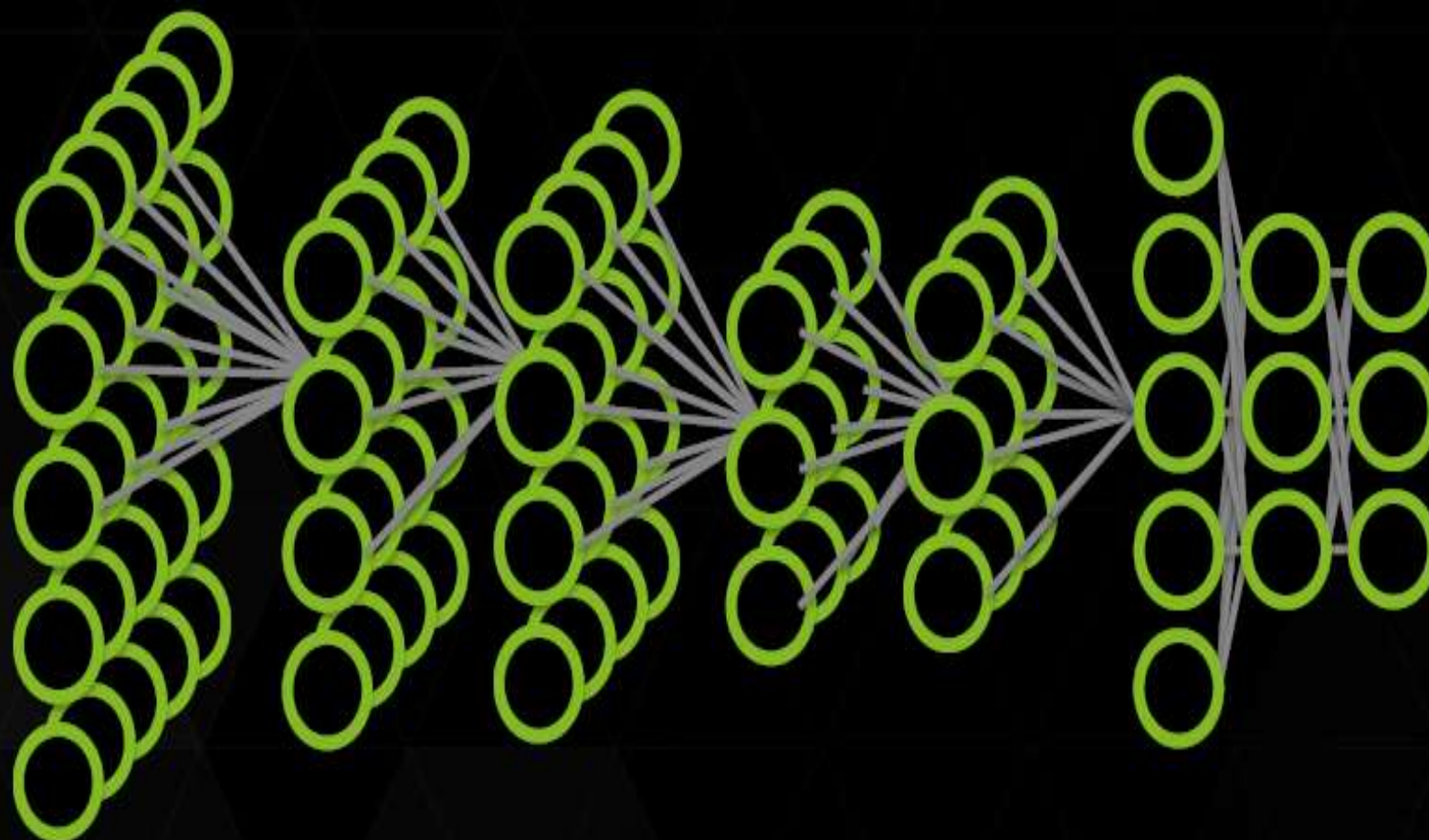


Center-surround receptive fields: emphasize edges.



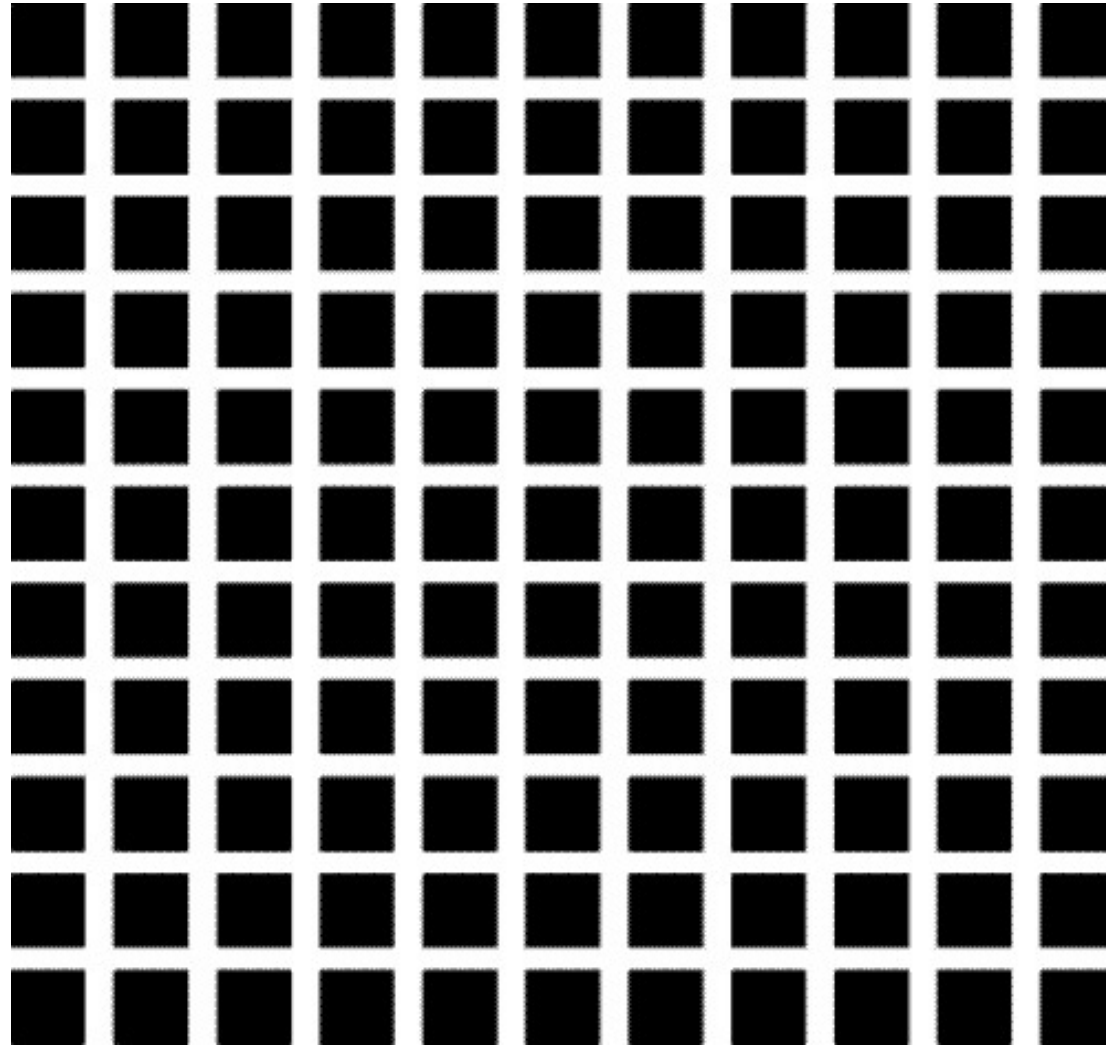
Deep Learning

Input

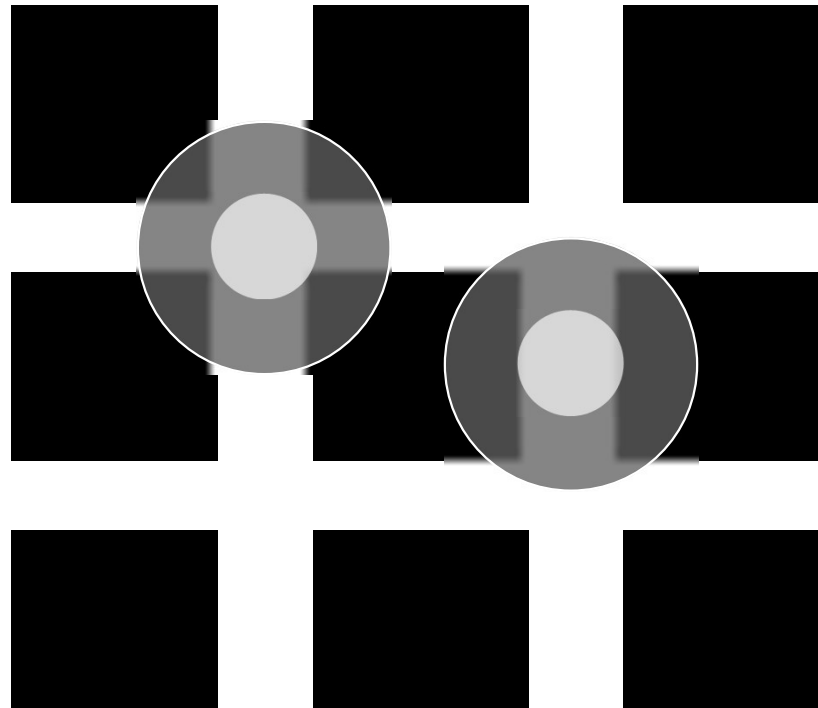


Result

Hermann Grid Illusion



Hermann Grid Illusion: Explanation



When the image is on the first receptive field there is more light falling on the surround (inhibitory) than in the second position

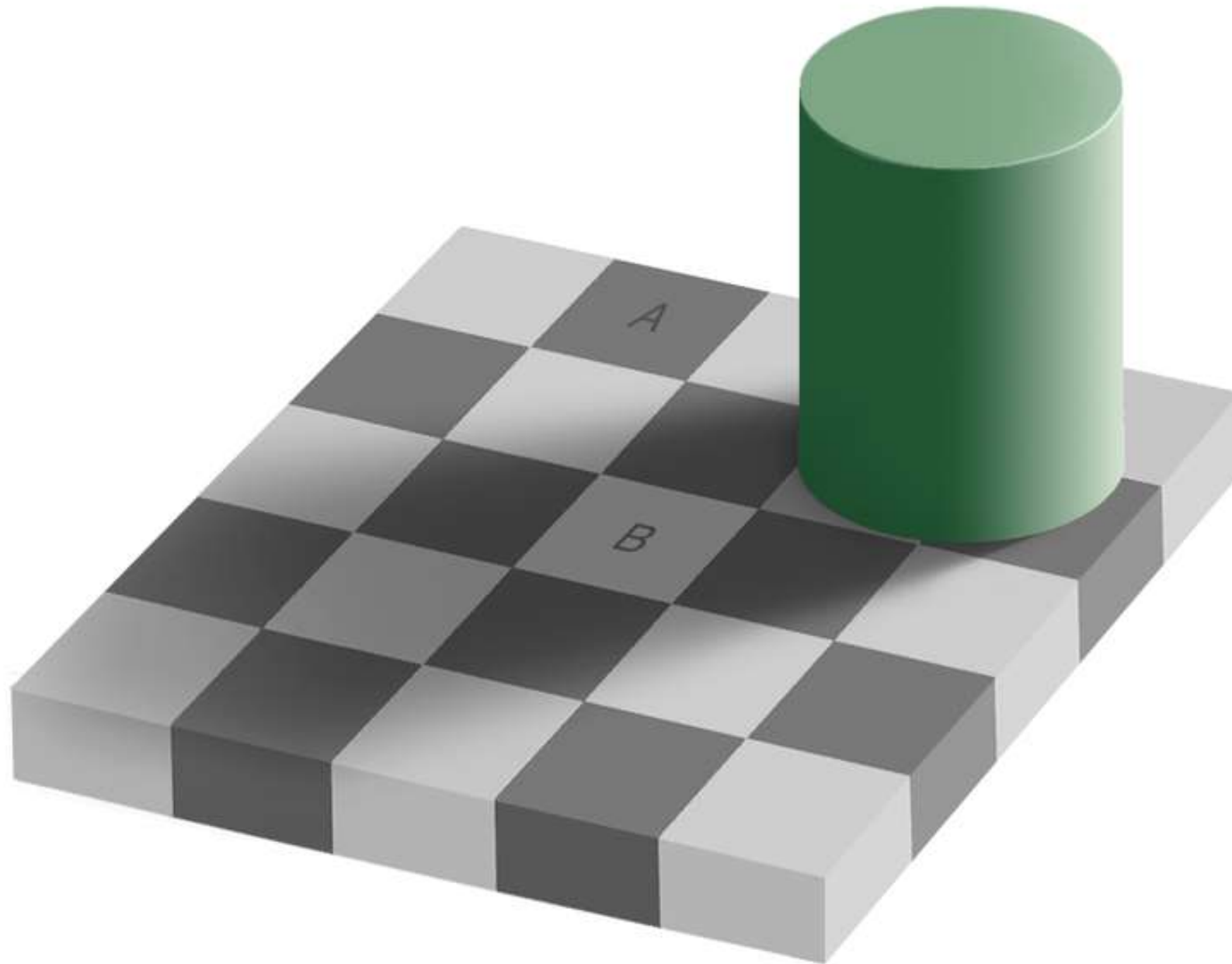
So there is more suppression and the illusion of a dark spot at the first location

Simultaneous Lightness Contrast

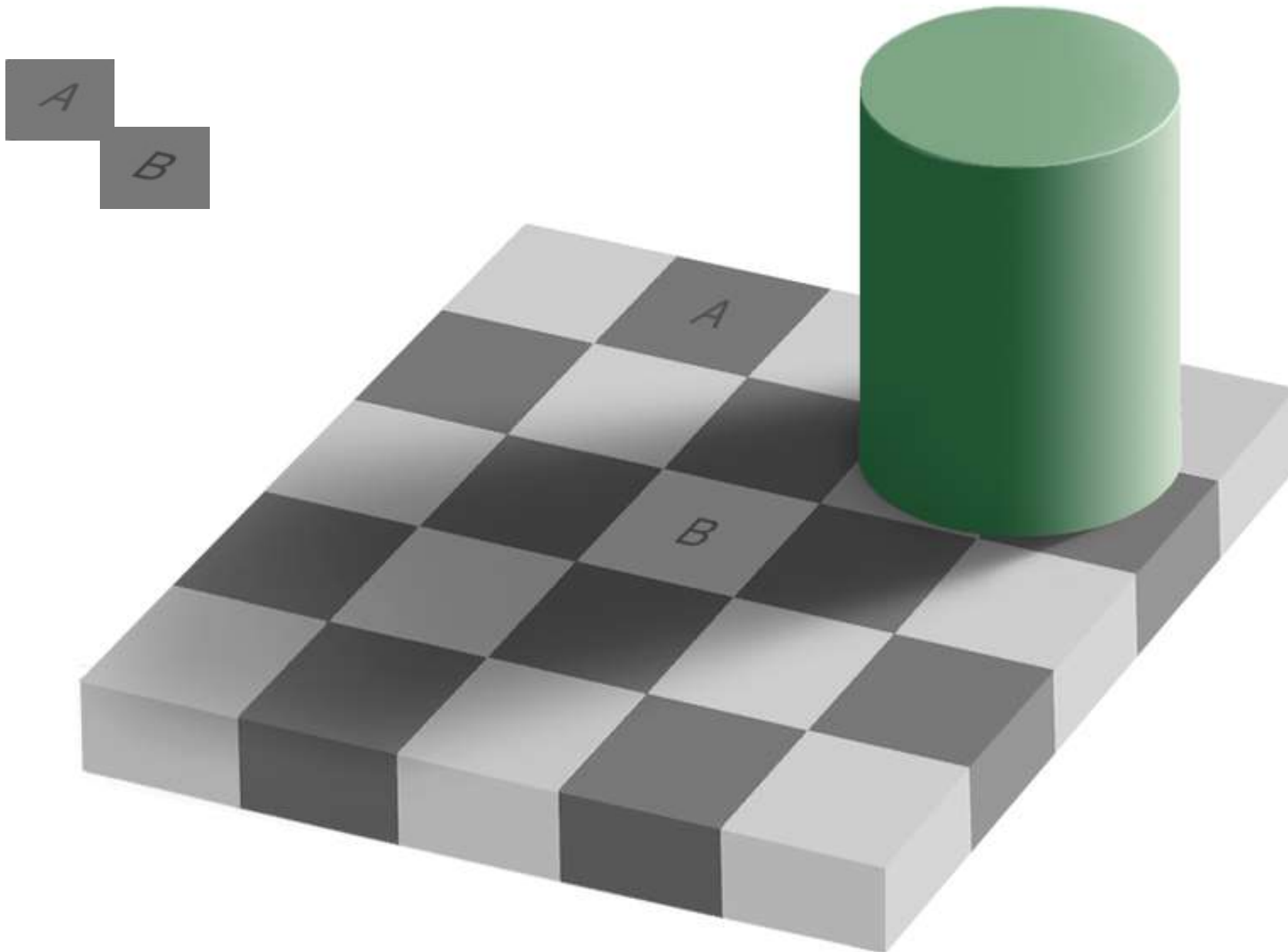


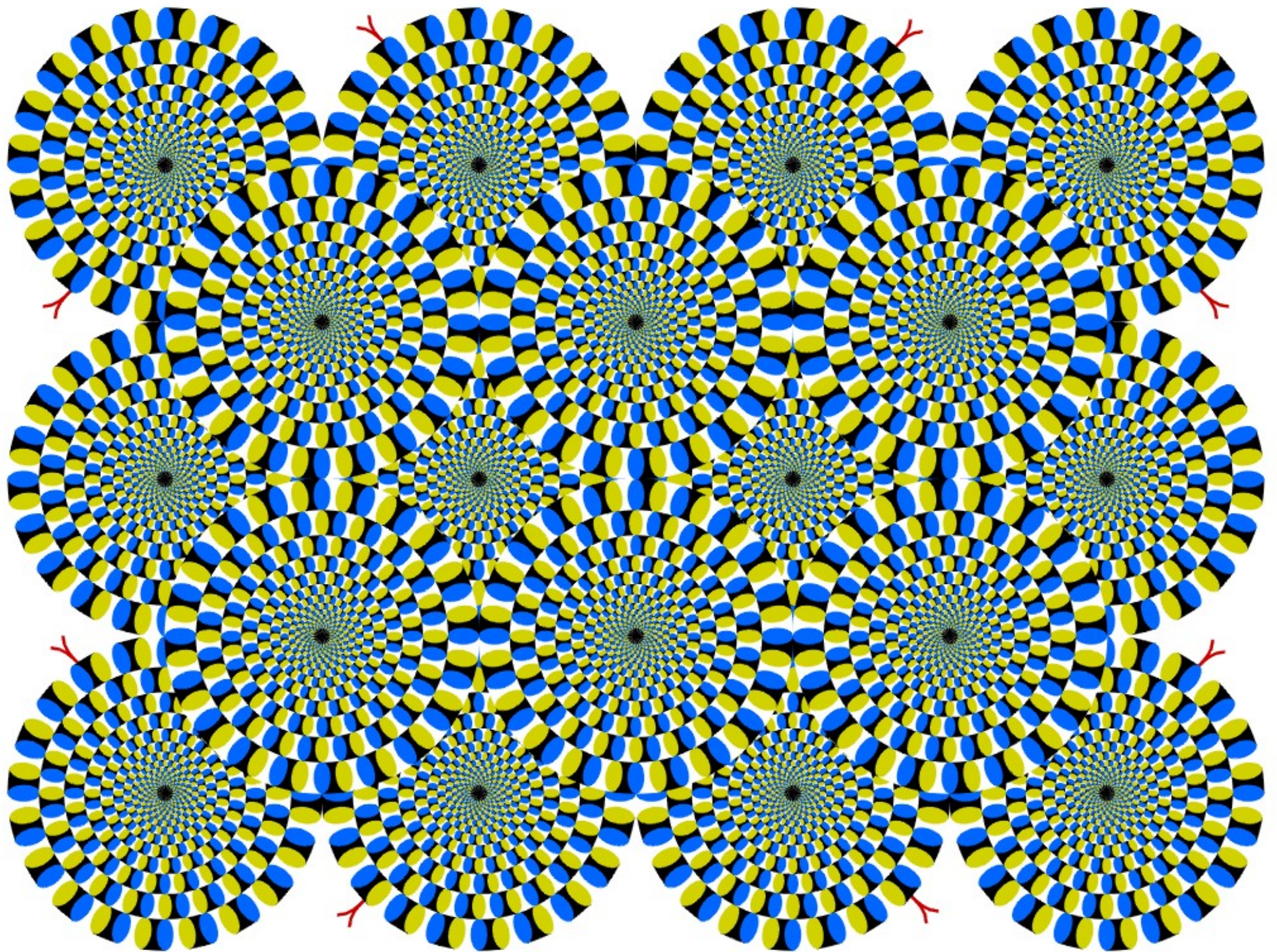
- Occurs when the lightness of an area is influenced by neighboring regions
- Our perception of lightness is not objective, but depends on the surrounding area
- The center square on the right looks lighter because the surrounding area is a darker gray

Perception of Intensity



Perception of Intensity





Illusions!

<http://www.michaelbach.de/>

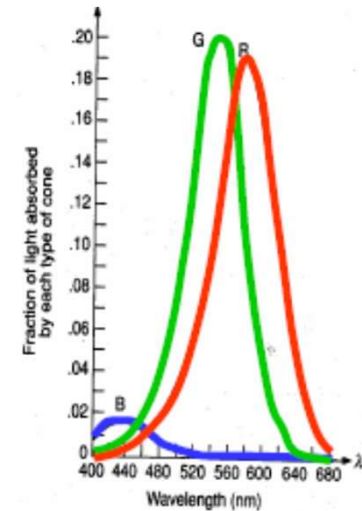
<http://www.ritsumei.ac.jp/~akitaoka/index-e.html>

Outline

- Camera Model
 - A pinhole camera
 - Geometric image formation
 - Photometric image formation
 - Image representation
- Colour
 - Physical and Biological Model
 - Light source
 - Object
 - Observer
 - Tristimulus Theory
 - Colour Systems

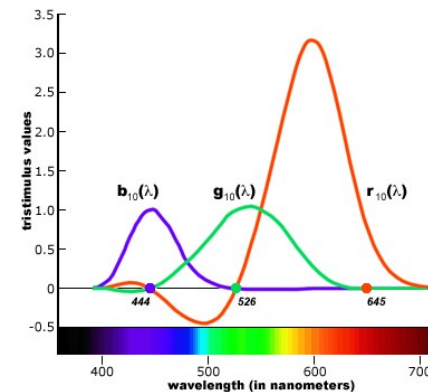
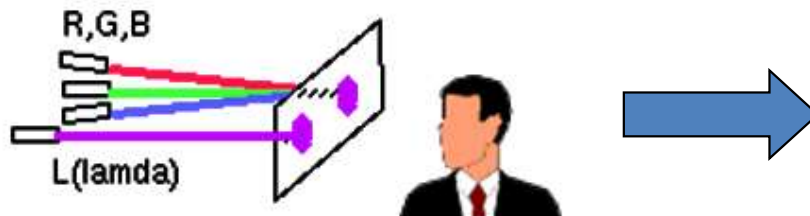
Tristimulus of Color Theory

Spectral-response functions of each of the three types of cones



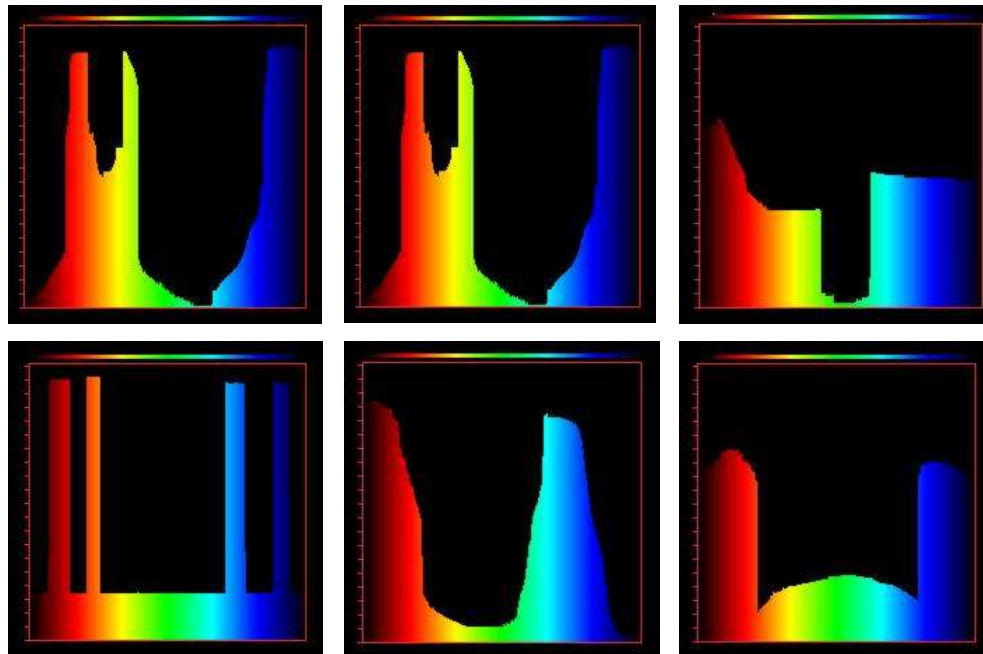
Color matching function based on RGB

- any spectral color can be represented as a linear combination of these primary colors

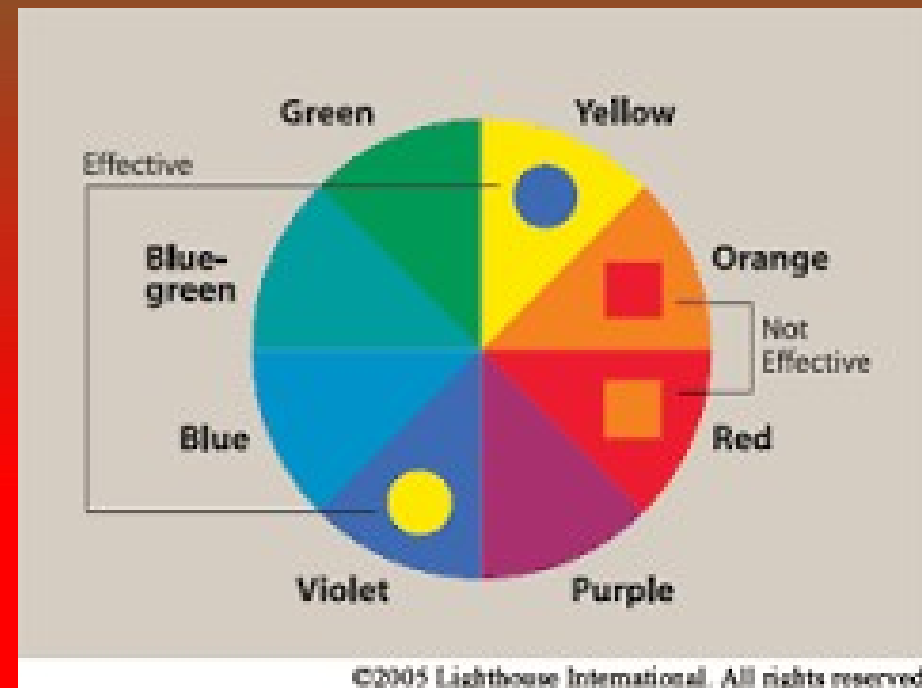
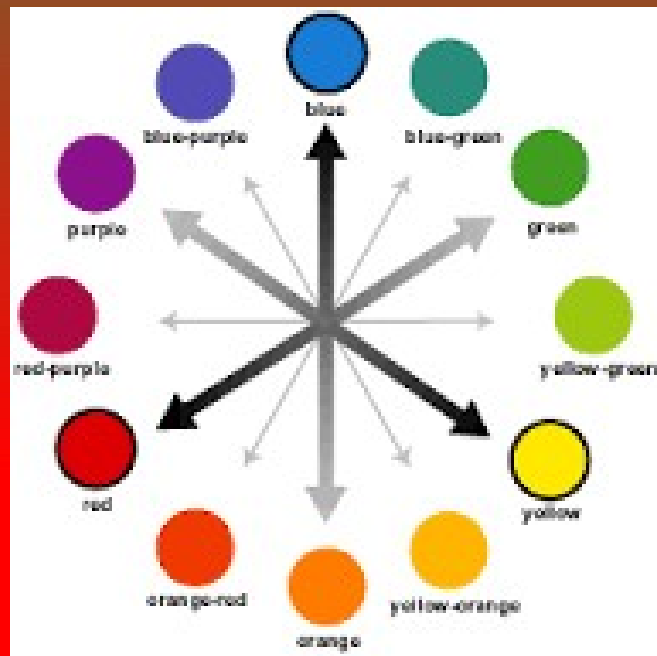


Spectral Energy Distribution

The six spectra below look the same purple to normal color-vision people



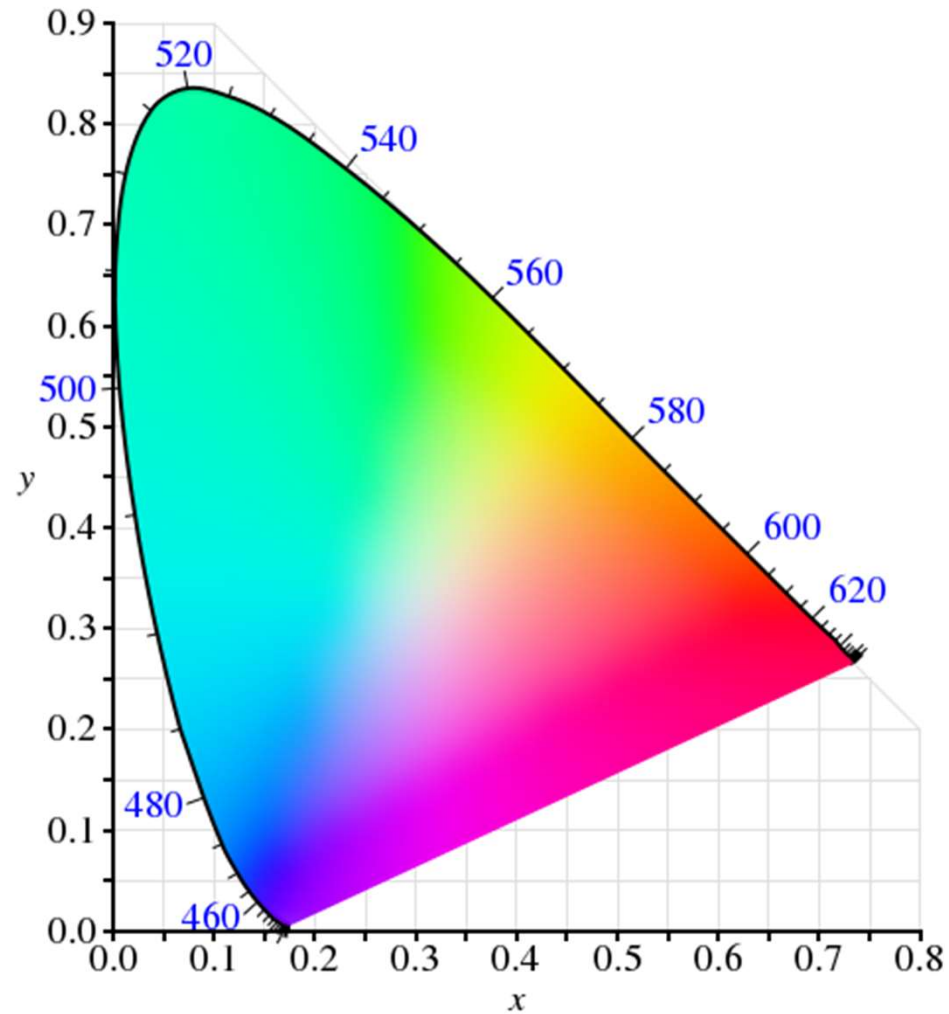
Opponent Colors



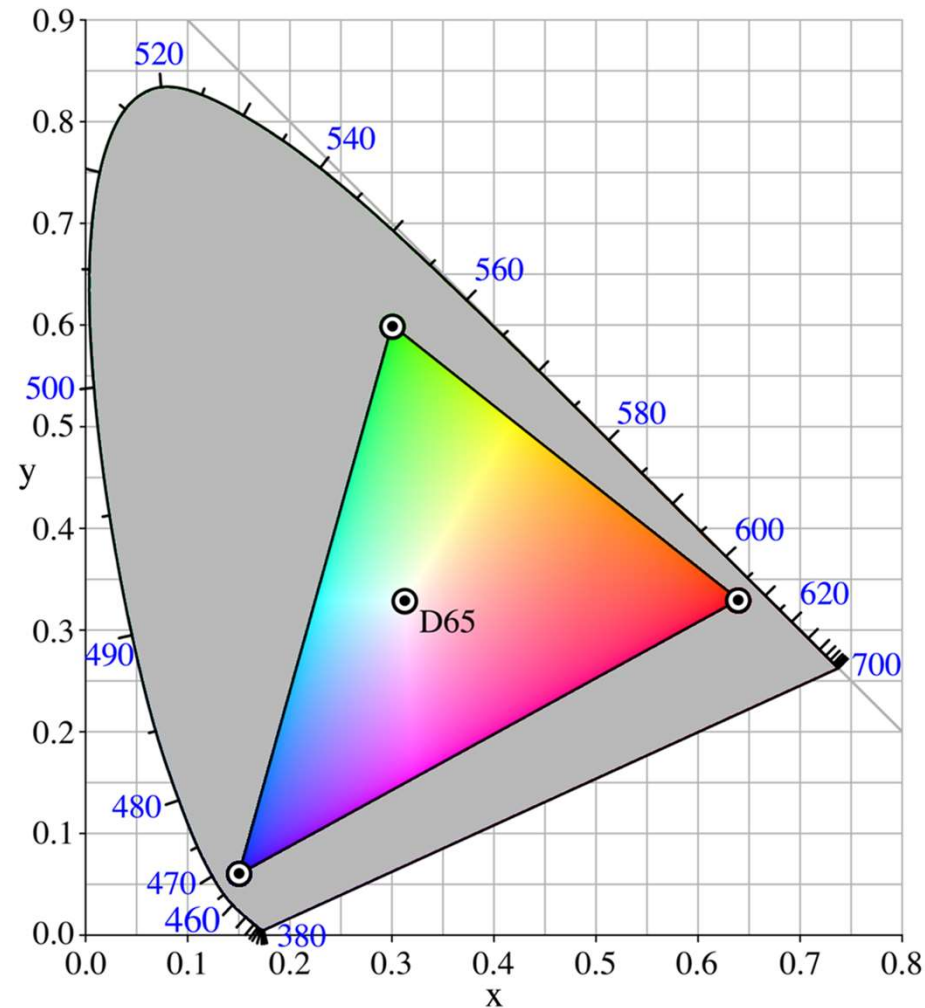
Van Gogh



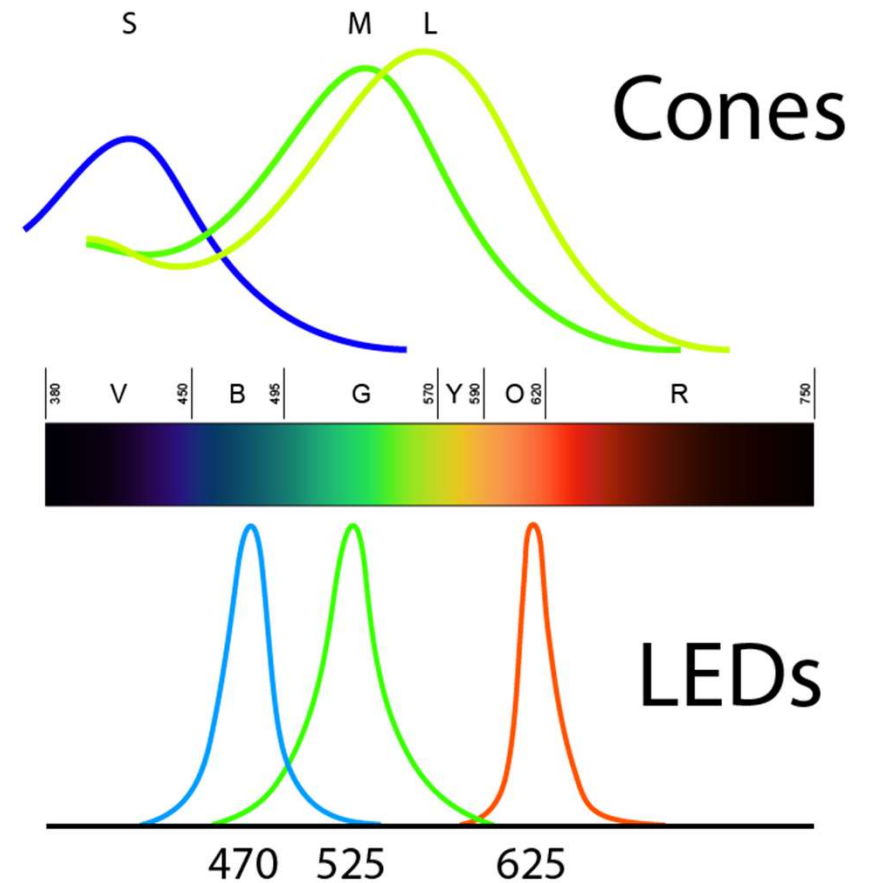
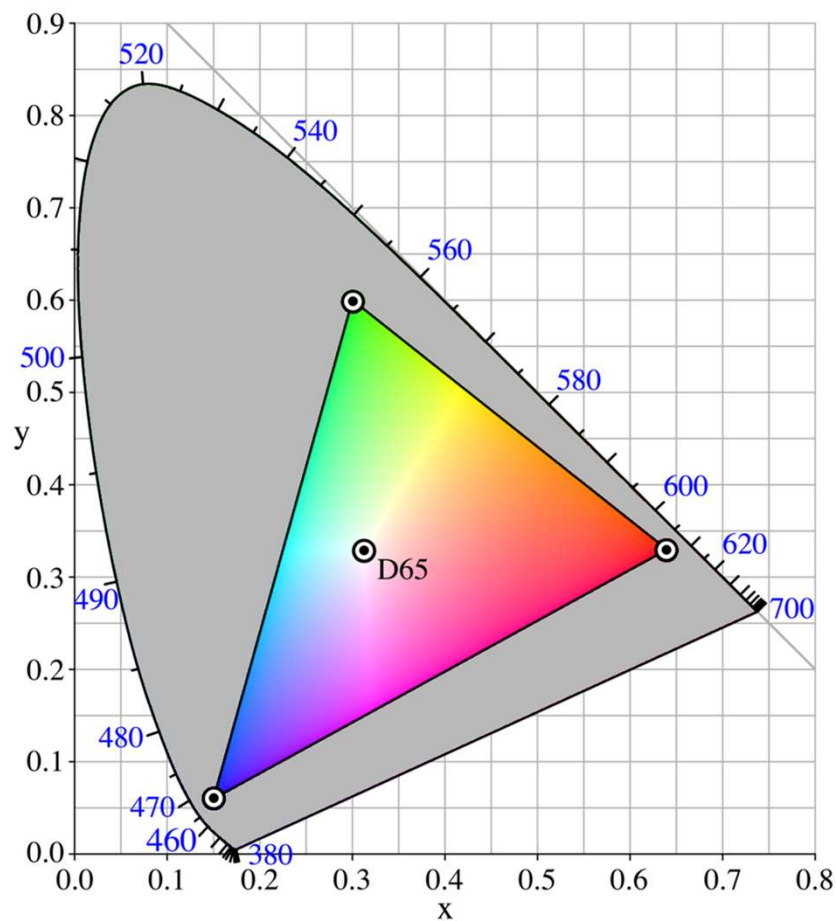
We can make a map of color



Practical sRGB primaries, MSFT 1996



sRGB: most widely used colorspace



What does this mean for computers?

- We represent images as grids of pixels
- Each pixel has a color, 3 components: RGB
- Not every color can be represented in RGB!
 - Have to go out in the real world sometimes
- RGB is made to trick humans, not be accurate
- sRGB is not actually linear, gamma compressed
 - Humans see differences between dark tones more than bright
 - Compress light tones, expand dark tones, more efficient
- Can represent color with 3 numbers
 - #ff00ff; (1.0, 0.0, 1.0); 255,0,255; etc....

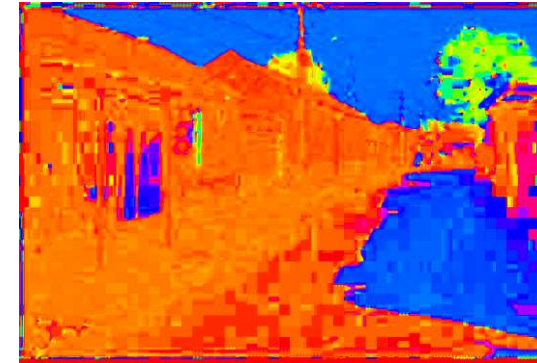
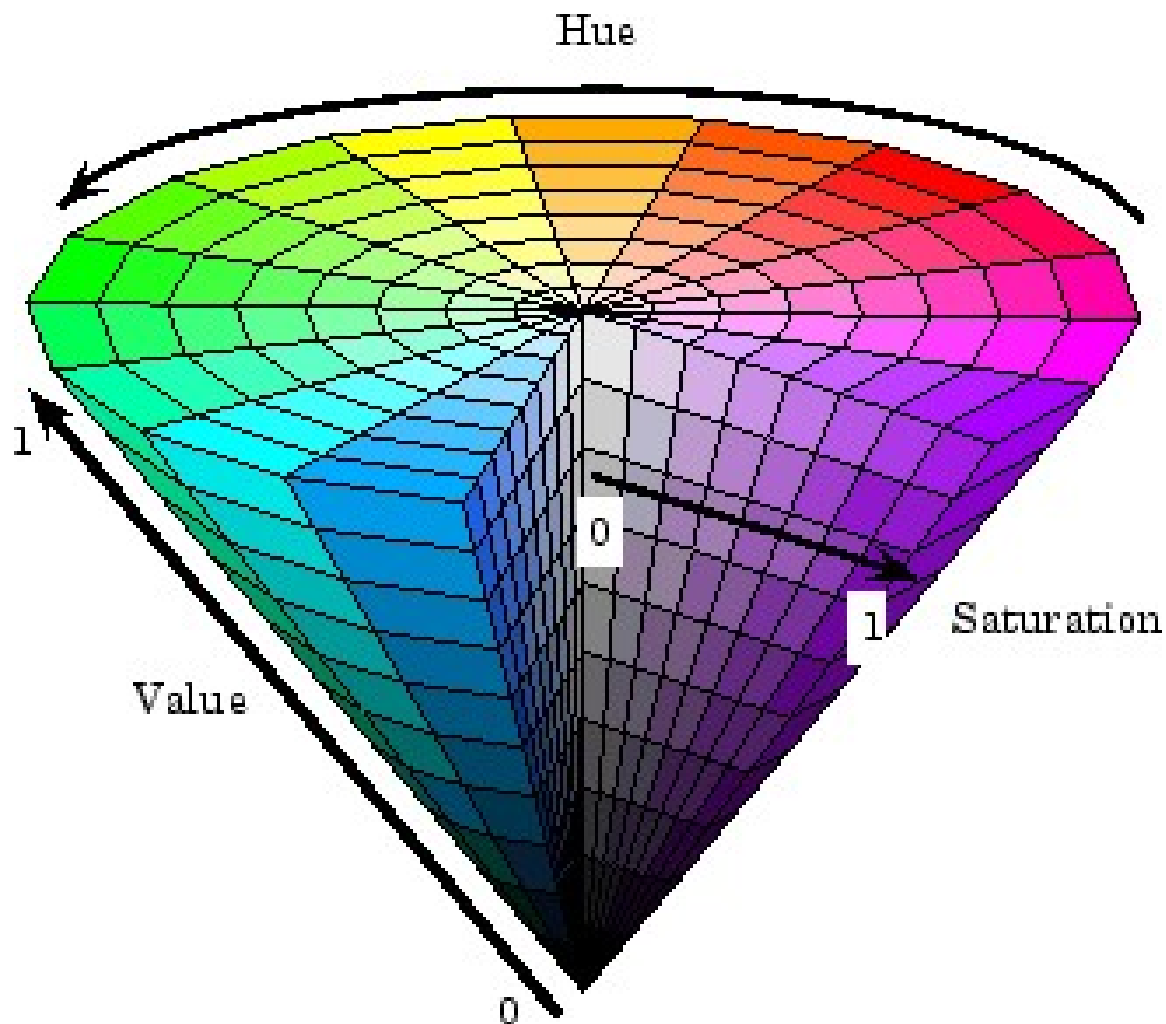
Outline

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 - Light source
 - Object
 - Observer
 - Tristimulus Theory
 - Colour Systems

Color Spaces: HSV



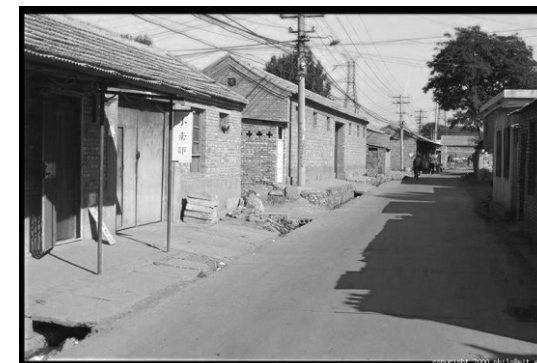
Intuitive color space



H
(S=1,V=1)

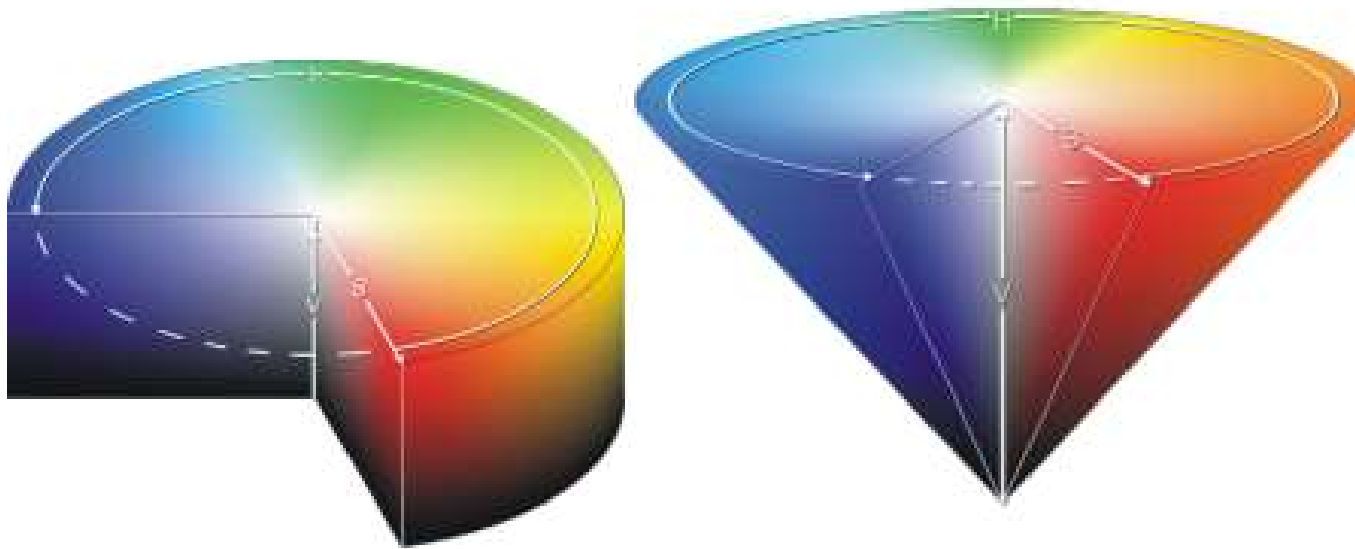
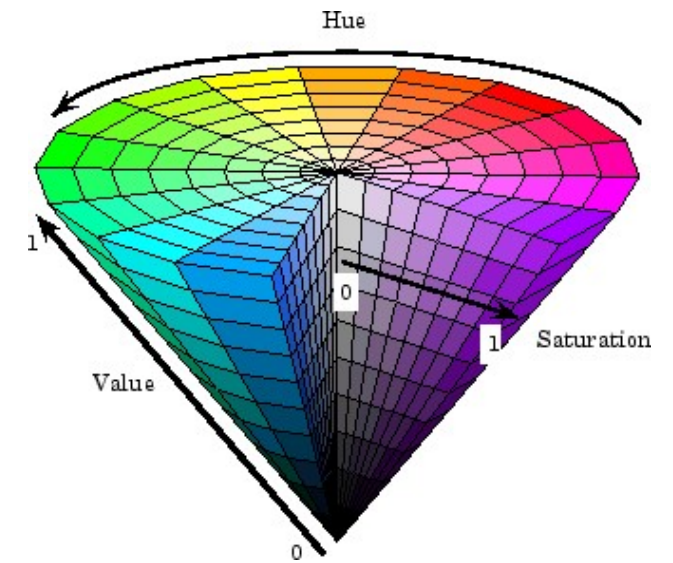


S
(H=1,V=1)

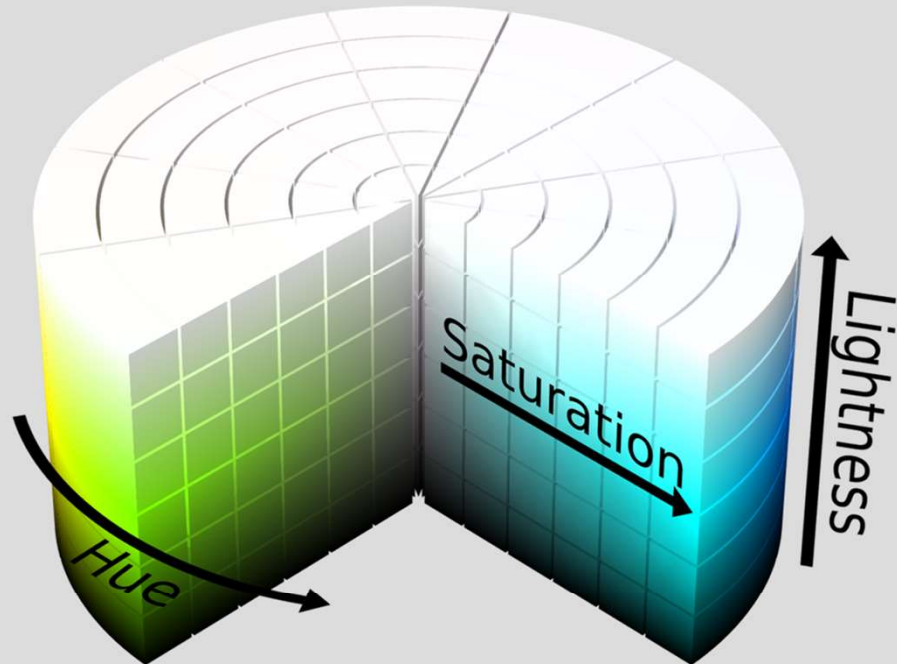


V
(H=1,S=0)

Color Spaces: HSV



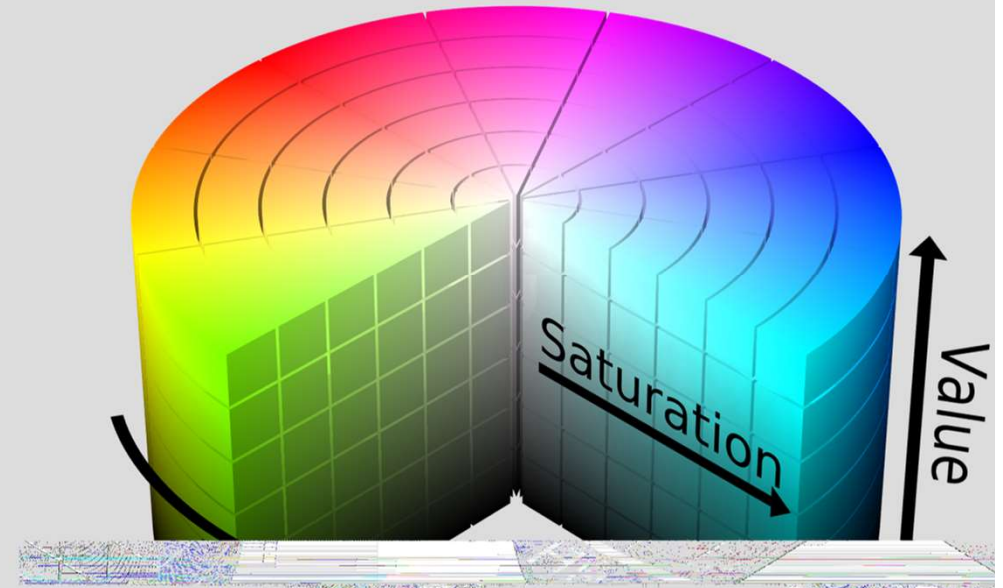
HSL



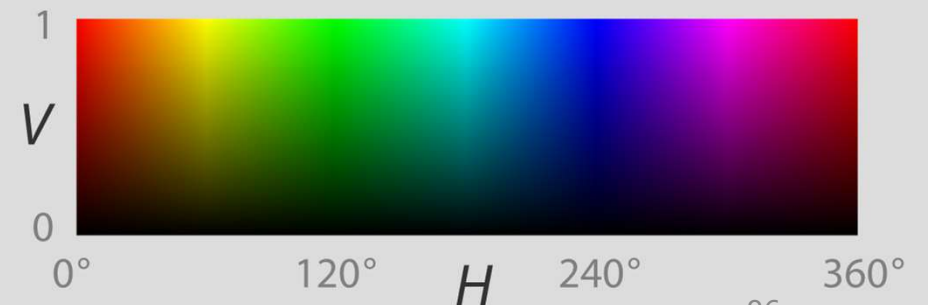
$$S_{HSL} = 1$$



HSV

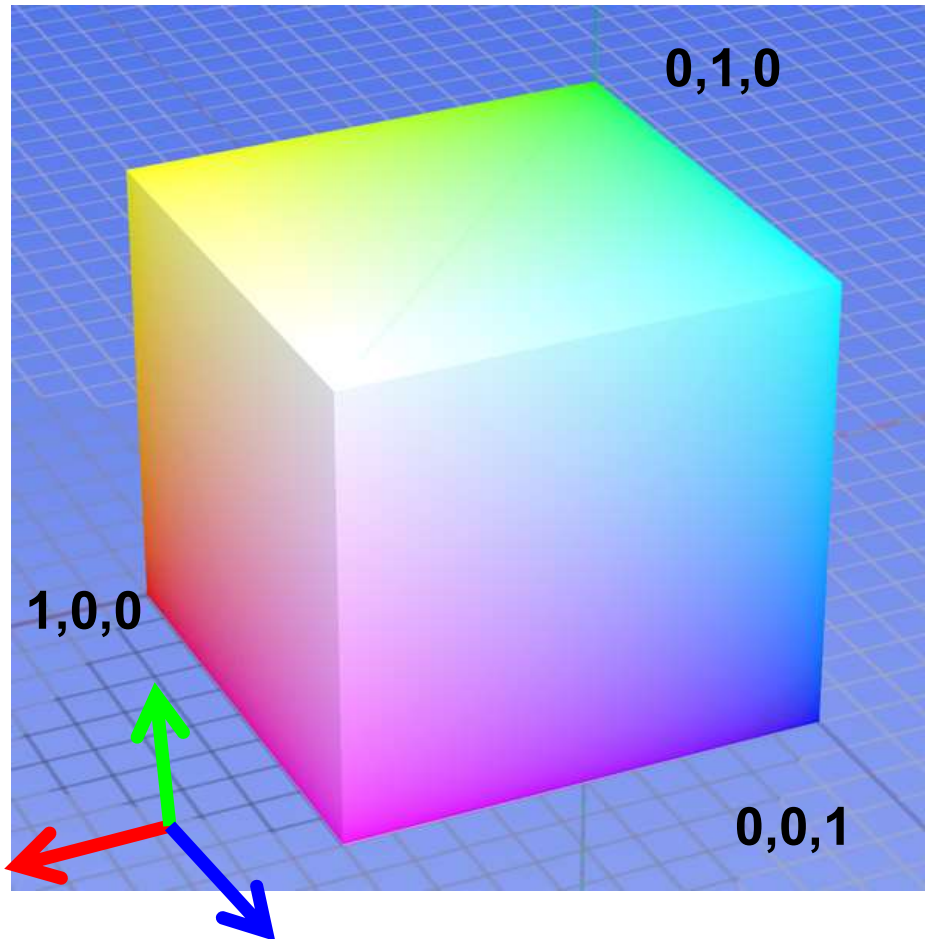


$$S_{HSV} = 1$$



RGB space

Default color space



R
($G=0, B=0$)



G
($R=0, B=0$)



B
($R=0, G=0$)

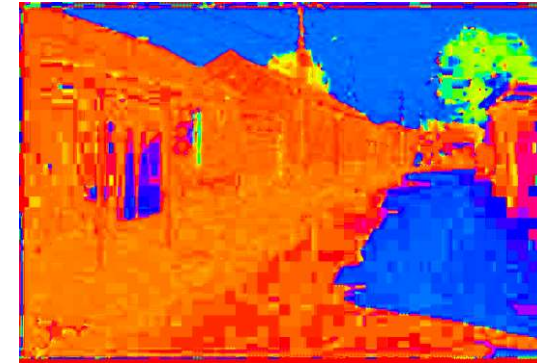
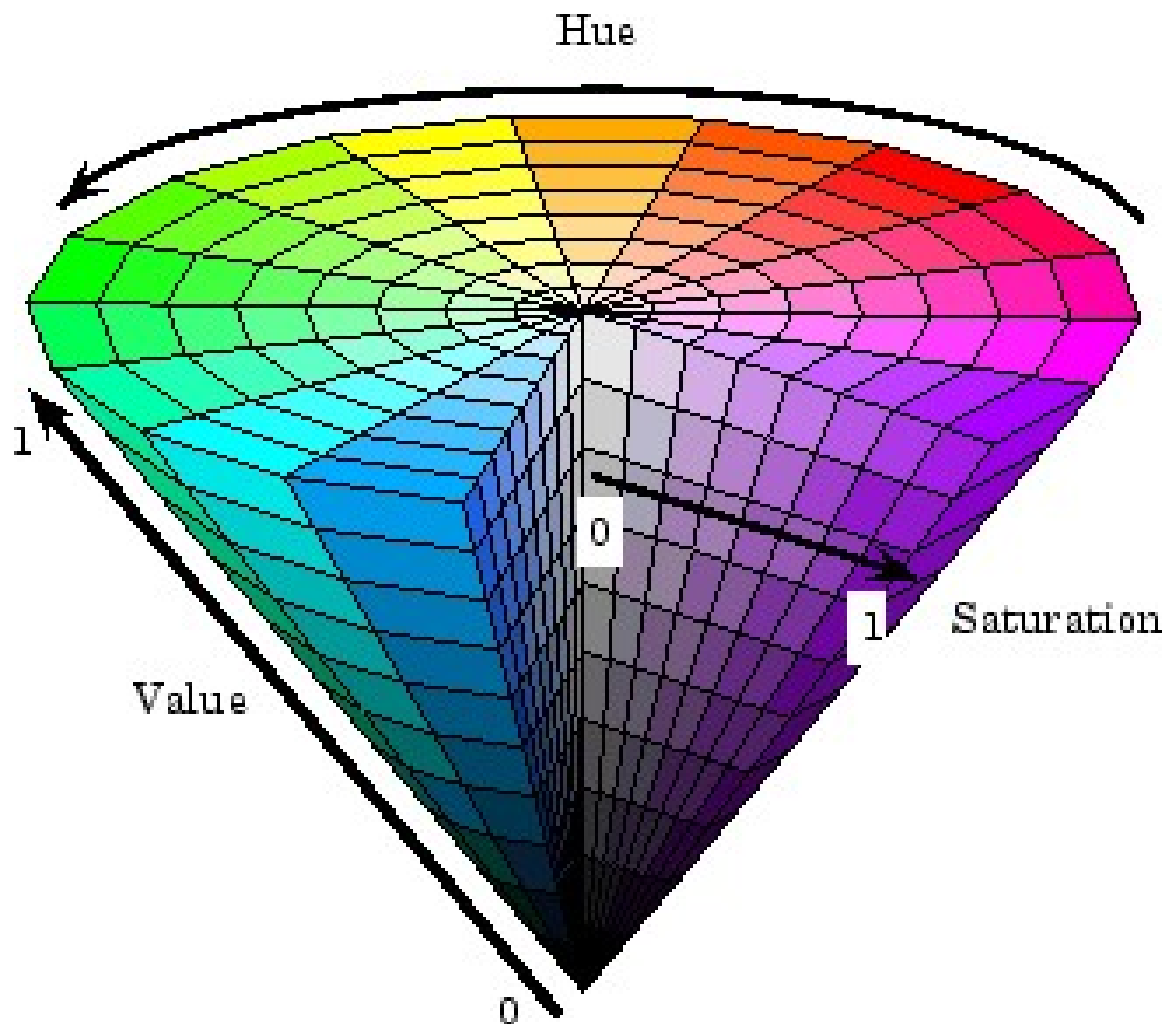
Some drawbacks

- Strongly correlated channels
- Not perceptually meaningful

Color Spaces: HSV



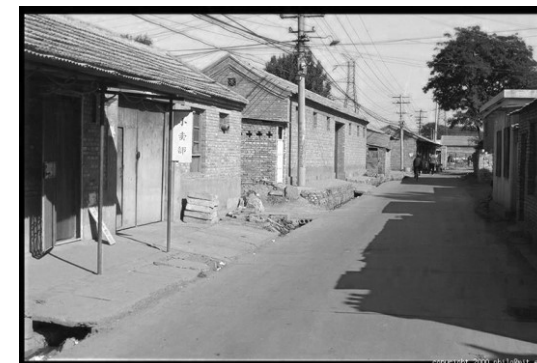
Intuitive color space



H
(S=1,V=1)



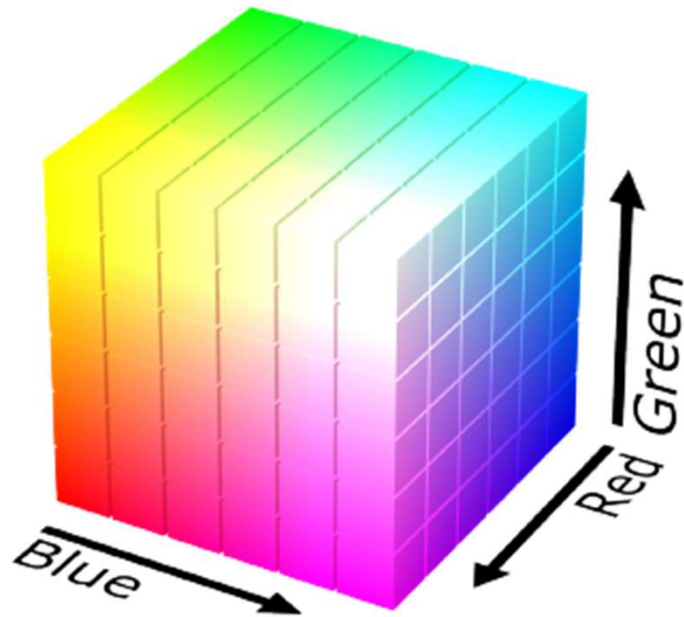
S
(H=1,V=1)



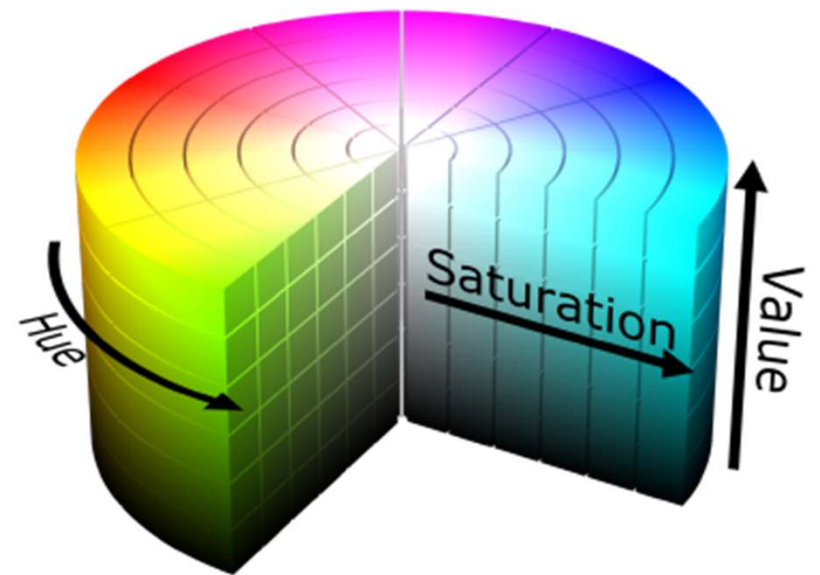
V
(H=1,S=0)

Playing around with colorspace!

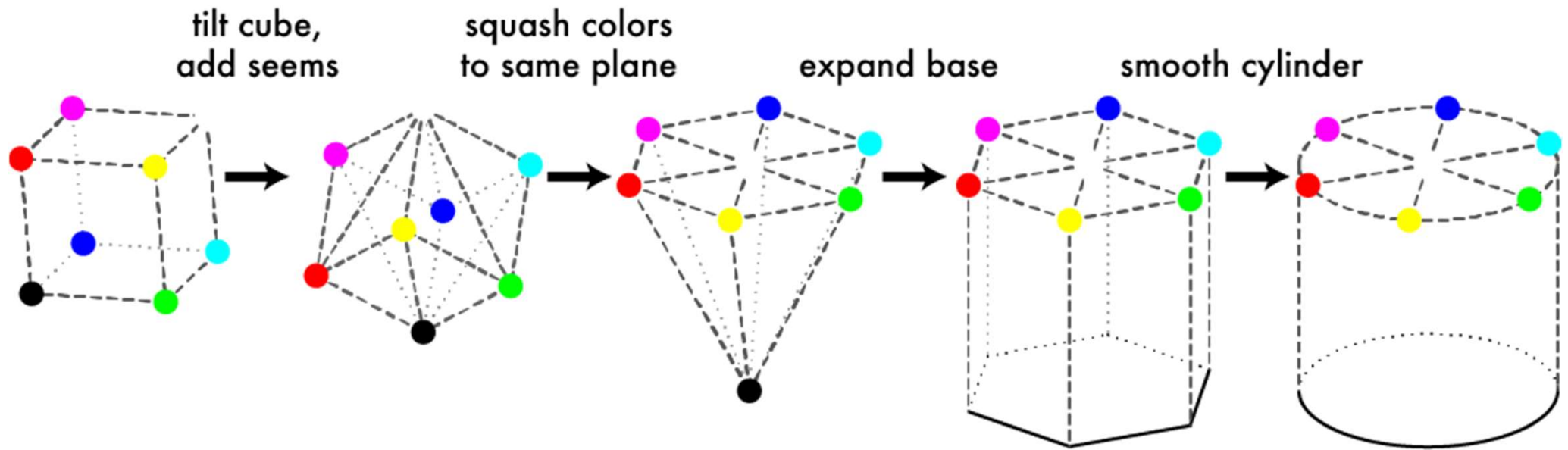
RGB



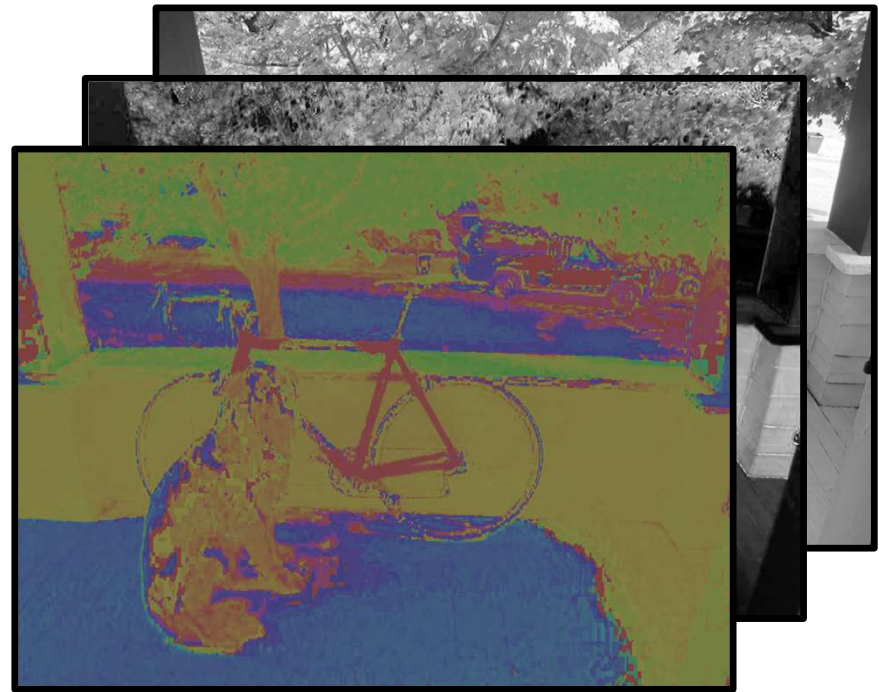
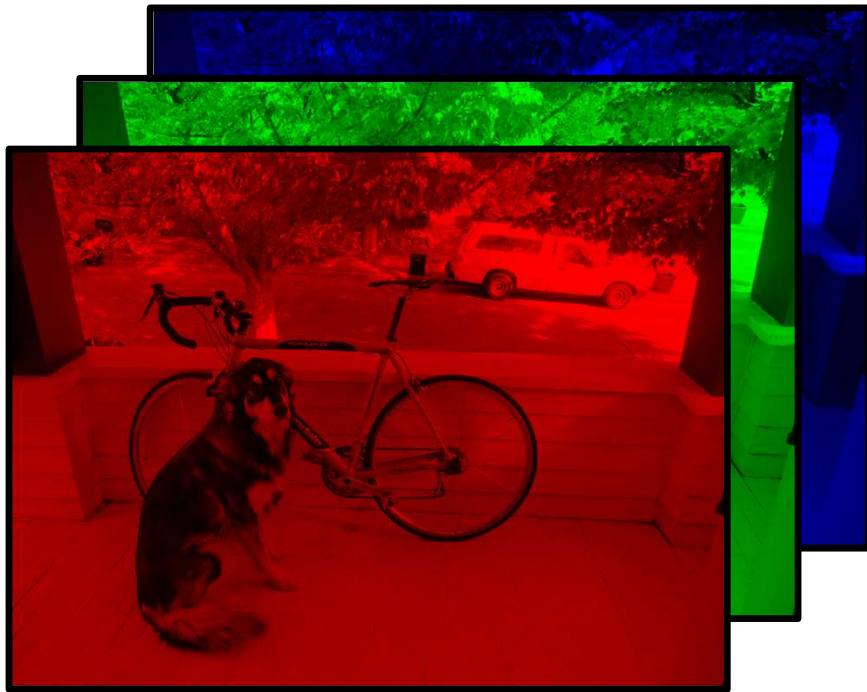
HSV



Geometric HSV to RGB:



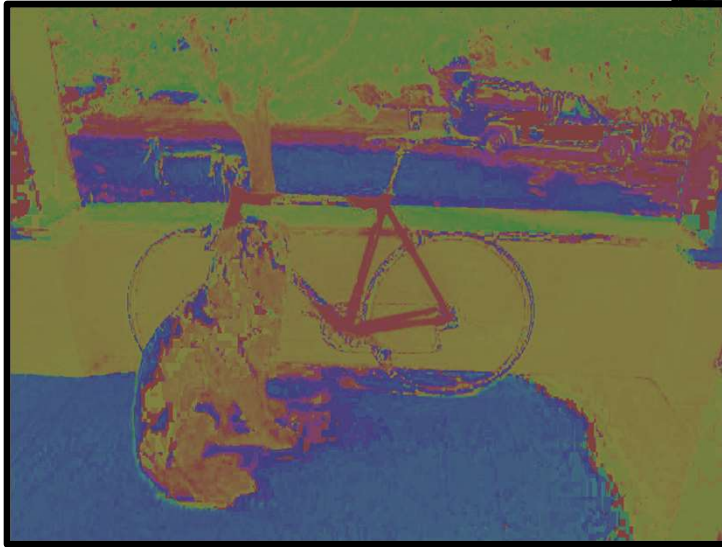
Still 3d tensor, different info



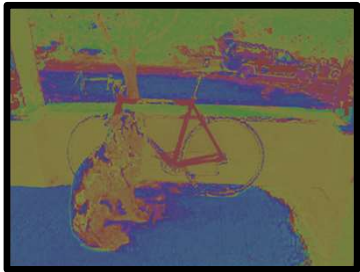
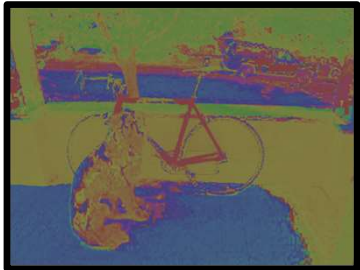
Hue

Saturation

Value



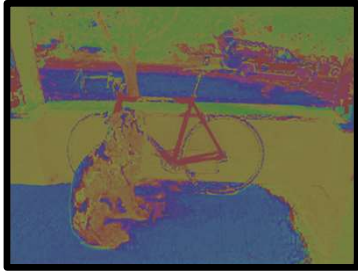
More saturation = intense colors



2x



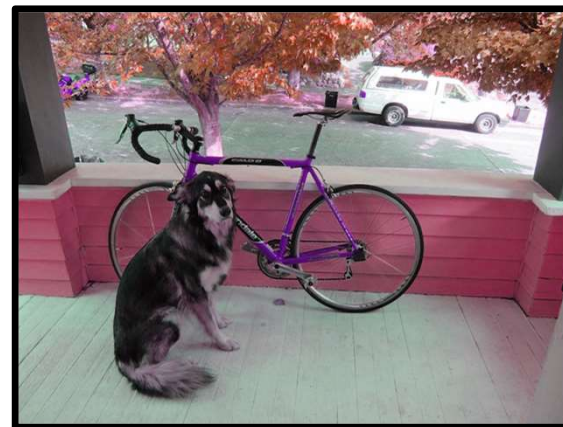
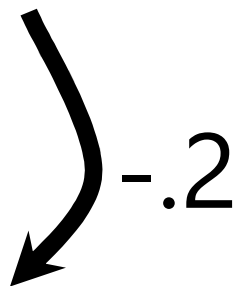
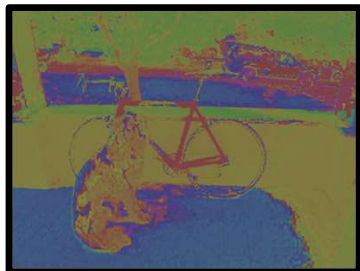
More value = lighter image



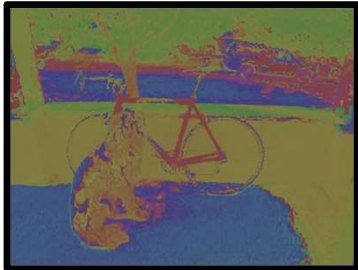
2x



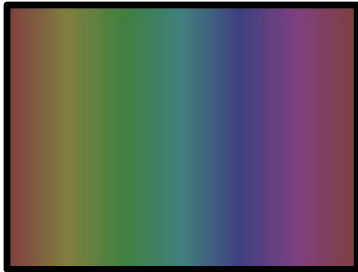
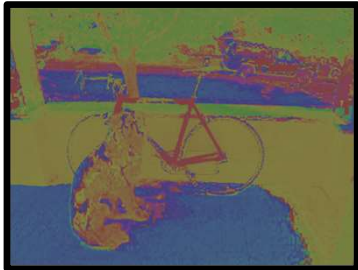
Shift hue = shift colors



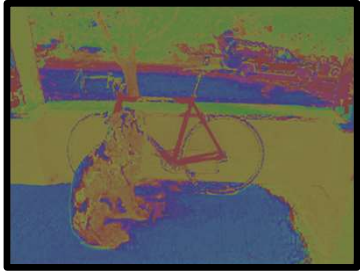
Set hue to your favorite color!



Or pattern...



Increase and threshold saturation





Colorimetry: CIE XYZ-system

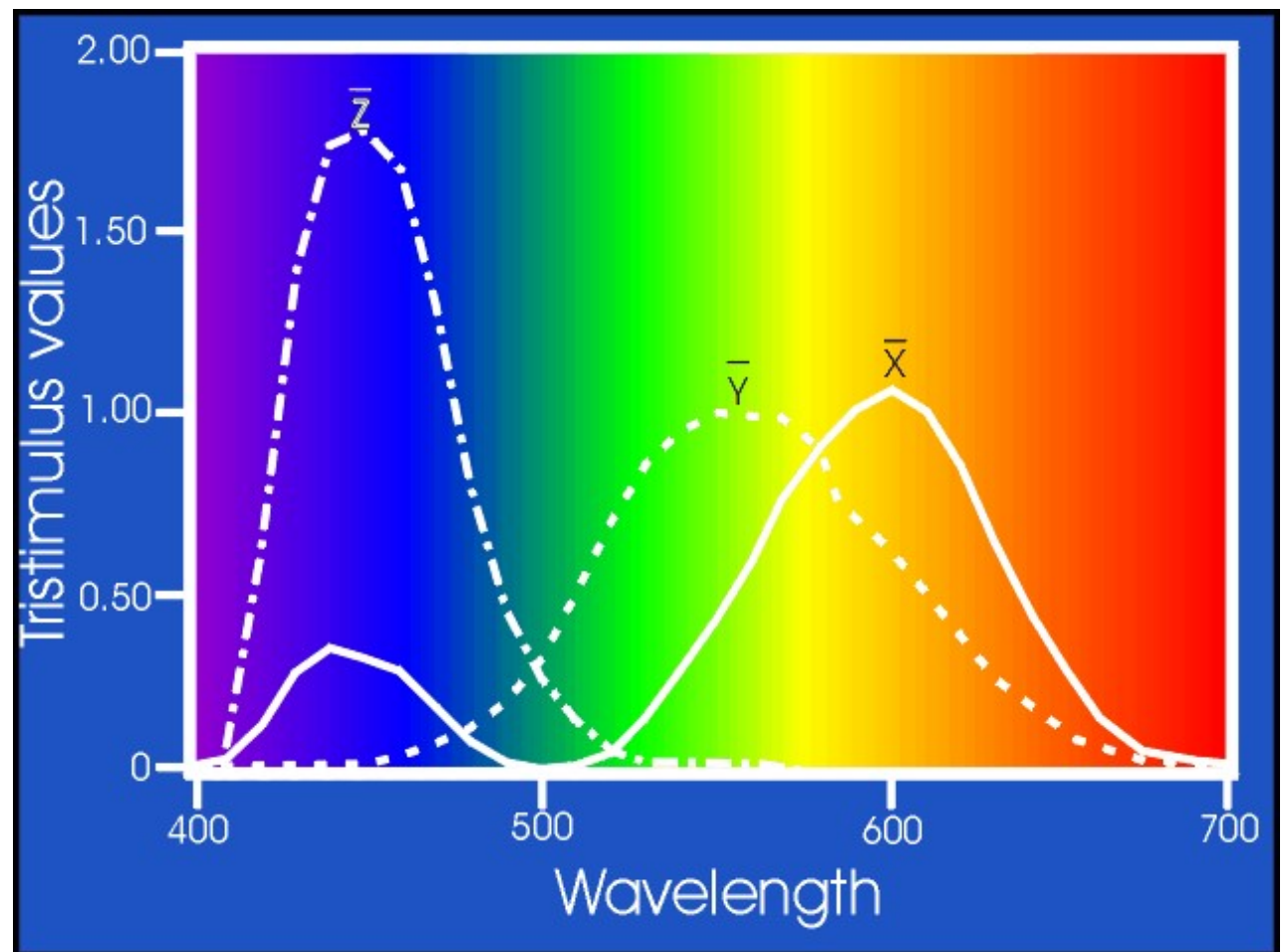
[International Commission on Illumination](#)

Commission Internationale de l'Éclairage (CIE)

$$X = \int_{\lambda} e(\lambda) \rho(\lambda) \bar{x}(\lambda) d\lambda$$

$$Y = \int_{\lambda} e(\lambda) \rho(\lambda) \bar{y}(\lambda) d\lambda$$

$$Z = \int_{\lambda} e(\lambda) \rho(\lambda) \bar{z}(\lambda) d\lambda$$

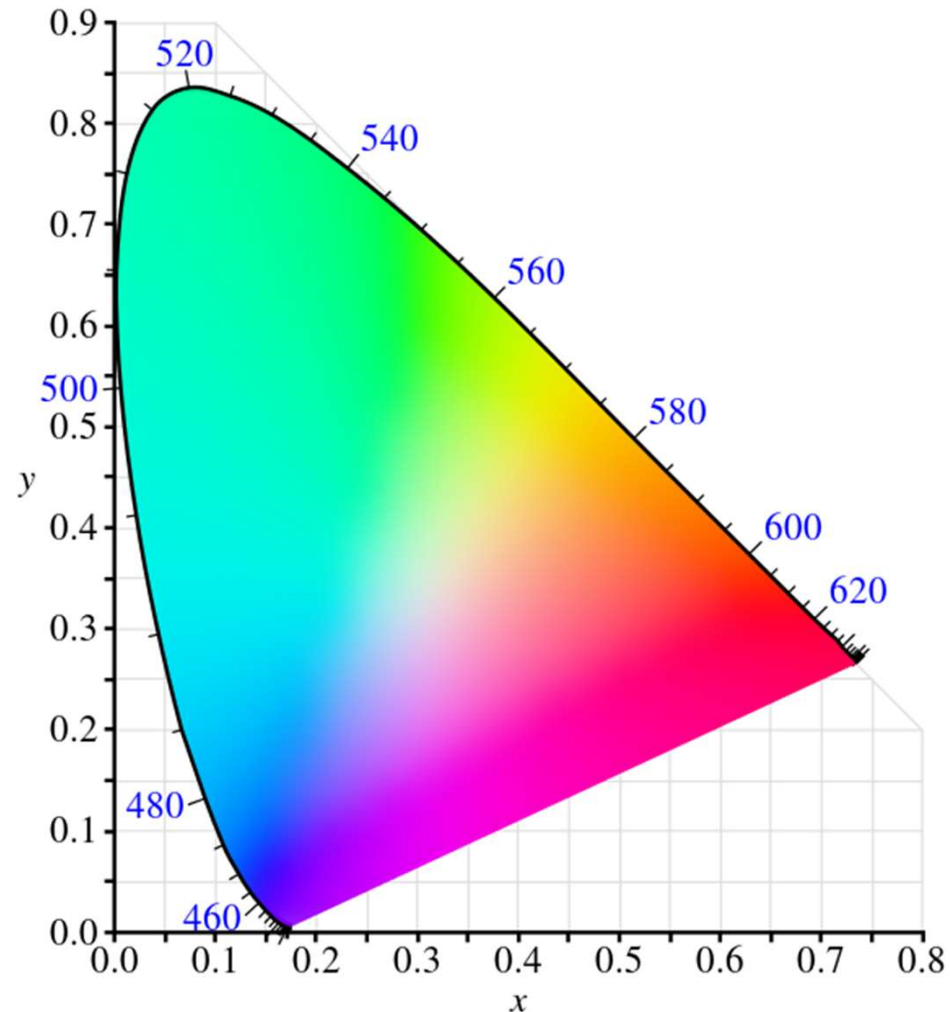


Colorimetry: CIE xy-system

$$x = \frac{X}{X + Y + Z}$$

$$y = \frac{Y}{X + Y + Z}$$

$$z = \frac{Z}{X + Y + Z}$$

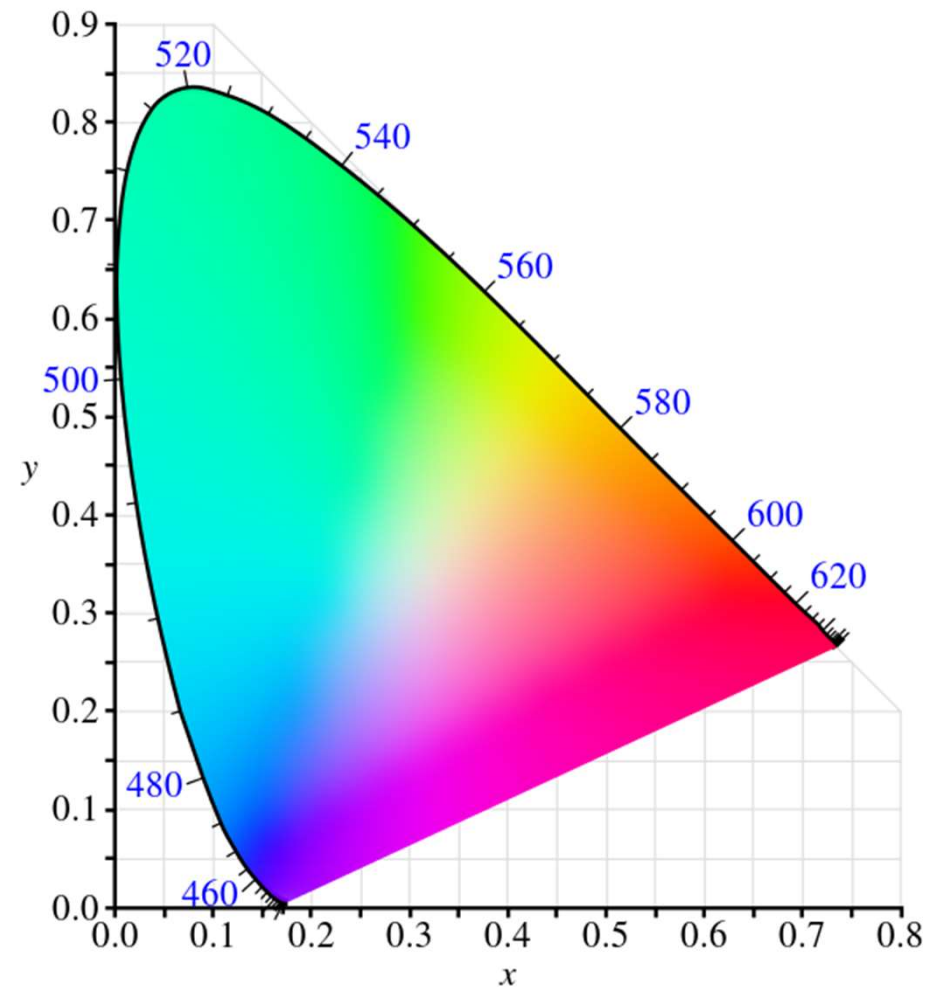


The shape of CIE chart

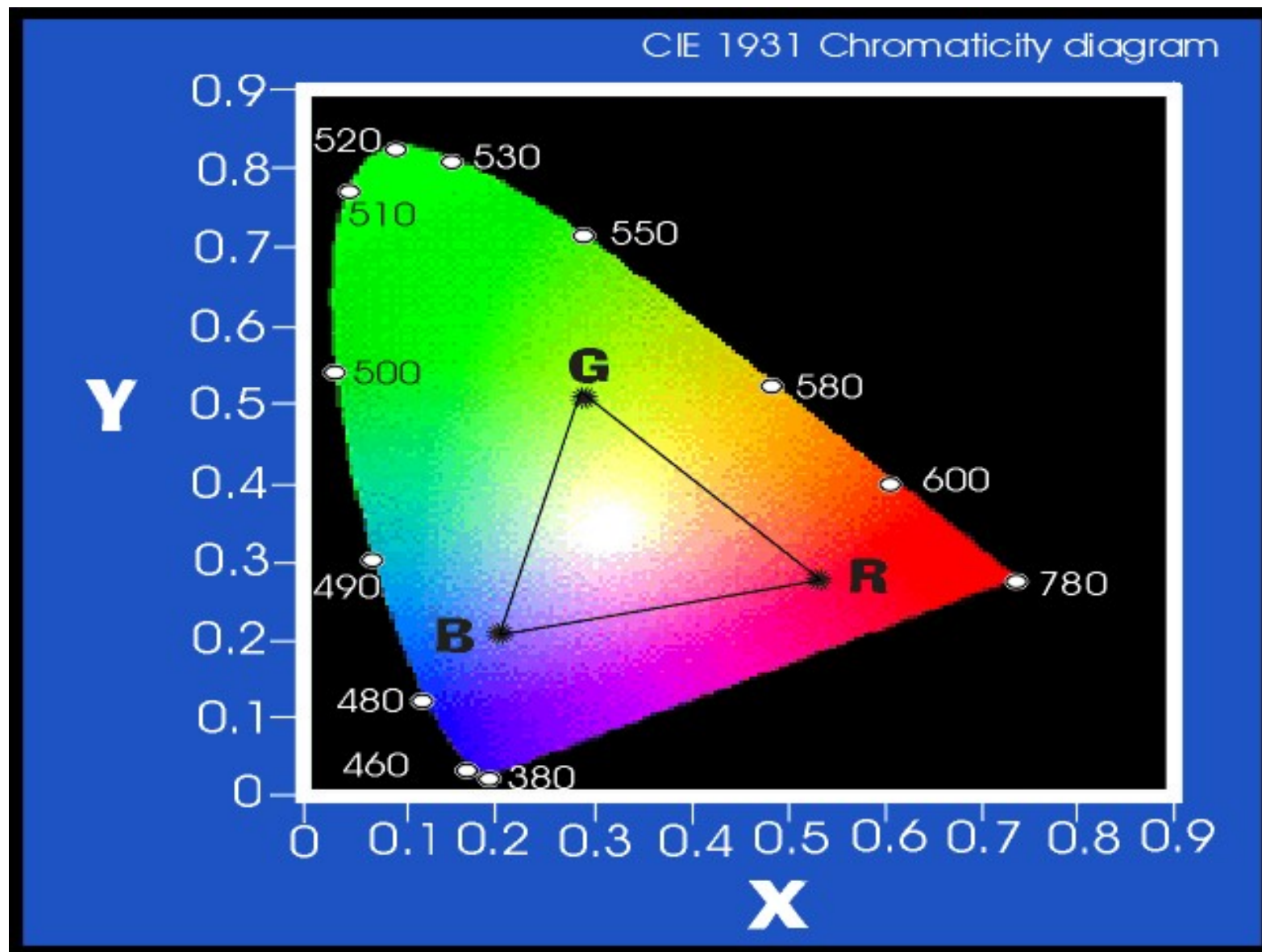
The concept of color can be divided into two parts: brightness and [chromaticity](#). For example, the color white is a bright color, while the color grey is considered to be a less bright version of that same white.

The CIE XYZ color space was deliberately designed so that the Y parameter is a measure of the [luminance](#) of a color.

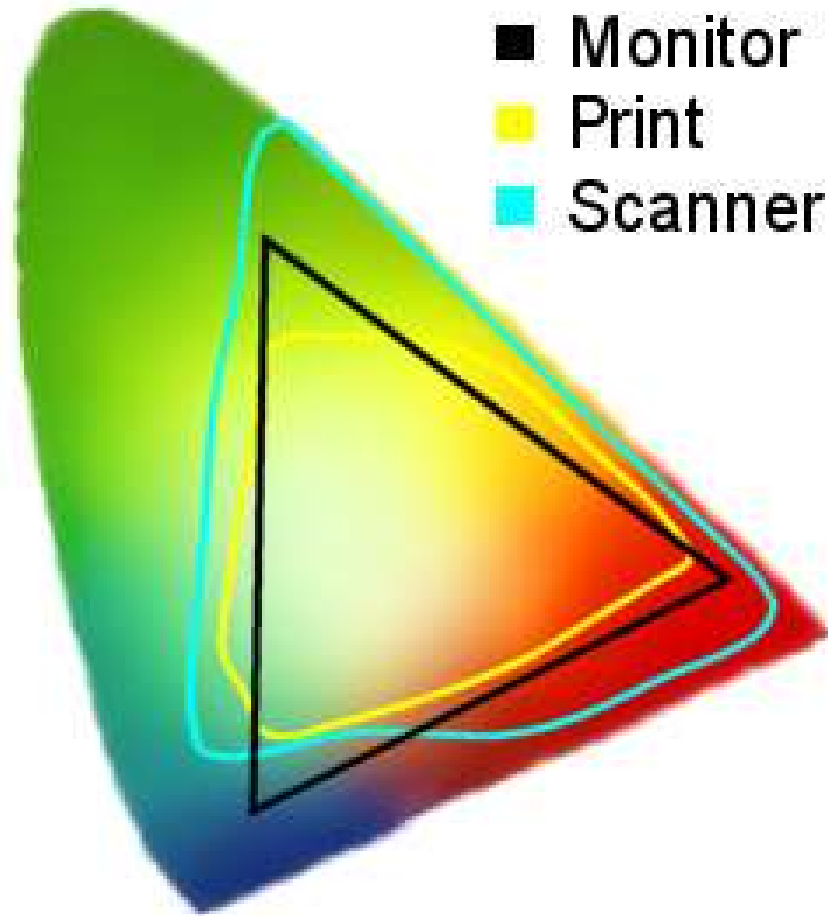
The chromaticity is then specified by the two derived parameters x and y .



Colour Gamuts in the xy-plane



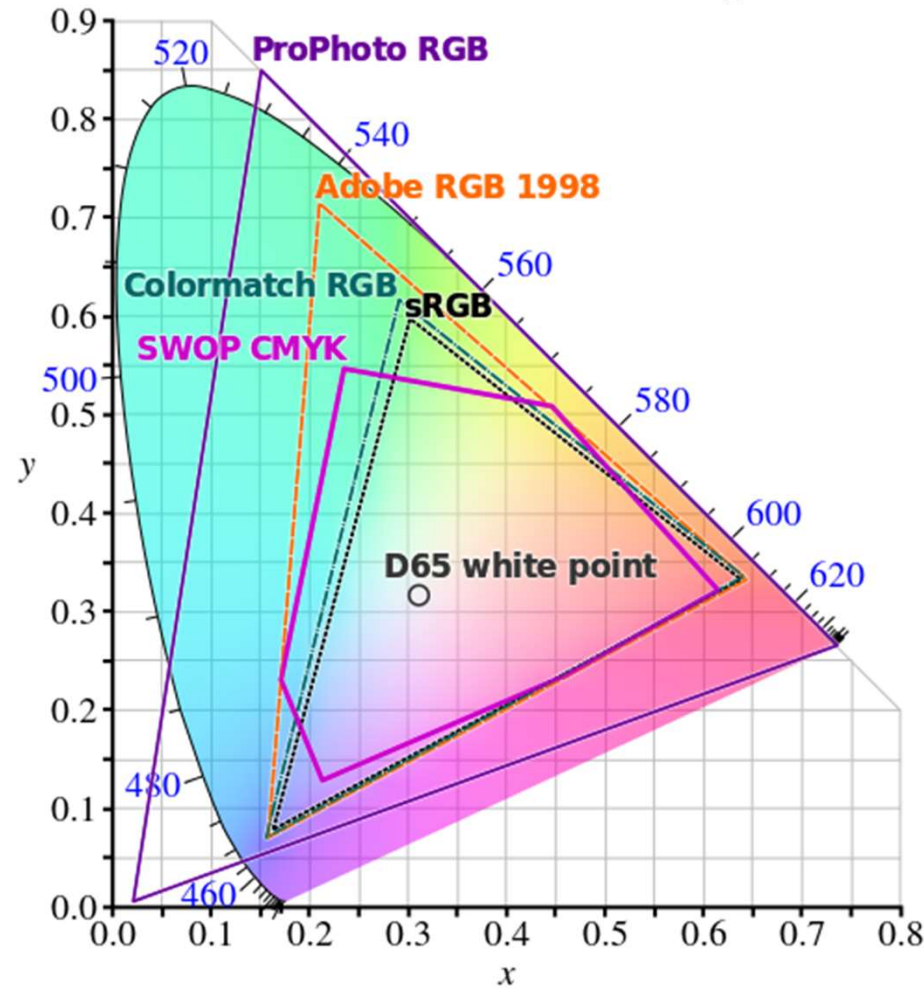
Monitor/Print/Scanner Gamut



The gamut is a triangle because all the color a device (for example a monitor with 3 types of LEDs, R, G, B respectively) can generate are positive linear combinations from those three light sources. This forms the gamut area as a triangle with the three light sources as vertices.

The three light sources are usually not pure color (single wavelength), so that the vertices are usually not on the boundary of the CIE-xy chart. In a physical color context, they are not pure color. However on the device, a single LED color is the purest color it can generate, so they are usually referred as pure color in such context.

MANY different colorspace



Action points

- Read the guidelines in this slides or on Canvas carefully. Feel free to ask Pieter and Seethu if it is not clear to you.
- Check the assignment and start working on it. Lecture 3 is the theory about the assignment.
- Check the exercises.

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